

Portfolio Construction and the Covariance Matrix

Why the optimiser is not the problem, and the matrix is

What this is. Portfolio construction is the step between having views and having positions. It is also the step where more real money has been lost to a correct-looking calculation than anywhere else in quantitative finance, because the calculation is trivial and the inputs are not. This document builds the subject from first principles: what the optimal portfolio is, why feeding it sample estimates destroys it, what the covariance matrix is actually doing in that formula, and what the working set of remedies is — with enough detail to implement one.

How to read it. §1–§2 are the conceptual core, and §2 is the one section nobody should skip: it is the mechanism behind every other section. §3 is history and §4 is the bibliography. §5 is the theory — the closed forms, the regression identity that explains the whole failure, and the random-matrix results that quantify it. §6 and §7 are the two surveys: ways to estimate a covariance matrix, and ways to turn one into weights. §8 shows that most of §6 and §7 are one method with five knobs. §9–§11 are engineering — implementation, costs, evaluation — and §12 synthesises.

If you want the single most important idea and nothing else, read §2.3, §5.4 and §8.2. If you are implementing, read §9 and §10 and treat the rest as reference. If you already know mean-variance theory and want the part that is not in the textbooks, start at §5.4.

Relationship to the other notes. This document is about the cross-sectional problem — given many assets, how much of each. A companion note, Trend-Following in Financial Markets, covers the scalar problem of how large a single position should be, and treats portfolio construction only as far as a trend system needs it. Another, Market Regimes and Machine Learning, argues in its §11.4 that the covariance matrix is where regime information most reliably pays; this document is the detail behind that claim. Where those notes ask what to forecast, this one assumes the forecasts exist and asks what to do with them. Each stands alone.

A warning about scope. [Practice] Nothing here is investment advice, and the numerical illustrations are simulations chosen to isolate a mechanism, not backtests of anything tradeable. Where a number comes from my own simulation rather than from a published study, I say so.

Epistemic tags. Claims are flagged by status:

- [Fact] — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- [Contested] — documented, but with live disagreement about magnitude, robustness, or cause.
- [Hypothesis] — a proposed mechanism, not decisively tested.
- [Practice] — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, derivations, or arithmetic — true by construction rather than by evidence.

Notation. N is the number of assets and T the number of observations in the estimation window. Their ratio $q = N/T$ recurs so often that it has a name: the aspect ratio.

$w \in \mathbb{R}^N$ is the vector of portfolio weights and $\mathbf{1}$ the vector of ones, so $\mathbf{1}'w = 1$ is the fully-invested constraint. $\mu \in \mathbb{R}^N$ holds expected excess returns and $\Sigma \in \mathbb{R}^{N \times N}$ their covariance matrix. Three matrices must be kept apart: Σ is the true covariance, unobservable; S is the sample covariance computed from T observations; and $\hat{\Sigma}$ is whatever estimator you actually use, which may be S , a shrunk version of it, a factor model, or something else. Hats denote estimates generally, so $\hat{\mu}$ is an estimate of μ .

σ_i is asset i 's volatility, $D = \text{diag}(\sigma_1, \dots, \sigma_N)$, and $C = D^{-1}\Sigma D^{-1}$ the correlation matrix, so $\Sigma = DCD$. $\bar{\rho}$ is the average pairwise correlation. $\sigma_p = \sqrt{w'\Sigma w}$ is portfolio volatility and σ^* a volatility target.

$\gamma > 0$ is risk aversion — never a significance level. Eigenvalues are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$ with eigenvectors v_i , so $\Sigma = V\Lambda V'$ with $\Lambda = \text{diag}(\lambda_i)$. $\kappa = \lambda_1/\lambda_N$ is a condition number.

Three parameters control regularisation and are kept distinct: $\delta \in [0, 1]$ is a shrinkage intensity toward a target matrix Φ ; $\nu \geq 0$ is a ridge or turnover penalty weight, and §10.3 shows those two uses are the same thing; and $\theta \in (0, 1)$ is an EWMA decay, whose effective sample size $T_{\text{eff}} = (1 + \theta)/(1 - \theta)$ is the T that belongs in q (§6.3).

In factor models, $F \in \mathbb{R}^{N \times K}$ is the loading matrix for K factors, Ω the $K \times K$ factor covariance, and Ψ the diagonal matrix of idiosyncratic variances, giving $\Sigma = F\Omega F' + \Psi$. R_i^2 always denotes the coefficient of determination from regressing

asset i 's return on the other $N - 1$ assets — the quantity of §5.4, and not a goodness-of-fit for anything else. SR is a Sharpe ratio. Note one near-collision held apart by typography throughout: C is the correlation matrix, $\mathcal{C}$ the constraint set of §8.1.

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1. What portfolio construction is

1.1 The wrong intuition

The intuition almost everyone starts with is:

Portfolio optimisation takes your forecasts and finds the best portfolio.

Every word of that is defensible and the sentence as a whole is badly wrong. It is wrong in a specific way that determines the shape of the entire field, so it is worth being precise about the failure.

An optimiser does not know which parts of its input are forecasts and which are errors. You hand it a vector $\hat{\mu}$ and a matrix $\hat{\Sigma}$; it treats every entry as exact. If the true expected return of asset 17 is 4% and your estimate says 45%, the optimiser does not hedge that possibility, discount it, or flag it — it buys asset 17 with both hands, because that is what the input said to do. And if a pair of assets has a true correlation of 0.90 and your sample says 0.97, the optimiser will construct a long-short position between them at significant leverage, because at 0.97 that spread looks nearly riskless.

So the honest description is the opposite of the intuition:

Optimisation is not a way of extracting value from your forecasts. It is a way of amplifying whatever is in them — signal and error alike, in proportion to how confident the input implies you are.

This is why the field's history (§3) is not a history of better optimisers. The optimisation itself was solved in 1952 and is, computationally, a first-year exercise. Every advance since has been about the inputs: estimating them better, imposing structure so that less has to be estimated, or arranging not to need them at all.

1.2 The right starting point

Portfolio construction is a decision made under uncertainty about the parameters of the decision problem itself.

That framing has three parts, and keeping them separate prevents most confusion:

	The object	Status
Choice variable	w , the weights	You control it exactly
Parameters	μ , Σ	You never observe them; you estimate them
Objective	some functional of $w'\mu$ and $w'\Sigma w$	You choose it, and the choice matters less than people think

The classical theory assumes the parameters are known and studies the choice. The modern theory accepts that they are not and studies the decision rule — the map from data to weights — and asks how well that map performs when it is handed a finite sample. Those are different questions with different answers, and the gap between them is what this document is about.

There is a useful way to state the gap. Write the decision rule as a function $\hat{w}(\text{data})$, and write w^* for the portfolio that is optimal at the true parameters — §1.3 gives its form. The classical question is “is w^* optimal?”, and the answer is yes by construction. The practical question is:

$$\mathbb{E}_{\text{data}}[U(\hat{w}(\text{data}); \mu, \Sigma)] \quad \text{versus} \quad U(w^*; \mu, \Sigma)$$

where the expectation runs over samples and the utility U is always evaluated at the true parameters. The first quantity is what you will actually experience; the second is what the optimiser reports. §2 is about how large that gap is, and it is much larger than intuition suggests.

1.3 The master form

Take the standard mean-variance objective: maximise expected return, penalised by variance, with risk aversion γ .

$$\max_w w' \mu - \frac{\gamma}{2} w' \Sigma w$$

Differentiate with respect to w and set to zero: $\mu - \gamma \Sigma w = 0$. Hence the master form of this entire document:

$$w^* = \frac{1}{\gamma} \Sigma^{-1} \mu$$

Three things about this equation deserve emphasis before anything else is said.

It is a ratio, not a product. The weight is expected return divided by risk, in a matrix sense. Division is where things blow up. Every pathology in §2 traces back to the fact that Σ appears inverted.

$1/\gamma$ is just a scalar. It sets leverage and nothing else. The direction of the portfolio — the relative weights, which is the interesting part — is $\Sigma^{-1} \mu$ regardless of risk aversion. This is Tobin's separation result, and it means the sizing question (§10 of the trend note) and the composition question are genuinely separable. You can get the direction right and the scale wrong, or vice versa, and they are different mistakes with different fixes.

Almost everything in the field is this formula with particular choices. Equal weighting, minimum variance, risk parity, maximum diversification, Black–Litterman, hierarchical risk parity — the survey in §7 works through them, and §8 shows they are all $\Sigma^{-1} \mu$ under a specific pair of assumptions about what μ is and how Σ has been regularised. Several of them look like they have abandoned optimisation entirely. None of them has.

1.4 The spine, stated once

Three claims generate the rest of the document. They are worth stating together, because each section is an elaboration of one of them. They are numbered C1–C3 here; §8.1's slots S1–S5 are a different numbering for a different purpose, and the two are not related.

C1 — the master form. Every portfolio construction method is $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ for some choice of $\hat{\mu}$ (often implicit, often deliberately degenerate) and some regularised $\hat{\Sigma}$. The honest exception is hierarchical risk parity (§7.11), which reaches a similar destination without ever forming the inverse.

C2 — the mechanism. Inverting $\hat{\Sigma}$ places the largest bets in the directions with the smallest estimated variance, and those are exactly the directions the estimate gets most wrong. The optimiser's confidence is highest precisely where it is least justified.

C3 — the diagnostic. How badly this goes is governed, to first order, by one number: the aspect ratio $q = N/T$. Not by N , not by T , but by their ratio. This governs the covariance half of the problem; the return view is limited by calendar span instead (§2.5), which is a separate and larger constraint.

And the payoff, which §8 makes precise:

Every repair in the field — shrinkage, factor models, position constraints, resampling, Bayesian priors, transaction-cost penalties, hierarchical clustering — does the same thing. Each adds structure that lifts the smallest eigenvalues of $\hat{\Sigma}$, and so suppresses the positions taken in those directions. They differ in where the structure comes from and how it is justified, not in what it does.

That last sentence is the reason this document is organised as a taxonomy rather than a catalogue. A practitioner who believes shrinkage, constraints and turnover penalties are three independent tools will apply all three at full strength and end up with a portfolio far more heavily regularised than intended.

1.5 What portfolio construction is not

Naming the neighbours prevents a lot of confusion, because several of them are routinely called by this name.

Not this	Why it is different
Signal generation	μ arrives from outside. Construction takes it as given and is not responsible for whether it is any good — though §2.5 shows it is responsible for how much damage a bad μ does
Position sizing	The scalar $1/\gamma$, or a volatility target. Orthogonal to composition (§1.3), and a different problem
Diversification	An outcome, sometimes, and a poor objective. A portfolio can be maximally diversified and terrible
Risk management	A constraint and monitoring layer around construction. Answers “what if I am wrong”, not “what is best”
Asset allocation	The same mathematics at $N \approx 10$, where estimation error is mild and the hard problems of §2 barely appear
Execution	How you get from current weights to target weights. Genuinely separate — except that §10 shows costs feed back into the target itself

The distinction that causes the most trouble in practice is the first. A disappointing backtest is nearly always blamed on the signal, when the construction step is at least as likely a culprit and is far easier to fix.

1.6 A worked instance, to fix ideas

Two assets, both with 20% annual volatility, correlation ρ , and expected excess returns of 6% and 5%. Risk aversion is set so the portfolio is fully invested. What does the master form give?

With $\Sigma = 0.04 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ and $\mu = (0.06, 0.05)'$, one line of algebra does it. Up to the positive factor $1/[0.04(1 - \rho^2)]$ in front of Σ^{-1} , which normalisation will kill anyway,

$$\Sigma^{-1}\mu \propto \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} = \begin{pmatrix} \mu_1 - \rho\mu_2 \\ \mu_2 - \rho\mu_1 \end{pmatrix},$$

whose two entries sum to $(\mu_1 + \mu_2)(1 - \rho)$. Dividing by that sum to make the weights add to one gives the fully-invested optimum:

$$w_1 = \frac{\mu_1 - \rho\mu_2}{(\mu_1 + \mu_2)(1 - \rho)}, \quad w_2 = 1 - w_1.$$

Tabulating:

ρ	w_1	w_2	Gross	Comment
0.00	0.55	0.45	1.0×	Mild tilt toward the better asset
0.50	0.64	0.36	1.0×	

ρ	w_1	w_2	Gross	Comment
0.90	1.36	-0.36	1.7×	Now short the second asset
0.95	2.27	-1.27	3.5×	An 18× spread trade on a 1% return difference
0.99	9.55	-8.55	18.1×	

This little table is the whole document in miniature. As the two assets become more alike, the optimiser stops allocating between them and starts arbitraging between them, and the leverage it applies goes to infinity. Nothing here is a bug — given exact inputs, these really are the optimal weights, and at $\rho = 0.99$ the spread really is nearly riskless.

The problem is that $\rho = 0.99$ is not something you know. It is something you estimated, and the large-sample standard error of a correlation estimated from T observations is roughly $(1 - \rho^2)/\sqrt{T}$. From two years of daily data ($T = 500$) that is 0.0009, so a two-standard-error band around a sample correlation of 0.99 runs from 0.988 to 0.992 — and the corresponding gross exposures run from 15.3× to 22.0×. A band narrow enough that you would happily round it away in any other context makes the trade at the top end 44% larger than the trade at the bottom.

Note what did not happen there. The correlation estimate was good — four significant figures, a tight confidence interval, no misspecification, no outliers, no regime change. The instability comes from the $1/(1 - \rho)$ in the denominator, which is a property of the map from parameters to weights, not of the estimate. Precision in the input does not imply stability in the output, and the optimiser reports no diagnostic that would tell you which regime you are in.

Scale that from two assets to five hundred, and you have \$2.

§1 Key takeaways

1. An optimiser cannot distinguish a forecast from an error; it amplifies both in proportion to the confidence its inputs imply. Better optimisation of bad inputs makes things worse, not better.
2. The master form is $w^* = \gamma^{-1} \Sigma^{-1} \mu$. Everything in the field is a choice of what to put in the two slots and how to regularise the inverse.
3. Composition and leverage separate cleanly: $1/\gamma$ sets scale, and $\Sigma^{-1} \mu$ sets direction. Treat them as two problems, because their failure modes are unrelated.
4. The classical question — is w^* optimal — has been settled since 1952 and is not interesting. The practical question is how a decision rule performs when fed a finite sample.
5. Inverting the covariance matrix means the optimiser's largest positions are spread trades between assets it believes are nearly identical. Those beliefs are the least reliable thing it has.
6. Portfolio construction is not signal generation, sizing, diversification, or risk management. Conflating it with signal generation is the most expensive of those errors, because it sends you to fix the wrong thing.

2. Why the naive answer fails

This section is the mechanism behind everything that follows. If you read one part of this document, read §2.3.

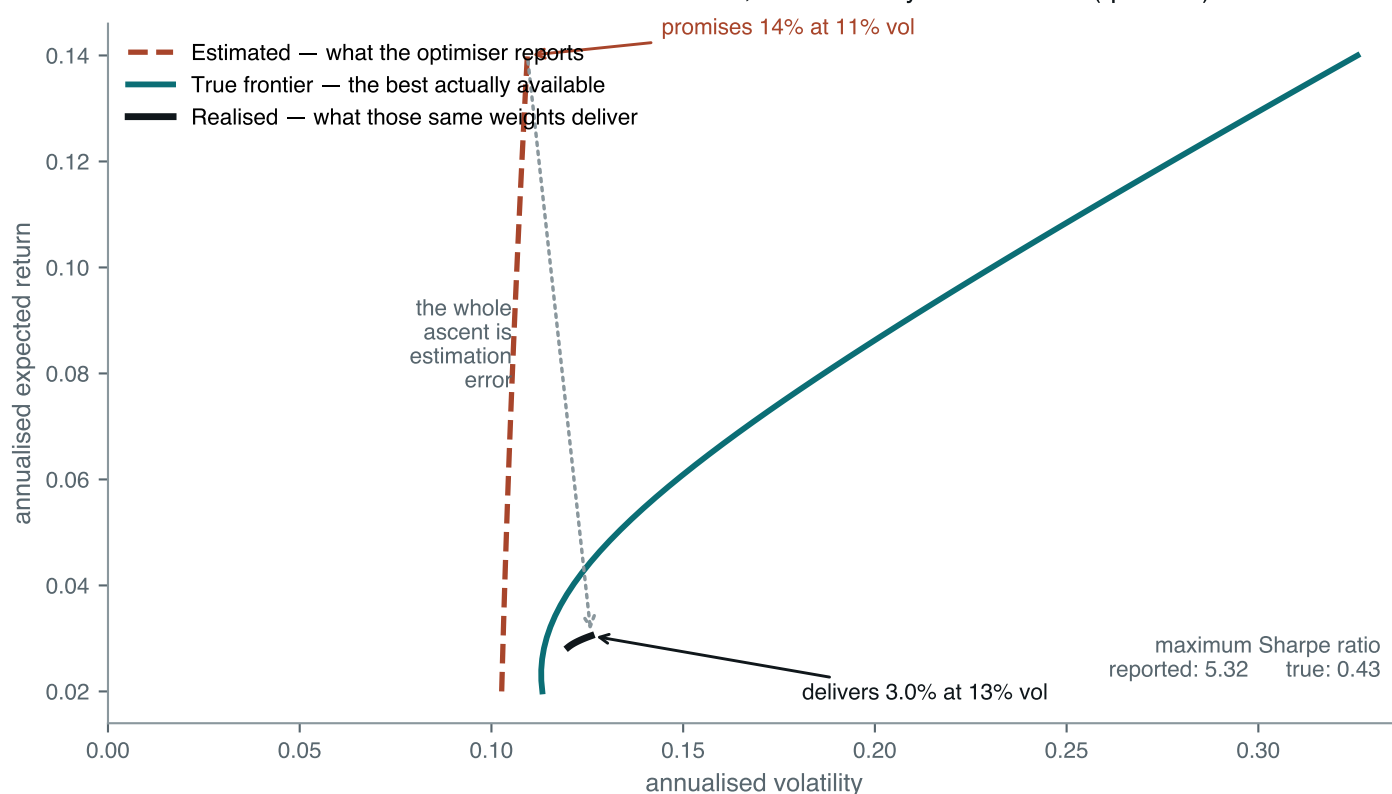
2.1 The demonstration

Here is the experiment, run under conditions maximally favourable to the optimiser. Fifty assets. Returns generated by a one-factor model — a market factor plus idiosyncratic noise — with betas between 0.6 and 1.4, idiosyncratic volatilities between 15% and 35%, and expected returns of 5% per unit of beta plus a small alpha. Two years of daily data, $T = 500$, so $q = N/T = 0.10$, which is a comfortable ratio by industry standards.

Crucially, the data are drawn from exactly the model the optimiser assumes. Returns are IID, multivariate normal, stationary, with no fat tails, no regime changes, no missing data, no non-synchronous trading and no misspecification of any kind. Whatever goes wrong is estimation error and nothing else.

Compute the sample mean $\hat{\mu}$ and sample covariance S , trace out the mean-variance frontier, and plot three curves: the frontier the optimiser reports (using $\hat{\mu}, S$), the frontier that is genuinely available (using μ, Σ), and what the optimiser's chosen weights actually deliver (the estimated weights, evaluated at the true parameters).

The frontier illusion — N = 50 assets, T = 500 daily observations (q = 0.10)



Returns are drawn from exactly the model the optimiser assumes, so every part of the gap is estimation error — none of it is misspecification. Ask the estimated frontier for more return and it hands you more leverage on noise: the realised curve barely moves.

The numbers, from my own simulation:

Quantity	Value
True maximum Sharpe ratio	0.43
Sharpe ratio the optimiser reports	5.32
True expected returns, range across assets	1.9% to 7.8%
Sample mean returns, range across assets	−38.9% to +44.9%
Estimated frontier at its top end	14.0% return, 10.9% volatility
What those weights actually deliver	3.0% return, 12.6% volatility

Three observations, in increasing order of importance.

The reported Sharpe ratio is off by a factor of twelve. The optimiser is not slightly optimistic; it believes it has found something twelve times better than the best thing in the universe it was shown. Any research process that treats an in-sample optimised Sharpe ratio as evidence is not measuring what it thinks.

The sample means are an order of magnitude more dispersed than the truth. The true spread of expected returns is about six percentage points; the sample spread is about eighty-four. Roughly speaking, essentially all of the cross-sectional variation in a two-year sample of mean returns is noise. This is not a small-sample curiosity — with daily data and annual returns of a few percent against annual volatility of 20%, the standard error of an annualised mean is the annual volatility divided by the square root of the number of years, so over two years it is $0.20/\sqrt{2} \approx 14\%$ — several times the quantity being estimated. [Fact] This ratio barely improves with more data, because the standard error of a mean falls as $1/\sqrt{T}$ in calendar time, not in sample size: sampling the same two years more finely does nothing at all (Merton, 1980).

The realised frontier is flat. This is the finding that should change behaviour. Ask the estimated frontier for 4% expected return and you get 2.8%; ask it for 14% and you get 3.0%. The entire ascent of the dashed curve — the part that looks like the optimiser trading risk for return, the part a committee would discuss — buys you two-tenths of one percent. What it actually purchases is leverage on noise.

2.2 Error maximisation

Michaud (1989) named the phenomenon: mean-variance optimisers are estimation-error maximisers. The argument is a selection argument and takes one sentence.

The optimiser overweights assets with high estimated return, low estimated variance, and low estimated correlation to the rest. Among a large group of assets, the ones whose estimates happen to be highest, lowest, and lowest are disproportionately those whose errors went up, down, and down. So the optimiser's largest positions are systematically concentrated in the assets whose inputs are most overstated, and this happens more aggressively the more assets you give it to choose among.

It is worth being precise about what kind of failure this is, because it is often described loosely as “overfitting” and that undersells it. Overfitting usually means a model with too many parameters fitting noise in a training set. Here there is no model and no fitting: $\Sigma^{-1}\mu$ has no free parameters and is not chosen to fit anything. The optimiser is a maximum, and a maximum over noisy inputs is biased upward by construction — the same reason the best-performing of a hundred coin-flippers looks skilled. What is being selected is not a parameter but a direction in portfolio space, and there are effectively as many directions to select from as there are assets.

2.3 The mechanism: what inversion actually does

The selection story above is correct but qualitative. The precise version is about eigenvalues, and it is worth working through slowly because everything in §6 and §7 is a response to it.

Any covariance matrix can be written in terms of its eigenvalues and eigenvectors:

$$\Sigma = \sum_{i=1}^N \lambda_i v_i v_i', \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N > 0.$$

Read this as a decomposition into eigen-portfolios. Each v_i is a particular combination of the assets — a portfolio, with weights summing to whatever they sum to — and, because eigenvectors are unit vectors, $v_i' \Sigma v_i = \lambda_i$ exactly: λ_i is that portfolio's variance. The first eigen-portfolio v_1 is typically “everything, long” — the market — and has the largest variance. The last, v_N , is the quietest combination available: some intricate long-short arrangement that hedges almost everything out.

Inverting flips the eigenvalues and leaves the directions alone:

$$\Sigma^{-1} = \sum_{i=1}^N \frac{1}{\lambda_i} v_i v_i'.$$

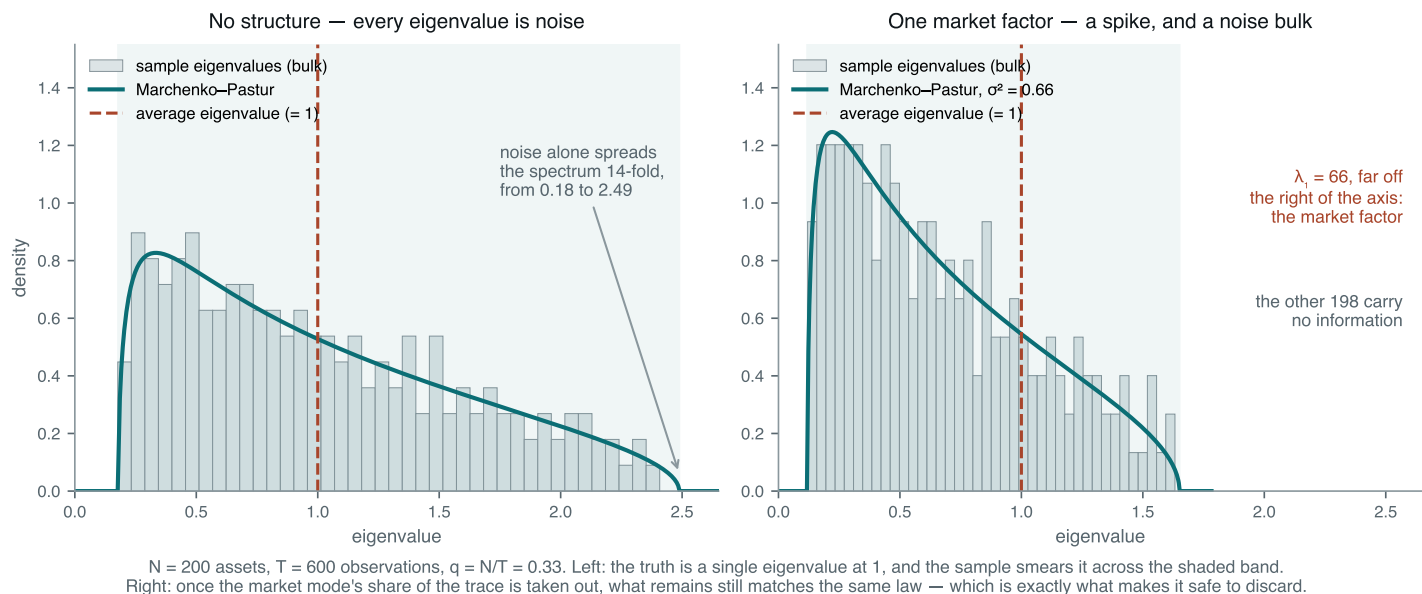
So the optimal portfolio decomposes as

$$w^* \propto \Sigma^{-1}\mu = \sum_{i=1}^N \underbrace{\frac{v_i' \mu}{\lambda_i}}_{\text{how much of } v_i} v_i.$$

Each eigen-portfolio is held in proportion to its expected return divided by its variance. That is exactly right, and exactly the problem: the quietest direction, v_N , is divided by the smallest number, so it receives the largest multiplier. The optimiser's biggest bets are, by construction, on the combinations of assets it believes are the least risky.

Now bring in what the sample gets wrong. When you estimate Σ from T observations, the eigenvalues do not come back with independent, symmetric errors. They come back spread apart in a specific, predictable way: the large ones biased up, the small ones biased down. This is the Marchenko–Pastur result, stated in §5.5. Its consequence here is stark. The sample's smallest eigenvalue is too small — so the optimiser divides by a number that is too small — so it takes a position that is too large — in the direction where its estimate of risk is least reliable.

The figure below shows the effect at $N = 200$, $T = 600$, on data with no structure at all: 200 independent series, so every true eigenvalue equals 1.



The truth is a single point at 1. The sample smears it from 0.18 to 2.49 — a fourteen-fold range. An optimiser handed this matrix will lever the “quietest” direction $1/0.18 = 5.6$ times, and the “noisiest” only $1/2.49 = 0.4$ times, a 14:1 spread of leverage across directions that are in truth identical. Every bit of that spread is noise, and the optimiser’s confidence in it is total.

Here is the summary worth memorising:

The optimiser allocates in inverse proportion to estimated variance, and estimated variance is least accurate exactly where it is smallest. So the largest positions carry the largest errors. Regularisation is not a statistical nicety here — it is the only thing standing between you and a portfolio built entirely out of the least reliable part of your data.

The right-hand panel makes the constructive point that saves the situation. Add one real market factor, and its eigenvalue leaps to 66 — far outside the noise band — while everything else stays inside a bulk that still has the Marchenko–Pastur shape once the factor’s share of variance is removed. Signal and noise separate cleanly in the spectrum. That observation is the basis of eigenvalue clipping (§6.7), factor models (§6.8), and in a sense every method in §6: the bulk is known to be noise in advance, so it can be replaced with something better-behaved at no cost in information.

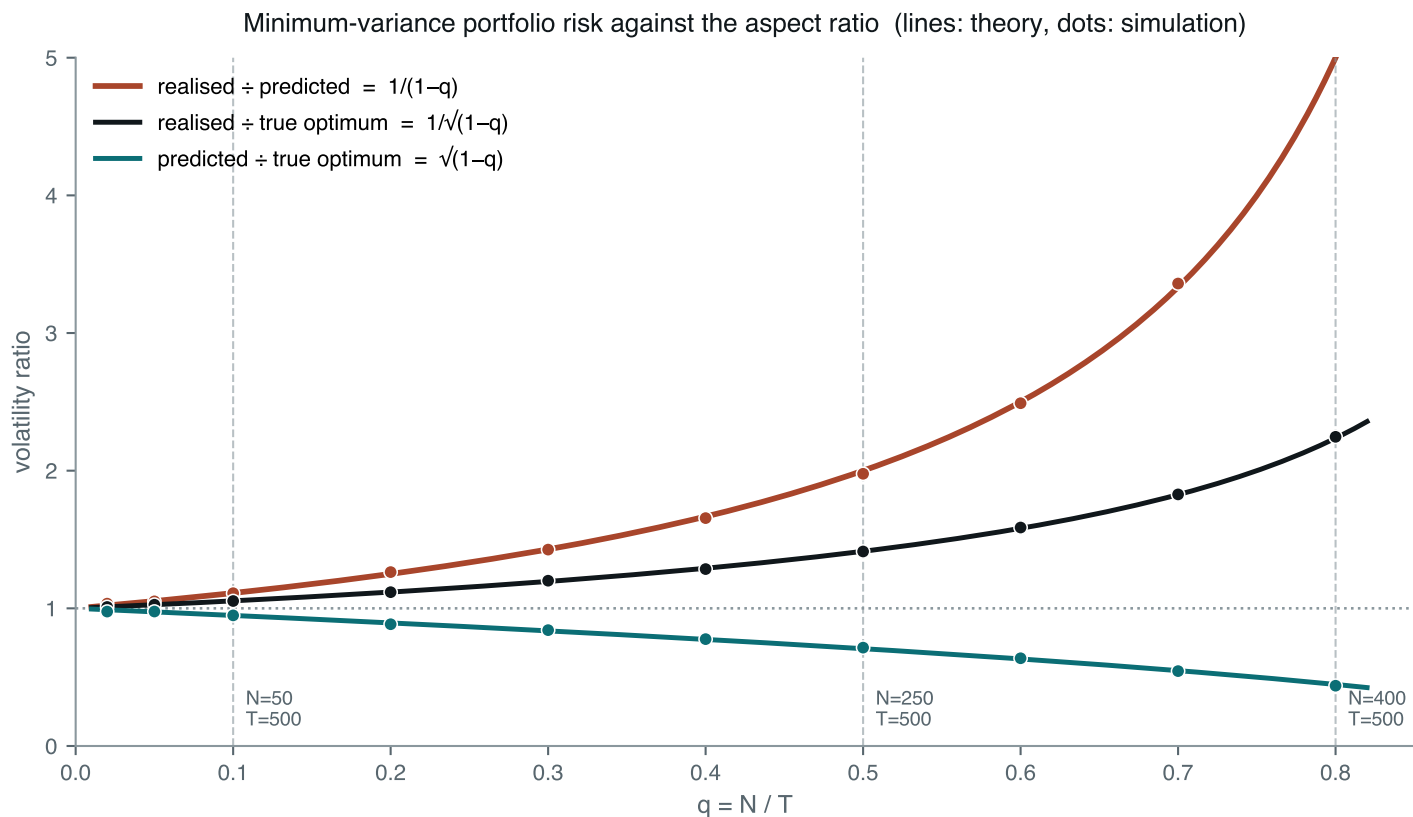
2.4 The diagnostic: everything is governed by $q = N/T$

The natural question after §2.3 is: how much data do I need? The answer is that the question is malformed, because the damage does not depend on T by itself. It depends on $q = N/T$.

For the global minimum-variance portfolio — the purest case, since it uses no μ at all — three quantities have clean large-sample limits. Writing σ_{opt}^2 for the variance of the true optimal portfolio:

$$\begin{aligned} \frac{\text{variance the optimiser predicts}}{\sigma_{\text{opt}}^2} &\rightarrow 1 - q, & \text{(it understates its own risk)} \\ \frac{\text{variance actually realised}}{\sigma_{\text{opt}}^2} &\rightarrow \frac{1}{1 - q}, & \text{(it takes more risk than optimal)} \\ \frac{\text{realised}}{\text{predicted}} &\rightarrow \frac{1}{(1 - q)^2}. & \text{(the two errors compound)} \end{aligned}$$

In volatility rather than variance units, the last line is simply $1/(1 - q)$, which is the number to carry around. The derivation is in §5.6; the figure below confirms it by simulation.



The optimiser is simultaneously understating its own risk and taking more of it. Both errors come from the same place, and they compound.

Read off the practical consequences:

N	T	q	Realised ÷ predicted volatility
50	500 (2 yr daily)	0.10	1.11×
100	500	0.20	1.25×
250	500	0.50	2.00×
400	500	0.80	5.00×
500	500	1.00	undefined — S is singular
500	2520 (10 yr daily)	0.20	1.25×

Two lessons follow, and both are counterintuitive enough to be worth stating flatly.

Doubling the universe is exactly as harmful as halving the history. Adding assets is not free diversification; it is a direct tax on the reliability of every position. A shop running 500 names on two years of data is in a qualitatively worse position than one running 50 names on the same data, and no amount of care elsewhere compensates.

At $q \geq 1$ the sample covariance matrix is singular and the optimiser is meaningless. With $N = 500$ names and two years of daily history there are literally infinitely many “zero-risk” portfolios — combinations that had exactly zero variance in-sample, purely by construction. The optimiser will find one and lever it without limit. If your code does not fail loudly in this case, it is silently doing something worse than failing. [Practice] Check q before you check anything else; it is one line and it invalidates entire research programmes.

2.5 Which input hurts most

Not all inputs are equally dangerous, and the ordering is lopsided enough to drive design decisions.

Chopra & Ziemba (1993) measured this by perturbing each input and computing the certainty-equivalent loss — the amount of money an investor would sacrifice by holding the perturbed portfolio instead of the true optimum. Their result, and it is one of the most useful numbers in the field: [Fact] at a risk tolerance of 50 — risk tolerance being their scale for the reciprocal of risk aversion — errors in means are roughly 11 times as costly as errors in variances, and roughly 22 times as costly as

errors in covariances. That is approximately a 22 : 2 : 1 ratio, and the dominance of means grows further as risk tolerance rises.

The mechanism is §2.1's second observation. A mean must be estimated in calendar time and its standard error is comparable to its own magnitude even over decades; a variance can be estimated to a few percent from a few months of daily data, because variance estimation benefits from finer sampling and means do not. [Fact] This asymmetry is structural, not a data problem, and it will not be fixed by a better dataset.

The design conclusion is direct, and it is the single highest-leverage decision in this document:

If you do not have a genuine, tested return forecast, do not put one in. Setting $\hat{\mu} \propto 1$ — that is, refusing to differentiate between assets on expected return — deletes the dominant error term outright. This is the entire justification for the risk-based portfolios of §7.2–§7.6, and it is why they perform so much better than their crudeness suggests they should.

Note what this argument does not say. It does not say expected returns are unforecastable, and it does not say you should never use a μ . It says the bar is high: a return forecast has to beat not just zero but the substantial benefit of removing the largest source of estimation error from the problem. A weak-but-real signal can easily make the portfolio worse.

2.6 The 1/N challenge, and what it actually showed

The most-cited empirical result in this literature is DeMiguel, Garlappi & Uppal (2009). Across seven datasets, they compared fourteen portfolio rules — sample mean-variance, Bayes–Stein, Black–Litterman-style methods, minimum variance, various constrained versions — against the naive rule of putting $1/N$ in each asset. [Fact] None of the fourteen was consistently better than $1/N$ on Sharpe ratio out of sample.

Their most quotable calculation is the estimation window that sample-based mean-variance would need in order to reliably beat $1/N$: about 3,000 months for 25 assets, and about 6,000 months for 50 — 250 and 500 years respectively. There is no dataset, and there will not be one.

This result is widely reported as “optimisation doesn't work”. That reading is wrong, and the correction matters.

Kritzman, Page & Turkington (2010) made the counter-argument directly. [Contested] The DGU comparison uses sample means as the return input. Replace them with almost anything else — a constant, a risk-premium estimate, a shrunk mean — and optimised portfolios beat $1/N$ comfortably over long samples. Their conclusion is that the finding is about the input, not the method.

I think Kritzman and co-authors have the better of this, and the resolution is already in §2.5:

DeMiguel, Garlappi and Uppal is a paper about $\hat{\mu}$, not a paper about optimisation. It demonstrates, very convincingly, that sample means are worse than useless as portfolio inputs. It does not show that the covariance matrix is useless — and indeed minimum-variance portfolios, which use Σ and discard μ , are among the strongest performers in their own tables and in most subsequent work.

[Contested] The live disagreement that remains is narrower and genuine: whether risk-based optimisation beats $1/N$ net of transaction costs and constraints in a realistic implementation, where turnover and the concentration of minimum-variance solutions eat much of the theoretical gain. That question is still open, is answered differently in different asset classes, and is the subject of §10 and §11.

§2 Key takeaways

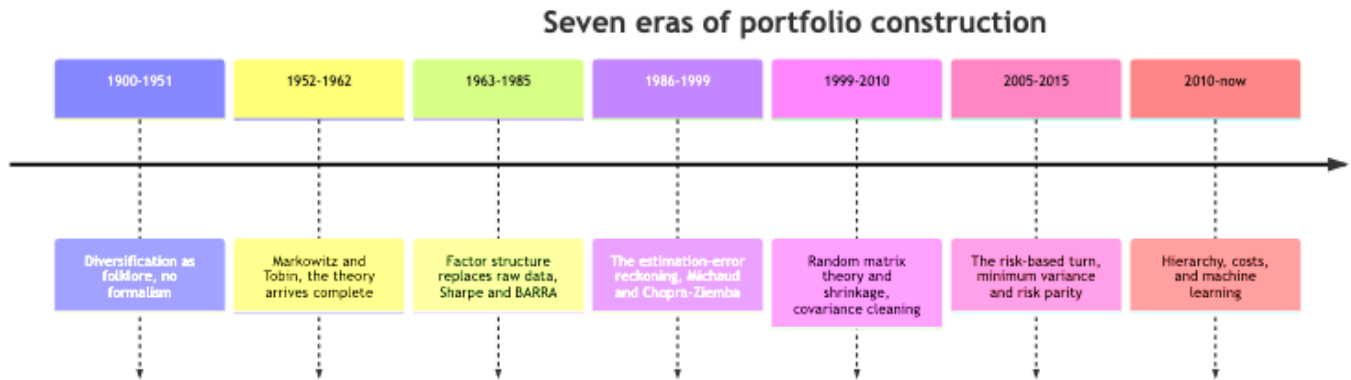
1. In a simulation where the optimiser's model is exactly correct, it reported a Sharpe ratio of 5.3 against a true attainable 0.43, and its realised frontier was flat: asking for 14% return rather than 4% delivered an extra two-tenths of a percent.
2. The mechanism is inversion. Eigen-portfolios are held in proportion to expected return over variance, so the smallest eigenvalue gets the largest bet — and the smallest eigenvalue is the one the sample gets most wrong.
3. Sample eigenvalues are biased apart in a known way: at $q = 1/3$, a truly flat spectrum comes back spread fourteen-fold. Because the distortion is predictable, it is also correctable — that is what §6 is for.
4. One number governs the damage: $q = N/T$. Realised volatility exceeds predicted volatility by $1/(1 - q)$. At $q = 0.5$ you take twice the risk you think you are taking; at $q \geq 1$ the problem is not defined at all.
5. Doubling the number of assets does exactly as much harm as halving the history. Universe size is a statistical decision, not just a business one.

6. Errors in means cost roughly $11\times$ errors in variances and $22\times$ errors in covariances. If you lack a real return forecast, omitting μ entirely is not a compromise — it is the best available move.
 7. The famous $1/N$ result is evidence against sample means, not against optimisation. Read it as licence to drop $\hat{\mu}$, not to drop $\hat{\Sigma}$.
-

3. How the field evolved

The history has one thesis, and it is visible in every era boundary:

The centre of gravity moved from “solve the optimisation” to “estimate the inputs” to “arrange not to need the inputs you cannot estimate.”



Era I — Diversification as folklore (1900–1951). Contribution: the idea that spreading holdings reduces risk, present in investment-trust practice and in Keynes’s writings on portfolio policy. What changed: nothing formal — there was no way to say how much diversification, or between what. Limitations: without a covariance concept, “don’t put all your eggs in one basket” cannot distinguish twenty correlated bets from five independent ones. Lasting influence: the intuition survives, and §7.2 shows the naive $1/N$ rule it implies is a far stronger competitor than the theory that displaced it.

Era II — The theory arrives complete (1952–1962). Contribution: Markowitz (1952) made risk a property of the portfolio rather than of its holdings, via the covariance matrix, and posed the problem as a quadratic programme. Tobin (1958) added the separation theorem: the risky portfolio’s composition is independent of risk aversion. What changed: everything conceptual. The word “risk” acquired a definition you could compute with. Limitations: Markowitz needed $N(N+1)/2$ covariance estimates and computers that could invert the result, and neither existed at scale; more importantly, nobody yet understood that the estimates were the binding constraint rather than the arithmetic. Lasting influence: total. The master form of §1.3 is unchanged, and every section below is a footnote to it.

Era III — Structure instead of data (1963–1985). Contribution: Sharpe (1963) proposed the single-index model, replacing $N(N+1)/2$ covariances with $2N+1$ parameters by routing all co-movement through one market factor. Barr Rosenberg’s commercial risk models (the firm that became BARRA) generalised this to multi-factor fundamental models. What changed: the recognition that a covariance matrix should be modelled, not measured. Limitations: factor structure is an assumption, and what it discards — genuine residual correlation within industries — is exactly what a statistical-arbitrage desk trades. Lasting influence: very large. Factor covariance models (§6.8) remain the default in institutional equity risk management, and §8.2 shows they are a form of shrinkage.

Era IV — The estimation-error reckoning (1986–1999). Contribution: a cluster of papers established that estimation error, not model error, was the binding constraint. Jorion (1986) applied Bayes–Stein shrinkage to the means; Michaud (1989) named the error-maximisation property; Best & Grauer (1991) computed the sensitivity of optimal weights to the means analytically; Chopra & Ziemba (1993) priced the damage; Black & Litterman (1992) sidestepped it by starting from market-implied returns. What changed: the field stopped treating the optimiser as a solved problem and started treating it as a hazard. Limitations: the diagnoses were sharper than the cures, most of which were tuning parameters without theory. Lasting influence: decisive. Everything after this is a response.

Era V — The physics import (1999–2010). Contribution: two independent literatures arrived at the same repair. Statistical physicists — Laloux, Cizeau, Bouchaud & Potters (1999) — showed that the bulk of a financial correlation matrix’s eigenvalue spectrum is indistinguishable from random-matrix noise, giving a principled rule for what to discard. Statisticians — Ledoit & Wolf (2003, 2004) — derived the optimal shrinkage intensity toward a structured target in closed form. Meanwhile Jagannathan & Ma (2003) proved that a no-short-sale constraint is mathematically identical to shrinking the covariance matrix, which explained why unsophisticated practice had been outperforming sophisticated theory. What changed: regularisation became principled rather than ad hoc, and the equivalences of §8.2 came into view. Limitations: almost all of it addresses Σ , and §2.5 says μ is the larger problem. Lasting influence: Ledoit–Wolf shrinkage is now the default in `scikit-learn` and most risk systems; this era’s ideas are the working toolkit.

Era VI — The risk-based turn (2005–2015). Contribution: practitioners drew the conclusion of §2.5 and abandoned μ .

Minimum-variance, maximum-diversification (Choueifaty & Coignard, 2008), and equal-risk-contribution (Maillard, Roncalli & Teiletche, 2010) portfolios became products. DeMiguel, Garlappi & Uppal (2009) supplied the negative result that justified the turn. What changed: “we do not forecast returns” went from an admission to a selling point. Limitations: [Contested] much of the measured outperformance is exposure to the low-volatility and value factors rather than to superior construction, and the products are capacity-constrained and crowded. Lasting influence: large and ongoing; risk-based allocation is now the default starting point for multi-asset portfolios.

Era VII — Hierarchy, costs, and learning (2010–present). Contribution: three strands. López de Prado (2016) proposed hierarchical risk parity, which allocates using the covariance matrix without ever inverting it. Gârleanu & Pedersen (2013) solved the dynamic problem with transaction costs in closed form, showing that the optimal portfolio is a weighted average of current holdings and a forward-looking target. Ledoit & Wolf (2017) extended shrinkage from a single intensity to a whole function of the spectrum. What changed: costs moved from a post-hoc adjustment into the objective, and regularisation became nonparametric. Limitations: [Contested] HRP’s empirical advantage over well-regularised alternatives is disputed and appears sensitive to the test design; end-to-end machine-learned allocation remains mostly promising rather than demonstrated. Lasting influence: too early to say for the learning strand; the cost-aware strand is already standard.

§3 Key takeaways

1. The optimisation problem was solved in 1952 and has never been the bottleneck. Every subsequent era is about inputs.
 2. Factor models (1963) were the field’s first regularisation, adopted for computational reasons and retained for statistical ones.
 3. The 1986–1999 cluster diagnosed estimation error precisely, but its cures were mostly untheorised tuning.
 4. Physics and statistics converged on the same repair around 2000: the noisy part of the spectrum is identifiable in advance and should be replaced.
 5. Jagannathan and Ma’s 2003 proof that constraints are shrinkage is the hinge of the whole history — it explained why crude practice beat refined theory, and unified two literatures that thought they disagreed.
 6. The risk-based turn was the field acting on Chopra–Ziemba: dropping the input you cannot estimate beats estimating it badly.
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4. Foundational references

Grouped by kind, because the kinds are read differently. Each entry says why it matters. Where a free copy exists it is the link; paywalled-only entries are marked.

4.1 The founding theory

- Markowitz, H. (1952). “Portfolio Selection.” *Journal of Finance* 7(1), 77–91. — The origin. Worth reading for how carefully Markowitz already hedges about the inputs; the field spent fifty years rediscovering his caution.
- Tobin, J. (1958). “Liquidity Preference as Behavior Towards Risk.” *Review of Economic Studies* 25(2), 65–86. — The separation theorem: leverage and composition are independent choices (§1.3).
- Sharpe, W. F. (1963). “A Simplified Model for Portfolio Analysis.” *Management Science* 9(2), 277–293. — The single-index model, and the first argument that a covariance matrix should be modelled rather than measured.
- Merton, R. C. (1980). “On Estimating the Expected Return on the Market.” *Journal of Financial Economics* 8(4), 323–361. — Why expected returns are hard in a way variances are not: the mean’s precision depends on calendar span, the variance’s on sampling frequency. The single most important paper for understanding §2.5.
- Stevens, G. V. G. (1998). “On the Inverse of the Covariance Matrix in Portfolio Analysis.” *Journal of Finance* 53(5), 1821–1827. — Underread, and the key to §5.4: Σ^{-1} decomposes into the coefficients and residual variance of regressing each asset on all the others. Link is the Fed working-paper version.

4.2 The estimation-error critique

- Jorion, P. (1986). “Bayes-Stein Estimation for Portfolio Analysis.” *Journal of Financial and Quantitative Analysis* 21(3), 279–292. [[paywalled]] — Shrinkage applied to the means, a decade before it reached the covariance.
- Michaud, R. O. (1989). “The Markowitz Optimization Enigma: Is ‘Optimized’ Optimal?” *Financial Analysts Journal* 45(1), 31–42. — Names error maximisation. Short, non-technical, and the best single statement of why §2 happens.
- Best, M. J. & Grauer, R. R. (1991). “On the Sensitivity of Mean-Variance-Efficient Portfolios to Changes in Asset Means.” *Review of Financial Studies* 4(2), 315–342. [[paywalled]] — The analytical companion to Michaud: how far weights move per unit of change in μ .
- Chopra, V. K. & Ziemba, W. T. (1993). “The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice.” *Journal of Portfolio Management* 19(2), 6–11. [[paywalled]] — The 22 : 2 : 1 result of §2.5. If you take one number from this document, take this one.
- Merton (1980), above, belongs here as much as in §4.1.

4.3 Estimating the covariance matrix

- Laloux, L., Cizeau, P., Bouchaud, J.-P. & Potters, M. (1999). “Noise Dressing of Financial Correlation Matrices.” *Physical Review Letters* 83(7), 1467–1470. — Four pages; showed that most of a financial correlation matrix’s spectrum is indistinguishable from noise. Also on arXiv.
- Ledoit, O. & Wolf, M. (2003). “Improved Estimation of the Covariance Matrix of Stock Returns with an Application to Portfolio Selection.” *Journal of Empirical Finance* 10(5), 603–621. — Shrinkage toward a single-index target, with the optimal intensity derived rather than tuned.
- Ledoit, O. & Wolf, M. (2004). “Honey, I Shrunk the Sample Covariance Matrix.” *Journal of Portfolio Management* 30(4), 110–119. — The practitioner-facing version. Start here; it is the most readable entry point to the whole subject.
- Ledoit, O. & Wolf, M. (2004). “A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices.” *Journal of Multivariate Analysis* 88(2), 365–411. — The theory behind the previous two.
- Ledoit, O. & Wolf, M. (2017). “Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks.” *Review of Financial Studies* 30(12), 4349–4388. — Shrinking each eigenvalue by its own amount rather than all by one intensity (§6.6).
- Ledoit, O. & Wolf, M. (2022). “The Power of (Non-)Linear Shrinking: A Review and Guide to Covariance Matrix Estimation.” *Journal of Financial Econometrics* 20(1), 187–218. — Fifteen years of their own work, reviewed with practical guidance on which variant to use when. The efficient entry point if you only read one item in this subsection.
- Bun, J., Bouchaud, J.-P. & Potters, M. (2017). “Cleaning Large Correlation Matrices: Tools from Random Matrix Theory.” *Physics Reports* 666, 1–109. — The definitive technical review of the RMT strand. Long, and the first thirty pages are the useful ones.
- Engle, R. F. (2002). “Dynamic Conditional Correlation: A Simple Class of Multivariate GARCH Models.” *Journal of Business and Economic Statistics* 20(3), 339–350. — Time-varying correlation, estimated in two tractable steps (§6.10).

- Fan, J., Liao, Y. & Mincheva, M. (2013). “Large Covariance Estimation by Thresholding Principal Orthogonal Complements.” *Journal of the Royal Statistical Society, Series B* 75, 603–680. — POET: a factor model plus a thresholded residual matrix, which recovers the within-industry correlation that a pure factor model discards (§6.9).
- Higham, N. J. (2002). “Computing the Nearest Correlation Matrix — a Problem from Finance.” *IMA Journal of Numerical Analysis* 22(3), 329–343. — What to do when your matrix is not positive semi-definite. You will need this (§9.5).
- Epps, T. W. (1979). “Comovements in Stock Prices in the Very Short Run.” *Journal of the American Statistical Association* 74(366), 291–298. [[paywalled]] — Measured correlation falls as the sampling interval shrinks. The reason §9.3 tells you to estimate correlations weekly and volatilities daily.

4.4 Constraints, and why they work

- Jagannathan, R. & Ma, T. (2003). “Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps.” *Journal of Finance* 58(4), 1651–1684. — The hinge result of the field: a no-short constraint is exactly a shrinkage of the covariance matrix. Read it next to Ledoit–Wolf and the two literatures collapse into one. Link is the NBER working paper.
- DeMiguel, V., Garlappi, L., Nogales, F. J. & Uppal, R. (2009). “A Generalized Approach to Portfolio Optimization: Improving Performance by Constraining Portfolio Norms.” *Management Science* 55(5), 798–812. [[paywalled]] — Generalises Jagannathan–Ma: constraining any norm of w is equivalent to a corresponding regularisation, which makes the constraint/shrinkage identity a family rather than a curiosity.

4.5 Risk-based portfolios

- Choueifaty, Y. & Coignard, Y. (2008). “Toward Maximum Diversification.” *Journal of Portfolio Management* 35(1), 40–51. — The diversification ratio and the portfolio that maximises it. [Contested] The authors’ firm sells this strategy; the mathematics is sound and the performance claims carry an interest.
- Maillard, S., Roncalli, T. & Teiletche, J. (2010). “The Properties of Equally Weighted Risk Contribution Portfolios.” *Journal of Portfolio Management* 36(4), 60–70. — Risk parity done properly, with the existence and uniqueness results and the proof that ERC sits between $1/N$ and minimum variance.
- Clarke, R., de Silva, H. & Thorley, S. (2011). “Minimum-Variance Portfolio Composition.” *Journal of Portfolio Management* 37(2), 31–45. — Analytic characterisation of which assets a minimum-variance portfolio ends up holding, and why so few of them. The best cure for surprise at a concentrated solution.
- Asness, C. S., Frazzini, A. & Pedersen, L. H. (2012). “Leverage Aversion and Risk Parity.” *Financial Analysts Journal* 68(1), 47–59. — The economic argument for why risk parity should earn anything at all. [Contested] AQR runs risk-parity products.
- López de Prado, M. (2016). “Building Diversified Portfolios that Outperform Out of Sample.” *Journal of Portfolio Management* 42(4), 59–69. — Hierarchical risk parity: allocate using the covariance matrix without inverting it (§7.11).
- Meucci, A. (2009). “Managing Diversification.” *Risk*, 74–79. — The effective number of bets, which is the right way to measure whether a portfolio is actually diversified (§11.4).

4.6 Costs and dynamics

- Gârleanu, N. & Pedersen, L. H. (2013). “Dynamic Trading with Predictable Returns and Transaction Costs.” *Journal of Finance* 68(6), 2309–2340. — The closed-form multi-period solution. Its result — trade partway toward a forward-looking aim portfolio, not toward today’s target — is the most useful single fact in §10.
- Almgren, R. & Chriss, N. (2000). “Optimal Execution of Portfolio Transactions.” *Journal of Risk* 3, 5–39. — The execution layer beneath §10; how a given trade list is worked, as distinct from how it is chosen.
- Constantinides, G. M. (1986). “Capital Market Equilibrium with Transaction Costs.” *Journal of Political Economy* 94(4), 842–862. [[paywalled]] — The origin of the no-trade region, and of the result that its width scales as the cube root of the cost rather than linearly (§10.4).

4.7 The critiques

A bibliography listing only a field’s successes is propaganda. These belong next to everything above.

- DeMiguel, V., Garlappi, L. & Uppal, R. (2009). “Optimal Versus Naive Diversification: How Inefficient is the $1/N$ Portfolio Strategy?” *Review of Financial Studies* 22(5), 1915–1953. — Fourteen rules, seven datasets, and none beats equal weighting. Read with §2.6’s correction in hand.
- Kritzman, M., Page, S. & Turkington, D. (2010). “In Defense of Optimization: The Fallacy of $1/N$.” *Financial Analysts Journal* 66(2), 31–39. [[paywalled]] — The rebuttal: the problem is sample means, not optimisation. I find this

persuasive (§2.6).

- Michaud (1989) and Chopra & Ziemba (1993), above, are critiques as much as contributions.
- Lo, A. W. (2002). “The Statistics of Sharpe Ratios.” *Financial Analysts Journal* 58(4), 36–52. [[paywalled]] — Where the standard error of §11.5 comes from, plus what serial correlation does to an annualised Sharpe ratio. Read before believing any backtest’s headline number.

4.8 Books

- Meucci, A. (2005). *Risk and Asset Allocation*. Springer. — The most complete single treatment of the estimation-and-allocation problem together. Demanding, and worth it.
- Roncalli, T. (2013). *Introduction to Risk Parity and Budgeting*. Chapman and Hall. — The definitive treatment of risk budgeting, with the algorithms written out.
- Grinold, R. C. & Kahn, R. N. (1999). *Active Portfolio Management*, 2nd ed. McGraw-Hill. — The practitioner’s frame: information ratio, breadth, the fundamental law, and the transfer coefficient that measures how much of a signal survives construction (§11.4).
- Michaud, R. O. & Michaud, R. O. (2008). *Efficient Asset Management*, 2nd ed. Oxford. — Resampled efficiency (§7.9), by its inventors. [Contested] The method is patented and the authors sell it; the diagnosis in the first chapters is excellent regardless.

4.9 If you only read six things

In order:

1. Michaud (1989) — why the problem exists, in fifteen readable pages.
2. Chopra & Ziemba (1993) — which input to worry about, quantified.
3. Ledoit & Wolf (2004), “Honey, I Shrunk...” — the standard repair, explained for practitioners.
4. Jagannathan & Ma (2003) — the result that unifies constraints and shrinkage, and retroactively justifies most of what desks were already doing.
5. DeMiguel, Garlappi & Uppal (2009) then Kritzman, Page & Turkington (2010) — the field’s central controversy, both sides, in that order.
6. Ledoit & Wolf (2022) — what to actually use, from the people who built it.

Notice what is not on that list: nothing about optimisation algorithms. That is the point of §3.

5. The mathematics

This section is the formal backing for §2. §5.4 is the part that is not in most textbooks and is worth the time even if you skip the rest.

5.1 The two canonical problems

Almost every allocation rule in §7 is one of two optimisations, or a constrained version of one.

Problem A — maximum Sharpe (tangency). With μ measured as excess return over cash:

$$\max_w \frac{w' \mu}{\sqrt{w' \Sigma w}} \implies w_{\text{tan}} \propto \Sigma^{-1} \mu.$$

Scale is undetermined because the Sharpe ratio is scale-invariant — which is §1.3's separation result appearing as a property of the objective rather than of the investor.

Problem B — global minimum variance (GMV). Use no return forecast at all:

$$\min_w w' \Sigma w \quad \text{s.t.} \quad 1' w = 1 \implies w_{\text{gmv}} = \frac{\Sigma^{-1} 1}{1' \Sigma^{-1} 1}.$$

Note the shape of the second: it is $\Sigma^{-1} \mu$ with $\mu = 1$, normalised to sum to one. The minimum-variance portfolio is the mean-variance portfolio of an investor who believes every asset has the same expected return. That is not an approximation or an analogy; it is the same formula. It is the first entry in the equivalence table of §8.2 and the reason §7's “risk-based” methods are not a separate family.

5.2 The frontier, and two-fund separation

Adding a target return to Problem B and solving with Lagrange multipliers gives the full efficient frontier. Define the three scalars

$$a = 1' \Sigma^{-1} 1, \quad b = 1' \Sigma^{-1} \mu, \quad c = \mu' \Sigma^{-1} \mu, \quad d = ac - b^2.$$

The minimum-variance portfolio achieving expected return m is

$$w(m) = \Sigma^{-1} \left[\frac{c - bm}{d} 1 + \frac{am - b}{d} \mu \right],$$

with variance $\sigma^2(m) = (am^2 - 2bm + c)/d$. Two consequences matter.

The frontier is a hyperbola in (σ, m) space, with its left vertex at the GMV portfolio, $m_{\text{gmv}} = b/a$ and $\sigma_{\text{gmv}}^2 = 1/a$.

Two-fund separation: $w(m)$ is a combination of the same two fixed vectors $\Sigma^{-1} 1$ and $\Sigma^{-1} \mu$ for every m , with coefficients affine in m . Normalise those two vectors to sum to one — they become the GMV portfolio $\Sigma^{-1} 1/a$ and the tangency portfolio $\Sigma^{-1} \mu/b$ — and the coefficients on them sum to one for every m . Every efficient portfolio is a weighted average of two fixed portfolios. This is why §2.1's realised frontier could be flat: the optimiser was sliding along a line between two portfolios, one of which — the $\Sigma^{-1} \hat{\mu}$ end — was almost pure noise. Moving further along that line adds leverage on noise and nothing else.

The quantity $c = \mu' \Sigma^{-1} \mu$ is the squared maximum Sharpe ratio available. Keep it in view: it is the thing estimation error inflates most violently, and $\sqrt{\hat{c}}$ was the 5.32 of §2.1.

5.3 What the covariance matrix contributes

Before inverting anything, it is worth separating the two jobs Σ does, because they are estimated with wildly different precision. Write $\Sigma = DCD$ with D the diagonal matrix of volatilities and C the correlation matrix.

Component	Parameters	Estimation difficulty
Volatilities D	N	Easy. Realised volatility from daily data is accurate within weeks; it is the most forecastable quantity in finance
Correlations C	$N(N-1)/2$	Hard. Quadratically many parameters, each individually noisy, and the errors interact through the inverse

[Fact] Essentially all of the difficulty is in C , and essentially all of the N^2 growth is too. This suggests — correctly — that the highest-value regularisation acts on the correlation matrix while leaving the volatilities alone, and §9.5 makes that a concrete implementation rule.

5.4 What Σ^{-1} actually is: the regression identity

Here is the result that makes everything else intuitive. It is standard in multivariate statistics and was brought into portfolio theory by Stevens (1998), where it remains underused.

For each asset i , run the regression of its return on all the other assets:

$$r_i = a_i + \sum_{j \neq i} \beta_{ij} r_j + \varepsilon_i,$$

and let $s_i^2 = \text{Var}(\varepsilon_i)$ be the residual variance and R_i^2 the coefficient of determination, so that $s_i^2 = \sigma_i^2 (1 - R_i^2)$. Read this regression as: what is the best portfolio of the other $N - 1$ assets for replicating asset i , and how well does it do?

Then the entries of the inverse covariance matrix are exactly

$$(\Sigma^{-1})_{ii} = \frac{1}{s_i^2}, \quad (\Sigma^{-1})_{ij} = -\frac{\beta_{ij}}{s_i^2} \quad (j \neq i).$$

The off-diagonal expression looks asymmetric — it carries s_i^2 , not s_j^2 — but it is not: $\beta_{ij}/s_i^2 = \beta_{ji}/s_j^2$ holds identically, which is the same statement as Σ^{-1} being symmetric.

Substituting into $w \propto \Sigma^{-1} \mu$ and collecting terms:

$$w_i \propto \frac{1}{s_i^2} \left(\mu_i - \sum_{j \neq i} \beta_{ij} \mu_j \right) = \frac{\alpha_i}{s_i^2} = \boxed{\frac{\alpha_i}{\sigma_i^2 (1 - R_i^2)}}$$

where $\alpha_i \equiv \mu_i - \sum_{j \neq i} \beta_{ij} \mu_j$ is the residual alpha: asset i 's expected return net of what you could have got by holding its replicating portfolio instead. Take expectations through the regression and you will see that α_i is nothing other than the intercept a_i above — alpha in the ordinary sense of the word, measured against a benchmark that is the rest of your own universe.

This one line reorganises the whole subject. Four readings:

1. The optimiser does not care about expected returns. It cares about expected returns net of replication. An asset whose forecast return is exactly what its replicating portfolio already delivers has $\alpha_i = 0$ and receives zero weight, correctly, however wonderful the headline number looks. This is why adding a highly-correlated asset to a universe can send existing weights wild: you have changed every other asset's α_i and R_i^2 simultaneously.

2. Every position is a hedged bet, sized by its own Sharpe ratio. The form α_i/s_i^2 is precisely the sizing you would apply to a single stand-alone asset with mean α_i and variance s_i^2 — its own Sharpe ratio α_i/s_i , divided once more by s_i to turn a ratio into a position size. So mean-variance optimisation is not doing anything exotic — it is hedging each asset against all the others, then sizing each hedged residual independently. That is a much better mental model than “it balances risk and return across the portfolio.”

3. $1/(1 - R_i^2)$ is the leverage amplifier, and it is unbounded. As an asset becomes more replicable, $R_i^2 \rightarrow 1$ and the weight diverges. The two-asset example of §1.6 is exactly this: regressing one asset on the other gives $R^2 = \rho^2$, so

at $\rho = 0.99$ the amplifier is $1/(1 - 0.99^2) \approx 50$ — acting not on the 6% headline return but on a residual alpha of $\alpha_1 = 0.06 - 0.99 \times 0.05 = 0.0105$. That product, divided by $\sigma_1^2 = 0.04$ and normalised so the weights sum to one, is where the 18× gross exposure came from.

4. And here is the punchline. R_i^2 is estimated by regressing on $N - 1$ regressors using T observations. Under the null hypothesis that asset i is genuinely unrelated to the others, the expected in-sample R^2 is not zero — it is

$$\mathbb{E}[\hat{R}_i^2] = \frac{N - 1}{T - 1} \approx q.$$

So the sample believes every asset is about q -replicable when none of them is replicable at all. Put $\hat{R}_i^2 = q$ into the boxed formula in place of the true $R_i^2 = 0$: the denominator shrinks by a factor of $1 - q$, so the weight is inflated by roughly $1/(1 - q)$ — the same constant §5.5 read off the Marchenko–Pastur spectrum and §5.6 will get from the Wishart inverse.

It is worth being exact about what that agreement is and is not. These are not three independent confirmations. The regression identity above is an algebraic fact about any positive-definite matrix, so applying it to S and applying the inverse-Wishart expectation to S are the same statement written in two coordinate systems — one asset by asset, one over the whole matrix. The spectral integral is that same statement a third time, in the eigenbasis. One fact, three faces.

The value is not corroboration, then, but interpretation: the spectral face tells you where the bias lives (the small eigenvalues), and the regression face tells you which assets carry it (the replicable ones) — and only the second of those gives you something to compute per asset, which is the diagnostic below.

This also says what the fix looks like. The textbook correction for an inflated R^2 is the adjusted R^2 , which inflates $1 - R^2$ by $(T - 1)/(T - N)$ — the same factor to leading order, and enough to bring the expected value of one-minus-adjusted- R^2 back to exactly 1 under the null. Shrinkage (§6.5), factor models (§6.8) and position constraints (§9.6) are all, in this light, ways of not believing your own $\hat{R}_i^2$.

A diagnostic you can run today. [Practice] For each asset, regress its returns on the rest of the universe and record R_i^2 . Sort descending. Any asset with $\hat{R}_i^2$ above roughly 0.9 — or, more carefully, well above the q you would expect from noise alone — is one whose optimiser weight is numerically unstable and whose position will swing between rebalances. This costs a few lines of code, needs no optimiser, and finds the assets that will hurt you before they do. I know of no cheaper diagnostic in this whole document.

5.5 Random matrix theory: what noise looks like

§2.3 asserted that sample eigenvalues spread in a predictable way. The precise statement is the Marchenko–Pastur law.

Take T independent observations of N uncorrelated, unit-variance series, so the true correlation matrix is the identity and every true eigenvalue is 1. Let $N, T \rightarrow \infty$ with $q = N/T$ fixed in $(0, 1)$. Then the sample eigenvalues do not converge to 1. Their empirical distribution converges to a density $f(\lambda)$ supported on $[\lambda_-, \lambda_+]$ with

$$\lambda_{\pm} = (1 \pm \sqrt{q})^2, \quad f(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi q \lambda}.$$

Three things make this useful rather than merely elegant.

It is a prediction, not a description. The noise spectrum is known before you look at your data. Anything inside $[\lambda_-, \lambda_+]$ is consistent with pure noise; anything outside is not. That converts “how much of my correlation matrix is real?” from a judgement call into a test, which is what Laloux and co-authors (1999) exploited.

The spread is severe at realistic q . The implied condition number of a pure noise matrix is $\kappa = \lambda_+/\lambda_- = ((1 + \sqrt{q})/(1 - \sqrt{q}))^2$:

q	λ_-	λ_+	κ (noise only)
0.10	0.47	1.73	3.7
1/3	0.18	2.49	13.9
0.50	0.086	2.91	34.0
0.80	0.011	3.59	322

q	λ_-	λ_+	κ (noise only)
0.90	0.0026	3.80	1442

At $q = 0.9$ — five hundred names on 555 days of history, a little over two years — noise alone manufactures a condition number of 1,442 in a matrix whose truth is the identity. Any optimiser inverting that is placing its largest bets on directions whose estimated variance is smaller than the truth by a factor of nearly 400.

Real matrices separate cleanly. Financial correlation matrices have a few eigenvalues far outside the bulk — the market mode, then sector modes — and a bulk that matches the law once the outliers' share of the trace is removed. That is the right-hand panel of §2.3's figure, and it is what licenses discarding the bulk (§6.7).

What the law says about the inverse. The optimiser does not use λ ; it uses $1/\lambda$ (§2.3). So integrate the law against $1/\lambda$, which has a closed form:

$$\int_{\lambda_-}^{\lambda_+} \frac{f(\lambda)}{\lambda} d\lambda = \frac{1}{1-q}.$$

The average eigenvalue of a pure-noise correlation matrix is exactly 1, and the average of its reciprocals is $1/(1-q)$. That is the whole bias, read straight off the spectrum, and it is the constant §5.6 will derive again by a different route.

One distinction matters here and is easy to lose, because the two numbers differ a lot. $1/(1-q)$ is an average over the spectrum: at $q = 1/3$ it is 1.5, a 50% inflation. The worst single direction is a different and much larger quantity, $1/\lambda_- = (1 - \sqrt{q})^{-2}$, which at the same q is 5.6. §2.3's 14:1 leverage spread is about the extremes; the $1/(1-q)$ of §2.4 and §5.6 is about the average. Both are consequences of the same spreading, but a portfolio concentrated in the quietest direction suffers the extreme, while a diversified one suffers something closer to the average — which is why §5.6's ratio and §9.8's measured 1.155 are mild compared with the alarm of §2.3's figure.

5.6 Why the minimum-variance portfolio misstates its own risk

Now derive the $(1-q)$ results quoted in §2.4. Let S be the sample covariance from T observations, so that $TS \sim W_N(\Sigma, T)$ — a Wishart matrix with T degrees of freedom, which is the count when the mean is known; subtracting an estimated mean costs one, and changes nothing below. The key fact is the inverse-Wishart expectation, valid whenever $T > N + 1$:

$$\mathbb{E}[S^{-1}] = \frac{T}{T - N - 1} \Sigma^{-1}.$$

The inverse is biased upward, by a factor of about $1/(1-q)$. Not the covariance — the sample covariance is unbiased — but its inverse, which is the thing the optimiser actually uses. Inversion is convex, so Jensen's inequality guarantees a bias, and the aspect ratio sets its size.

The GMV portfolio built on S reports the variance $\hat{\sigma}_{\text{pred}}^2 = 1/(1'S^{-1}1)$, which is just §5.2's $1/a$ computed from the sample. Taking expectations in the denominator, and using $\sigma_{\text{opt}}^2 = 1/(1'\Sigma^{-1}1)$ from the same place,

$$\mathbb{E}[1'S^{-1}1] = \frac{T}{T - N - 1} 1'\Sigma^{-1}1 \approx \frac{1}{1-q} \cdot \frac{1}{\sigma_{\text{opt}}^2},$$

so to leading order $\hat{\sigma}_{\text{pred}}^2 \approx (1-q)\sigma_{\text{opt}}^2$. That last step swaps $\mathbb{E}[1/X]$ for $1/\mathbb{E}[X]$, which is legitimate here because $1'S^{-1}1$ concentrates around its mean as N and T grow together. The optimiser understates its risk by the factor $1-q$.

The realised variance of those same weights, evaluated at the true Σ , goes the other way:

$$\mathbb{E}[\hat{w}'_{\text{gmV}} \Sigma \hat{w}_{\text{gmV}}] \approx \frac{\sigma_{\text{opt}}^2}{1-q}.$$

Both statements are leading-order in the large- (N, T) limit with q fixed; the simulation in §2.4's figure tracks them closely up to $q \approx 0.8$. Dividing one by the other gives the headline: realised variance exceeds predicted variance by $1/(1 - q)^2$, or in volatility terms by $1/(1 - q)$.

The asymmetry deserves a sentence of its own, because it is what makes this dangerous rather than merely inaccurate. The two errors do not offset. The optimiser takes more risk than is optimal and reports less risk than it is taking, and both errors have the same cause. A risk report generated from the same covariance matrix used to build the portfolio will confirm the portfolio is safe, using precisely the numbers that made it unsafe. §11.2 turns this into a monitoring rule.

5.7 The condition number as a working diagnostic

Numerical analysis supplies a bound that turns out to be a good practical guide. For $w = \Sigma^{-1}\mu$, a relative perturbation of size ϵ in Σ produces a relative change in w of up to

$$\frac{\|\Delta w\|}{\|w\|} \lesssim \kappa(\Sigma) \epsilon, \quad \kappa(\Sigma) = \frac{\lambda_1}{\lambda_N}.$$

The condition number is the amplification factor from input error to output error. [Practice] The rules of thumb worth internalising:

κ	Interpretation
< 100	Comfortable. Weights are stable under resampling
10^2 – 10^4	Workable, but regularise, and expect visible turnover
10^4 – 10^8	Weights are dominated by the smallest eigenvalues. Do not trust them
$> 10^8$	Approaching double-precision limits. The inverse is numerically meaningless

Because κ costs one `eigvalsh` call and needs no forecast, it belongs in production monitoring next to q . A jump in κ between rebalances is an early warning that two assets in your universe have become near-duplicates — which happens routinely when a merger is announced, when two share classes converge, or when a market stops trading and prices go stale (§9.3).

§5 Key takeaways

1. Minimum variance is mean-variance with $\mu = 1$. Risk-based portfolios are not a separate family; they are choices of μ together with choices about how much structure to impose on Σ .
2. Every efficient portfolio is a mix of two fixed portfolios. Moving along the frontier moves you toward $\Sigma^{-1}\hat{\mu}$, which is the noisiest object in the problem.
3. The regression identity is the intuition that unlocks the subject: $w_i \propto \alpha_i / [\sigma_i^2(1 - R_i^2)]$, where α_i is expected return net of replication and R_i^2 measures how replicable the asset is.
4. Read that as: mean-variance optimisation hedges each asset against all the others, then sizes each hedged residual by its own Sharpe ratio. Nothing more mysterious is happening.
5. In-sample R_i^2 is inflated to about q by overfitting alone, which inflates weights by $1/(1 - q)$ — the same constant random-matrix theory gives. Two routes, one phenomenon.
6. The sample covariance is unbiased but its inverse is not, by a factor of about $1/(1 - q)$. The optimiser uses the inverse.
7. The two errors compound and never offset: you take more risk than optimal and report less than you take, from the same cause. Never validate a portfolio's risk with the matrix that built it.
8. Two numbers belong in production monitoring, and both are one line of code: the aspect ratio q and the condition number κ .

6. Estimating the covariance matrix

6.1 The one axis that organises this section

There are a dozen named estimators below and they are not a dozen ideas. Every one of them answers a single question:

How much structure am I willing to impose in exchange for less estimation error?

At one end sits the sample covariance: no structure, no bias, maximal variance. At the other sits a single number applied to everything: total structure, large bias, no variance. Every useful estimator is somewhere between, and the choice is a bias-variance trade-off in which — because of §2 — the variance term is almost always the one that is binding.

It helps to know where an estimator applies its structure. Recall $\hat{\Sigma} = V\Lambda V'$: eigenvectors V say what the risk directions are, eigenvalues Λ say how large each one is. That gives three families:

Family	What it modifies	Members
Rotationally invariant	Eigenvalues only; keeps the sample's eigenvectors	Nonlinear shrinkage, RMT clipping, linear shrinkage to a scaled identity
Structure-imposing	Eigenvectors too; asserts where risk lives	Factor models, constant correlation, hierarchical — and linear shrinkage toward any of them
Reweighting	Which observations count	EWMA, DCC, and any window choice

The families are composable and routinely combined — a factor model estimated on EWMA-weighted data with shrunk residuals uses all three. §8.3 gives the ordering of what actually matters.

A note on the fields below. Each estimator gets the same six: intuition, definition, assumptions, cost, failure modes, and when to prefer it. The failure modes field carries the most practical value and gets the most room.

6.2 Sample covariance

Intuition. Measure what happened. Impose nothing.

Definition. $S = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})(r_t - \bar{r})'$.

Assumptions. Returns IID with finite fourth moments, and $T > N$ for invertibility. Stationarity over the window.

Cost. $O(N^2T)$ to form, $O(N^3)$ to invert or factorise.

Failure modes. Singular whenever $T \leq N$, and near-singular well before that — §5.5's table is the quantitative version. The eigenvalue spreading of §2.3 is at its maximum here because nothing counteracts it. Highly sensitive to outliers, since a single 10-sigma day enters squared and can dominate a covariance entry. Silently mis-estimates when assets trade in different time zones (§9.3).

When preferred. When $q < 0.05$ or so — a genuinely small universe with a long history — and as the baseline every other estimator must beat. Never in production above $q \approx 0.1$ without regularisation. [Practice] It remains the right thing to compute, since most other estimators are functions of it.

6.3 Exponentially weighted (EWMA / RiskMetrics)

Intuition. Recent data is more relevant. Weight it more.

Definition. With decay $\theta \in (0, 1)$, $\hat{\Sigma}_t = (1 - \theta) r_{t-1} r_{t-1}' + \theta \hat{\Sigma}_{t-1}$. RiskMetrics popularised $\theta = 0.94$ for daily data.

Assumptions. Covariance drifts smoothly; the recent past is the best guide. Implicitly, that a single decay rate suits every entry of the matrix.

Cost. $O(N^2)$ per update — the cheapest thing here, and the reason it is everywhere in real-time risk systems.

Failure modes. This one has a trap that catches people repeatedly. The effective sample size of an EWMA with decay θ — the number of equally-weighted observations that would carry the same estimation variance — is

$$T_{\text{eff}} = \frac{1 + \theta}{1 - \theta}.$$

At the standard $\theta = 0.94$ that is 32 observations. So a RiskMetrics covariance matrix on a universe of 100 assets has an effective aspect ratio of $q_{\text{eff}} = 100/32 = 3.1$ — hopelessly singular — no matter how many years of history you feed it. [Practice] The matrix will still invert, because the arithmetic runs on all T rows, but it is numerically meaningless and the optimiser will happily use it. Compute N/T_{eff} , not N/T . For a universe of 100 names you need $\theta \gtrsim 0.995$ ($T_{\text{eff}} \approx 400$) before the matrix is usable for optimisation at all.

When preferred. Excellent for volatilities, where responsiveness genuinely helps and N parameters are cheap to estimate. Dangerous for correlations at any realistic N . [Practice] The sensible hybrid is common and rarely written down: EWMA the volatilities with a short decay, estimate the correlation matrix on a much longer window, and recombine as $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$.

6.4 Constant correlation

Intuition. Estimate one correlation number instead of $N(N-1)/2$ of them.

Definition. $\hat{C}_{ij} = \bar{\rho}$ for $i \neq j$, where $\bar{\rho}$ is the average sample pairwise correlation; $\hat{C}_{ii} = 1$. Then $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$ with sample volatilities on the diagonal.

Assumptions. All pairs are equally related — plainly false, and deliberately so.

Cost. Trivial. The inverse has a closed form via Sherman–Morrison and never needs a decomposition.

Failure modes. Discards all sector and industry structure, which for equities is a large and genuinely tradeable part of the truth. Any strategy whose edge lives in relative value within a sector is destroyed by it.

When preferred. Rarely on its own — but it is the single most useful shrinkage target (§6.5), and it is a surprisingly strong benchmark. Constant-correlation and single-index targets are the two that Ledoit and Wolf’s own work keeps returning to, because both capture the dominant market mode while costing almost no parameters.

6.5 Linear shrinkage (Ledoit–Wolf)

Intuition. The sample matrix is unbiased but noisy; a structured target is biased but stable. Take a weighted average, and derive the weight rather than guessing it.

Definition.

$$\hat{\Sigma}_{\text{LW}} = \delta \Phi + (1 - \delta) S, \quad \delta \in [0, 1],$$

where Φ is a low-parameter target — commonly the identity scaled to the average variance, the constant-correlation matrix, or a single-index matrix. Ledoit & Wolf (2003, 2004) derive the δ minimising expected squared Frobenius distance to the true Σ , in closed form from the data. Roughly, δ^* grows with the sampling noise in S and shrinks with the target’s bias — so it rises with q automatically.

Assumptions. That the chosen target is a reasonable central tendency. The optimal δ is derived under a quadratic loss on the matrix, which is not the same as the loss you care about (portfolio variance) — see failure modes.

Cost. $O(N^2T)$, dominated by forming S . In `scikit-learn` as `LedoitWolf / OAS`; a handful of lines otherwise.

Failure modes. The optimal intensity is optimal for matrix estimation, not for portfolio performance, and the two objectives differ: portfolio variance weights the small-eigenvalue directions far more heavily than Frobenius loss does. [Contested] In practice the Frobenius-optimal δ is usually somewhat too small for portfolio use, and shrinking harder often does better out of sample. Shrinking toward the identity distorts the volatility structure — an asset with genuinely low volatility gets pulled up — which is why a correlation-space target is generally better (§9.5). Finally, a single δ applies one affine map to the whole spectrum, but §5.5 says the distortion is not uniform: the largest eigenvalue needs almost no correction and the smallest needs a great deal. That observation is exactly what §6.6 fixes.

When preferred. [Practice] This is the default. If you do one thing to your covariance matrix, do this. It is well-tested, cheap, captures most of the available improvement, and its intensity is derived rather than guessed — which is not the same as being beyond your control. Treat δ^* as a floor rather than a final answer, for the reason given under failure modes, and see §9.8 for what raising it buys. Everything after it in this section is a refinement with a worse effort-to-benefit ratio.

6.6 Nonlinear shrinkage

Intuition. §6.5 applies one correction to all eigenvalues. But the bias depends on where in the spectrum you are. So give each eigenvalue its own correction.

Definition. Keep the sample eigenvectors and replace each sample eigenvalue λ_i with a corrected value $\tilde{\lambda}_i$: $\hat{\Sigma} = \sum_i \tilde{\lambda}_i v_i v_i'$. The map $\lambda_i \mapsto \tilde{\lambda}_i$ is estimated nonparametrically from the spectrum itself, targeting the oracle that minimises loss given the sample eigenvectors (Ledoit & Wolf, 2017). The shape is intuitive: large eigenvalues barely move, mid-spectrum ones shrink toward the mean, and the smallest are pushed up substantially — undoing the downward bias that §2.3 identified as the source of the damage.

Assumptions. That the sample eigenvectors are worth keeping, which random-matrix theory supports for the bulk but not for the leading directions in small samples. Large N and T with q fixed.

Cost. $O(N^3)$ for the eigendecomposition, plus a numerical inversion of the Marchenko–Pastur relation. Implementations exist in R (`nlshrink`) and Python; writing one from scratch is a real project.

Failure modes. Needs a reasonably large N for the spectral estimation to be meaningful — below roughly 50 assets it has little to work with and linear shrinkage is as good. The theory assumes IID observations, so it is sensitive to the same non-stationarity everything else is. [Contested] The out-of-sample advantage over well-tuned linear shrinkage is real but modest in most published comparisons, and can vanish once transaction costs are included.

When preferred. Large universes ($N \gtrsim 100$) with q between roughly 0.2 and 1, where you are optimising rather than merely reporting risk, and where a few basis points of variance reduction justifies the implementation effort. Ledoit and Wolf’s own 2022 review is the guide to which variant to pick.

6.7 Random-matrix eigenvalue clipping

Intuition. §5.5 tells you exactly which eigenvalues are consistent with pure noise. Those carry no information, so stop pretending they differ: replace them all with their average.

Definition. Compute the sample correlation matrix’s eigenvalues. Fit the Marchenko–Pastur edge λ_+ , allowing for the fact that outlier eigenvalues absorb part of the trace (the iterative fit used in §2.3’s figure). Keep every $\lambda_i > \lambda_+$ unchanged; replace all the rest by their common mean, preserving the trace. Rebuild from the same eigenvectors.

Assumptions. Noise is IID across assets and time within the bulk. That the eigenvectors of the retained directions are estimated well.

Cost. One eigendecomposition, $O(N^3)$, plus a short fitting loop.

Failure modes. The hard threshold is arbitrary at the margin — an eigenvalue just inside the edge is treated as pure noise and one just outside as pure signal, though they are nearly indistinguishable. Nonlinear shrinkage (§6.6) is the smooth version and generally dominates it. Fat tails inflate the empirical edge, so a naive fit over-clips during turbulent periods. [Practice] Clipping also destroys genuine small-eigenvalue structure such as tightly cointegrated pairs — which is fatal if that structure is what you trade.

When preferred. When you want the transparency: clipping tells you how many real risk factors your data supports, which is a useful diagnostic in its own right and is easy to explain to a risk committee. As an estimator it has largely been superseded by §6.6.

6.8 Explicit factor models

Intuition. Assets move together because they share exposures. Model the exposures, and the N^2 correlations follow from a handful of factors.

Definition. $\hat{\Sigma} = F\Omega F' + \Psi$, with F the $N \times K$ matrix of loadings, Ω the $K \times K$ factor covariance, and Ψ diagonal. Loadings come from observable characteristics — sector, size, value, momentum, duration, country — which is what distinguishes this from §6.9. Parameter count drops from $O(N^2)$ to $O(NK)$.

Assumptions. The factor set spans the systematic risk, and residuals are genuinely uncorrelated. The second assumption is the one that fails.

Cost. $O(NK^2 + K^3)$. The inverse has a closed form via the Sherman–Morrison–Woodbury identity that never requires an $N \times N$ decomposition — a large practical advantage at N in the thousands.

Failure modes. Ψ diagonal is a strong claim: two airlines have correlated residuals after every standard factor, and a pairs strategy trading that residual will be told by its risk model that the position is riskless. [Fact] This is the classic way a factor risk model understates the risk of a statistical-arbitrage book. Loadings are themselves estimated and drift. And because the model can only express K systematic risk directions, the remaining $N - K$ eigenvalues are pinned inside the range of the idiosyncratic variances on Ψ 's diagonal — a form of very aggressive implicit shrinkage which is usually helpful and occasionally badly wrong.

When preferred. Large equity universes; anywhere the factor structure is economically well-understood; anywhere you must attribute risk as well as measure it, since factor models answer “why is the portfolio risky” and rotationally-invariant estimators do not. [Practice] The dominant choice in institutional equity risk management for forty years, and deservedly.

6.9 Statistical factor models and POET

Intuition. Don't specify the factors — let the data find them. Then patch the factor model's worst assumption.

Definition. Take the leading K principal components of S as factors, with K chosen by the Marchenko–Pastur edge (§6.7) or an information criterion. POET (Fan, Liao & Mincheva, 2013) adds the important refinement: after removing the K principal components, threshold the residual covariance rather than forcing it diagonal — keep residual correlations that are large enough to be real, zero the rest. That recovers the within-industry structure §6.8 discards while still regularising.

Assumptions. That the leading eigenvectors are estimated well — true when the corresponding eigenvalues are far outside the bulk, false for marginal ones. That residual correlation is sparse.

Cost. $O(N^2T + N^3)$; the thresholding step is $O(N^2)$.

Failure modes. Statistical factors are not interpretable and rotate over time, so risk attribution becomes unstable and period-to-period comparisons are hard. Choosing K badly is costly in both directions: too small and real risk is dumped into the “idiosyncratic” bucket, too large and you have re-imported the noise you were removing. The thresholding level in POET is a genuine tuning parameter.

When preferred. When you lack a credible characteristic set — futures, crypto, a cross-asset book — or when you suspect your explicit factors miss something. POET specifically when you trade residual relationships and §6.8's diagonal- Ψ assumption is the thing that would hurt you.

6.10 Dynamic conditional correlation

Intuition. Volatilities and correlations both move over time, but they move differently. Model them separately.

Definition. Engle (2002) in two steps: fit a univariate GARCH to each asset to get $\hat{D}_t$; then fit a low-parameter autoregressive process to the standardised residuals' correlation to get $\hat{C}_t$; combine as $\hat{\Sigma}_t = \hat{D}_t \hat{C}_t \hat{D}_t$. Two parameters govern the entire correlation dynamics regardless of N .

Assumptions. That every pair's correlation follows the same two-parameter dynamics — the “correlation targeting” restriction that makes it tractable.

Cost. N univariate GARCH fits plus a small joint optimisation. Feasible to a few hundred assets; beyond that the composite likelihood becomes awkward.

Failure modes. The two-parameter restriction is severe and empirically rejected. The correlation-targeting step is biased in high dimensions. [Contested] Its practical advantage over a well-shrunk static estimator for portfolio construction, as opposed to risk forecasting, is disputed — correlation persistence is real but the portfolio gain from tracking it is smaller than the turnover it generates. Combining DCC with nonlinear shrinkage (DCC-NLS) addresses the dimension problem and is the modern form.

When preferred. When correlation dynamics are themselves the object — tail-risk work, stress testing, anything where the crisis-correlation spike matters. See the regime note's §11.4 for the same argument from a different angle. Less compelling when you simply need a matrix to invert once a month.

6.11 Hierarchical and clustering-based estimators

Intuition. Assets cluster — by sector, by asset class, by geography. Estimate the cluster structure, and impose it.

Definition. Convert the correlation matrix to a distance, $d_{ij} = \sqrt{2(1 - \rho_{ij})}$, cluster hierarchically, and use the resulting tree to regularise: either by averaging correlations within and between clusters (producing a block-structured matrix) or by using the tree for allocation directly without ever inverting anything (§7.11).

Assumptions. That a tree is a reasonable model of the dependence structure — a strong assumption, since real assets belong to several groupings at once.

Cost. $O(N^2 \log N)$ for the clustering.

Failure modes. Hierarchical clustering is notoriously unstable: small perturbations to the correlation matrix can reorganise the tree entirely, which propagates into weights. The linkage choice (single, average, ward) is a free parameter with real consequences and no principled selection rule. A tree cannot represent an asset that belongs to two clusters.

When preferred. When the universe has genuine, stable, known group structure and you want that reflected explicitly. As an ingredient in §7.11 rather than as a covariance estimator in its own right.

6.12 Robust estimation

Intuition. Covariance is a squared quantity, so one bad print moves it a lot. Down-weight extremes.

Definition. Several routes: winsorise or truncate returns before computing S ; use a robust scatter estimator such as the Minimum Covariance Determinant; or fit a multivariate t and use its scatter matrix, which down-weights observations by their Mahalanobis distance.

Assumptions. That extreme observations are contamination rather than information. In finance this is often false — the crisis days are exactly the ones you need the risk model to know about.

Cost. Winsorising is free; MCD is expensive and scales badly in N .

Failure modes. The assumption above is the failure mode. Robustify a risk model and you may have built one that is blind to precisely the days that matter. Robust estimators also interact badly with shrinkage, since both are pulling toward the centre and the combined effect is easy to overdo.

When preferred. For data errors — bad prints, stale quotes, mis-scaled corporate actions — where the outlier is genuinely spurious. [Practice] The right frame is that this is a data-cleaning step (§9.2–§9.3), not a modelling step. Clean the errors robustly, then estimate on the cleaned series without further robustifying.

6.13 Comparison

Ratings are my assessment, not measurements.

Estimator	Structure imposed	Cost	Needs $T > N$	Main risk
Sample	None	Low	Yes	Unusable above $q \approx 0.1$
EWMA	Recency	Very low	Effectively	T_{eff} trap (§6.3)
Constant correlation	Extreme	Trivial	No	Destroys sector structure
Linear shrinkage	Moderate, tunable	Low	No	One δ for all eigenvalues
Nonlinear shrinkage	Moderate, adaptive	Medium	No	Implementation effort
RMT clipping	Moderate	Medium	No	Hard threshold; kills real spreads
Explicit factor	Strong, economic	Low	No	Diagonal Ψ is false
Statistical factor / POET	Strong, data-driven	Medium	No	K selection; uninterpretable
DCC	Dynamic	High	No	Two-parameter restriction
Hierarchical	Strong, tree	Low	No	Tree instability
Robust	Outlier down-weighting	Medium–high	Yes	Blind to the days that matter

§6 Key takeaways

1. Every estimator here answers one question: how much structure to impose in exchange for less estimation error. Because of §2, the variance term is almost always the binding one — err toward more structure.
2. Estimators modify eigenvalues (rotationally invariant), eigenvectors (structure-imposing), or observation weights (reweighting). Knowing which tells you what an estimator can and cannot fix.
3. Linear shrinkage is the default. It is cheap, untuned, well-tested, and captures most of the available gain. Do it before anything else in this section.
4. EWMA's effective sample size is $(1 + \theta)/(1 - \theta)$ — just 32 observations at the standard $\theta = 0.94$. A RiskMetrics correlation matrix is unusable for optimisation above about 30 assets, and it will not warn you.
5. Split the problem: volatilities are easy and benefit from responsiveness; correlations are hard and benefit from long windows. Estimate them separately and recombine.
6. A factor model's diagonal residual assumption is exactly the thing a relative-value book trades. If your edge lives in residual correlation, a standard factor risk model will tell you your position is riskless.
7. Robust estimation is a data-cleaning tool, not a modelling tool. In markets the outliers are usually the information.

7. From covariance to weights

7.1 How to read this section

Each rule below carries the same six fields, and one of them is unusual: what it implicitly assumes about μ and Σ . Every rule here — including the ones that loudly disclaim optimisation — is $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ for some choice, and naming that choice is the fastest way to understand what a rule will do when the world does not cooperate. Filling in that field for all eleven rules produces §8's taxonomy directly.

A rule that “doesn't use expected returns” is a rule that uses a particular vector of expected returns, chosen for its statistical stability rather than its plausibility. That is often a good trade. But it is a choice, not an abstention, and pretending otherwise is how people end up surprised by their own portfolios.

7.2 Equal weight ($1/N$)

Intuition. Refuse to estimate anything.

Definition. $w_i = 1/N$.

Implied beliefs. All assets have identical expected returns, identical variances, and identical pairwise correlations. Under those beliefs $1/N$ is exactly the mean-variance optimum — so this is not the absence of a model but an extremely strong one.

Cost. None.

Failure modes. Ignores obvious risk differences: an equal-weighted portfolio of a T-bill fund and a leveraged biotech is not balanced in any meaningful sense. Concentrates risk in whatever is most volatile. Requires rebalancing to maintain, which is a real cost that comparisons often omit. Its performance depends heavily on the universe handed to it — $1/N$ over a curated list of survivors is a different animal from $1/N$ over everything.

When preferred. As the benchmark every other rule must beat, always. And genuinely as the answer when N is small, the assets are broadly comparable, and q is large enough that nothing can be estimated — DeMiguel and co-authors' (2009) result is a real result, whatever §2.6 says about its interpretation.

7.3 Inverse volatility, and inverse variance

Intuition. Give each asset the same risk budget, ignoring how they interact.

Definition. Inverse volatility: $w_i \propto 1/\sigma_i$. Inverse variance: $w_i \propto 1/\sigma_i^2$. These are different rules and are routinely confused.

Implied beliefs. Both discard correlation information. Beyond that they differ: inverse variance is the minimum-variance portfolio of uncorrelated assets ($\mu \propto 1$), while inverse volatility is the maximum-Sharpe portfolio of assets with equal Sharpe ratios ($\mu \propto \sigma$) — under a diagonal Σ , and equally under any equicorrelated one (§7.6). Inverse volatility is therefore the more aggressive of the two, since it credits volatile assets with proportionally higher returns.

Cost. $O(N)$. Needs only the diagonal, which is the part of Σ that can actually be estimated well (§5.3).

Failure modes. Ignoring correlation is not a small approximation. Ten correlated European bank stocks and one gold miner receive eleven equal risk budgets, of which ten are the same bet. Volatility is also backward-looking, so the rule systematically underweights assets that were recently calm and are about to move.

When preferred. Very often, and more often than its crudeness suggests. [Practice] It is the sensible default when q is too large for any correlation estimate to be trusted, and it is the right first implementation of almost any system — get it working, then see whether correlations add anything.

7.4 Global minimum variance

Intuition. Make the portfolio as quiet as possible and decline to forecast returns.

Definition. $w = \Sigma^{-1} \mathbf{1} / (\mathbf{1}' \Sigma^{-1} \mathbf{1})$, usually with a long-only or box constraint attached.

Implied beliefs. $\mu \propto 1$: all assets have the same expected return. Note this is a strong claim, not a neutral one — it says a biotech startup and a utility have identical expected returns, which no one believes. It is adopted because §2.5 says a wrong-but-stable μ beats a noisily-estimated one.

Cost. One linear solve, $O(N^3)$, or a small QP with constraints.

Failure modes. Notoriously concentrated: unconstrained GMV typically puts large weights on a handful of low-volatility assets and shorts others heavily. Clarke, de Silva & Thorley (2011) characterise exactly which assets survive — broadly, those with low beta to the dominant factor and low idiosyncratic volatility — and the answer is usually a small fraction of the universe. It is the rule most exposed to §5.6's $1/(1-q)$ risk understatement, because it is dominated by the small-eigenvalue directions. Turnover is high without constraints.

When preferred. Whenever you have no return forecast and do have a usable correlation estimate. [Fact] Long-only minimum-variance portfolios have outperformed cap-weighted benchmarks on a risk-adjusted basis over long samples in most equity markets — though [Contested] much of that is attributable to the low-volatility factor rather than to construction skill.

7.5 Maximum diversification

Intuition. Maximise the gap between the weighted-average volatility of your holdings and the volatility you actually experience. That gap is diversification.

Definition. Maximise the diversification ratio $DR(w) = (w'\sigma)/\sqrt{w'\Sigma w}$, where σ is the vector of asset volatilities. The solution is $w \propto \Sigma^{-1}\sigma$ (Choueifaty & Coignard, 2008).

Implied beliefs. $\mu \propto \sigma$ — every asset has the same Sharpe ratio. This is a genuinely defensible prior, arguably more defensible than GMV's, since equal Sharpe ratios is roughly what an efficient market with risk-averse investors would produce.

Cost. One solve, same as GMV.

Failure modes. Shares GMV's concentration and Σ^{-1} instability. The diversification ratio can be gamed by adding assets that are near-duplicates of each other — the measure rewards apparent breadth. Long-only constraints bind frequently and change the character of the solution.

When preferred. When the equal-Sharpe prior is more comfortable than the equal-return prior, which for a cross-asset universe it usually is. [Practice] Worth computing alongside GMV: if the two disagree sharply, the disagreement is informative about which assets are carrying your risk.

7.6 Equal risk contribution (risk parity)

Intuition. Every asset should contribute the same amount of risk — not the same capital, and not the same standalone volatility, but the same share of portfolio variance after correlations are accounted for.

Definition. Asset i 's marginal contribution to risk is $\partial\sigma_p/\partial w_i = (\Sigma w)_i/\sigma_p$, so its total contribution is $RC_i = w_i(\Sigma w)_i/\sigma_p$. The ERC portfolio solves $RC_i = RC_j$ for all i, j , subject to $1'w = 1$, $w \geq 0$. There is no closed form in general; it is found numerically, and Maillard, Roncalli & Teiletche (2010) prove existence and uniqueness for the long-only case when Σ is positive definite — which is why ERC, despite never inverting anything, still needs $q < 1$.

Implied beliefs. Equal Sharpe ratios and equal pairwise correlations. Under exactly those two conditions ERC is the tangency portfolio. Under equicorrelation alone it reduces to inverse volatility (§7.3).

Cost. An iterative solve — cyclical coordinate descent converges reliably. More expensive than a single linear solve but not by much.

Failure modes. ERC's volatility always lies between the minimum-variance portfolio's and $1/N$'s, which is a useful guarantee and also a ceiling on how much it can help. Because it never shorts and never concentrates, it is robust — but the robustness comes from ignoring most of the information in Σ . [Contested] The levered version, where risk parity is scaled up to equity-like volatility, has a well-known dependence on the bond leg and on borrowing costs.

When preferred. Multi-asset allocation, where the assets have genuinely different volatilities and you want a defensible, explainable, low-turnover answer. [Practice] It is the best-behaved of the risk-based rules in production and the easiest to explain to someone who will not read §5.

7.7 Mean-variance with an explicit μ

Intuition. You actually have a return forecast. Use it.

Definition. $w \propto \hat{\Sigma}^{-1}\hat{\mu}$, subject to constraints, scaled to a volatility target.

Implied beliefs. That $\hat{\mu}$ is good enough to survive §2.5's $11\times$ penalty.

Cost. One solve.

Failure modes. Everything in §2. Specifically: the portfolio will concentrate on whichever assets have the most extreme $\hat{\mu}$, and those are disproportionately the ones whose forecast errors are largest. The damage scales with the dispersion of $\hat{\mu}$, not its accuracy, so a signal that is correctly ranked but badly scaled will still wreck the portfolio.

When preferred. When you have a real, out-of-sample-validated signal. And then, [Practice] almost always in a tamed form: rank-transform the signal, winsorise it, scale it to a sensible dispersion, and shrink it toward zero. The common industry practice of converting a signal to cross-sectional z-scores, clipping at ± 2 or ± 3 , and multiplying by a modest assumed information coefficient is not crude — it is a defensible response to the fact that the optimiser will exploit any scale error you leave in.

7.8 Black–Litterman

Intuition. Instead of forecasting returns from scratch, start from the returns the market's own weights imply, and move away from them only where you have a view.

Definition. Reverse-optimize the market portfolio to get equilibrium returns $\Pi = \gamma \Sigma w_{\text{mkt}}$ — the μ that would make the observed market portfolio optimal. Then combine Π with explicit views (expressed as a matrix P picking out portfolios, a vector Q of expected returns on them, and a confidence matrix) via a Bayesian update, producing a posterior μ_{BL} that is used in the ordinary master form.

Implied beliefs. That the market portfolio is a sensible prior — which is a statement about market efficiency, and is the model's real content.

Cost. Trivial arithmetic; the difficulty is entirely in specifying views and confidences.

Failure modes. [Practice] The confidence parameters are the model, and there is no principled way to set them, so in practice they become the knob that is tuned until the output looks acceptable. Requires a defensible market portfolio, which exists for global equities and barely exists for a futures book. The apparatus is often more elaborate than the underlying idea warrants.

When preferred. When you have a natural equilibrium reference and a small number of well-articulated views on specific portfolios rather than a full cross-sectional forecast. Its enduring contribution is conceptual: shrink your views toward a defensible prior, which is §2.5's advice with machinery attached.

7.9 Resampled efficiency

Intuition. You do not know μ and Σ ; you have one draw of them. So simulate many draws, optimise each, and average the resulting weights.

Definition. Bootstrap or parametrically resample returns; re-estimate $\hat{\mu}, \hat{\Sigma}$ and re-optimize for each sample; average the weight vectors across samples. Due to Michaud (1989).

Implied beliefs. That averaging over the sampling distribution approximates the decision a Bayesian would make. This is heuristic rather than derived.

Cost. B optimisations for B resamples — the most expensive rule here, and trivially parallel.

Failure modes. [Contested] It has no formal optimality justification, and the averaged portfolio is not optimal for any coherent set of beliefs. Averaging weights that were individually extreme produces a less extreme portfolio, which is the whole benefit — but the same effect is obtained more cheaply and more transparently by shrinkage. It also inherits any bias in the original estimates, since resampling from a bad estimate reproduces the bad estimate. Patented, which has limited its adoption.

When preferred. [Practice] Mostly superseded. The idea remains valuable as a diagnostic rather than an estimator: resample your inputs, re-optimize, and look at the spread of the resulting weights. If asset 12's weight ranges from -40% to $+60\%$ across resamples, you have learned something important that a point estimate will not tell you.

7.10 Norm-constrained and regularised optimisation

Intuition. Solve the ordinary problem, but forbid extreme answers.

Definition. Minimise $w' \hat{\Sigma} w$ subject to $\|w\|_1 \leq c$ (which limits gross exposure and induces sparsity) or $\|w\|_2 \leq c$ (which spreads weight out), or add the norm as a penalty. DeMiguel, Garlappi, Nogales & Uppal (2009) show these constraints nest $1/N$, minimum variance, and shrinkage as special cases of one family.

Implied beliefs. A prior that weights are small — the same content as ridge or lasso regularisation in regression, and via §8.2 the same content as shrinking $\hat{\Sigma}$.

Cost. A convex programme; ℓ_2 is a QP, and ℓ_1 becomes one once w is split into positive and negative parts, which makes the constraint linear. Both are fast at realistic N .

Failure modes. c is a tuning parameter and must be chosen out of sample, or you have simply relocated the overfitting. ℓ_1 produces sparse portfolios, which sounds appealing but means the solution jumps discontinuously as inputs change — bad for turnover.

When preferred. When you want explicit, explainable control over gross exposure and position size, which in an institutional setting you usually do. [Practice] This is the form regularisation most often takes in production, precisely because a leverage limit is easier to defend to a risk committee than a shrinkage intensity — even though §8.2 says they are the same thing.

7.11 Hierarchical risk parity

Intuition. The problem is the matrix inverse. So don't invert anything. Use the correlation matrix to build a tree, then split capital down the tree.

Definition. Three steps (López de Prado, 2016): convert correlations to distances and cluster hierarchically; reorder the covariance matrix so similar assets are adjacent (quasi-diagonalisation); then recursively bisect, splitting capital between the two halves in inverse proportion to each half's variance.

Implied beliefs. That the dependence structure is well-described by a tree, and that $\hat{\Sigma}$ can be trusted for its diagonal, for the tree ordering, and for each cluster's aggregate variance — but never for its inverse.

Cost. $O(N^2 \log N)$, no inversion, and it works when $T < N$ — a real advantage that none of §7.4–§7.6 shares.

Failure modes. Tree instability (§6.11) propagates directly into weights. The linkage method is an unprincipled free parameter. [Contested] The claimed out-of-sample advantage over well-regularised alternatives has not replicated uniformly, and several studies find it comparable to inverse-variance weighting once both are given the same shrinkage treatment — which, given that HRP's inner step is inverse-variance weighting, is not surprising.

When preferred. When $q \geq 1$ and you need something, since it degrades gracefully where every inversion-based rule fails outright. Also when the universe has real hierarchical structure you want respected.

7.12 Growth-optimal (Kelly)

Intuition. Maximise the long-run growth rate of wealth rather than a one-period utility.

Definition. Maximise $\mathbb{E}[\log(1 + w'r)]$. Expanding the logarithm to second order gives $w'\mu - \frac{1}{2}w'\Sigma w$, whose maximiser is exactly $w = \Sigma^{-1}\mu$ — the master form with $\gamma = 1$. The approximation is in the expansion, not in the optimisation (§8.2).

Implied beliefs. That you know μ and Σ , and that maximising long-run growth is the objective.

Cost. One solve.

Failure modes. Full Kelly is far too aggressive under parameter uncertainty: it is derived assuming known parameters, and its drawdowns are brutal even when they are. [Practice] Practitioners use fractional Kelly — typically a quarter to a half — which is exactly the master form with γ of 2 to 4, and is better understood as a response to estimation error than as a different theory.

When preferred. As a way of thinking about leverage, and as the answer to “how large can this be” rather than “how large should this be”. The composition question it answers is identical to §7.7's.

7.13 Comparison

Rule	Uses correlations	Implied μ	Concentration risk	Works at $q > 1$
$1/N$	No	$\propto 1$, with $\Sigma \propto I$	None	Yes
Inverse variance	No	$\propto 1$	Low	Yes
Inverse volatility	No	$\propto \sigma$	Low	Yes
Minimum variance	Yes	$\propto 1$	High	No
Maximum diversification	Yes	$\propto \sigma$	High	No
Equal risk contribution	Yes	$\propto \sigma$, equicorrelated	Low	No
Mean-variance	Yes	Explicit forecast	Very high	No
Black–Litterman	Yes	Views shrunk to equilibrium	Medium	No
Resampled	Yes	Explicit, averaged	Medium	No
Norm-constrained	Yes	$\propto 1$ or explicit	Controlled by c	Yes
Hierarchical risk parity	Ordering and cluster variance	$\propto 1$ within clusters	Low	Yes
Growth-optimal (Kelly)	Yes	Explicit forecast	Very high	No

§7 Key takeaways

1. Every rule here is $w \propto \hat{\Sigma}^{-1} \hat{\mu}$. A rule that “doesn’t forecast returns” has chosen a μ — usually 1 or σ — for stability rather than plausibility. That is often right, and it is still a choice.
2. $1/N$ implies identical means, variances and correlations. It is the most opinionated rule in the section, not the least.
3. Inverse volatility and inverse variance are different rules with different implied beliefs: equal Sharpe ratios versus equal expected returns. Know which one you are running.
4. Minimum variance and maximum diversification are the same object with $\mu \propto 1$ and $\mu \propto \sigma$. Both concentrate, both live in the fragile part of the spectrum, and both need constraints in production.
5. Equal risk contribution is guaranteed to sit between minimum variance and $1/N$ in volatility. That is both its safety property and its limit.
6. Black–Litterman’s durable idea is not its algebra but its instruction: shrink your views toward a defensible prior.
7. Resampling is better as a diagnostic than an estimator. Resample, re-optimize and look at the spread of weights — if a position swings from -40% to $+60\%$, you have learned more than any point estimate will tell you.
8. Hierarchical risk parity’s real advantage is that it degrades gracefully at $q \geq 1$, where every inversion-based rule simply fails.

8. Taxonomy and equivalences

8.1 The master form, with slots

Everything in §6 and §7 fits one expression. Writing it with its slots named is the most compact statement of the field available:

$$w \propto \left(\underbrace{\hat{\Sigma}}_{S1} + \underbrace{\Delta}_{S2} \right)^{-1} \underbrace{\hat{\mu}}_{S3} \quad \text{subject to} \quad w \in \underbrace{\mathcal{C}}_{S4}, \quad \text{scaled to} \quad \underbrace{\sigma^*}_{S5}.$$

Slot	What it is	Choices	Section
S1	Covariance estimator	Sample, EWMA, factor, DCC, hierarchical	§6
S2	Regulariser added to it	Shrinkage, clipping, ridge, none	§6.5–§6.7
S3	Return view	1, σ , an explicit forecast, a Bayesian posterior	§7
S4	Constraint set	Long-only, box, gross-exposure, turnover, cardinality	§7.10, §9.6
S5	Leverage	Risk aversion, volatility target, Kelly fraction	§1.3, §10

Δ is whatever the regulariser adds to the raw estimate: νI for a ridge, $\delta(\Phi - \hat{\Sigma})$ for shrinkage toward a target Φ (§6.5), zero for none.

A design is a coordinate in this five-dimensional space. Naming a method — “minimum variance”, “risk parity”, “HRP” — specifies some coordinates and leaves others implicit, which is why two shops running “risk parity” can hold different portfolios and both be right about the label.

8.2 The equivalences

This is the highest-value half page in the document. Each row states that two things which look different are the same, and marks whether the identity is exact or only approximate. A practitioner who does not know these will apply the same correction three times under three names.

Exact identities.

This	Is exactly	Because
Minimum variance	Mean-variance with $\mu \propto 1$	Same formula (§5.1)
Inverse variance	Minimum variance with diagonal Σ	$\Sigma^{-1}1$ has entries $1/\sigma_i^2$
Inverse volatility	Max-Sharpe, diagonal Σ , $\mu \propto \sigma$	$\Sigma^{-1}\sigma$ has entries $1/\sigma_i$
Maximum diversification	Max-Sharpe with $\mu \propto \sigma$	$w \propto \Sigma^{-1}\sigma$
$1/N$	Mean-variance with $\mu \propto 1$ and equal variances and correlations	$C^{-1}1 \propto 1$; $\Sigma \propto I$ is the special case
Equal risk contribution	Inverse volatility, under equicorrelation	Maillard et al. (2010)
Equal risk contribution	The tangency portfolio, under equal Sharpe ratios and equicorrelation	Same
Ridge on Σ : $\Sigma + \nu I$	Linear shrinkage to a scaled identity	Rearrange the convex combination; the leftover positive scalar drops out of w
...which is also	An ℓ_2 penalty on w	Add $\nu\ w\ ^2$ to the objective
...which is also	A Gaussian prior on w centred at zero	MAP with a zero-mean Gaussian prior of variance $\propto 1/\nu$
Long-only constraint on GMV	GMV on a shrunk covariance matrix	Jagannathan & Ma (2003)
An ℓ_1 or ℓ_2 constraint on w	A corresponding shrinkage of $\hat{\Sigma}$	DeMiguel, Garlappi, Nogales & Uppal (2009), §7.10

This	Is exactly	Because
Black–Litterman with no views Any efficient portfolio	Reverse-optimised market weights An affine combination of GMV and tangency	The posterior collapses to the prior Two-fund separation (§5.2); the two weights sum to one but may be negative

The bolded row is the most consequential result in the field's practical history. Jagannathan & Ma (2003) showed that when you solve minimum variance subject to $w \geq 0$, the Karush–Kuhn–Tucker conditions mean the solution is identical to the unconstrained solution on a modified matrix $\tilde{\Sigma} = \Sigma - \zeta 1' - 1\zeta'$, where $\zeta \geq 0$ holds the multipliers on the non-negativity constraints. Assets that want to be shorted have their covariances reduced — which is precisely shrinkage, arrived at by a completely different route.

Two things follow, and both are practical:

A no-short constraint is not a business restriction that costs you performance. It is a statistical estimator, and often a good one. This explains the otherwise puzzling empirical record in which constrained naive optimisation kept beating unconstrained sophisticated optimisation.

Regularisation compounds. If you shrink your covariance matrix, impose a long-only constraint, cap position sizes, and add a turnover penalty, you have applied four shrinkages, not one strategy plus three prudent safeguards. The combined effect is frequently a portfolio far closer to $1/N$ than anyone intended — and, more insidiously, the individual components each look conservative in isolation. [Practice] Measure the total: compare your final weights to the unregularised solution and to $1/N$, and see which you are nearer.

Approximate identities.

This	Approximately equals	Caveat
Michaud resampling	Shrinkage toward a less concentrated portfolio	No formal correspondence; empirically similar
A turnover penalty	Shrinkage toward the current portfolio	Exact for a quadratic penalty (§10.3)
RMT eigenvalue clipping	Nonlinear shrinkage with a step function	Clipping is the hard-threshold special case
EWMA with decay θ	A rolling window of $(1 + \theta)/(1 - \theta)$ observations	Different weight profile, same effective sample size
Kelly	Mean-variance with $\gamma = 1$	Exact only in the small-return limit
Fractional Kelly at f	Mean-variance with $\gamma = 1/f$	Same caveat

8.3 Which slot actually matters

This is the payoff of the taxonomy, and the ordering is my assessment rather than a measured result — though §2.5's evidence supports the top of it strongly.

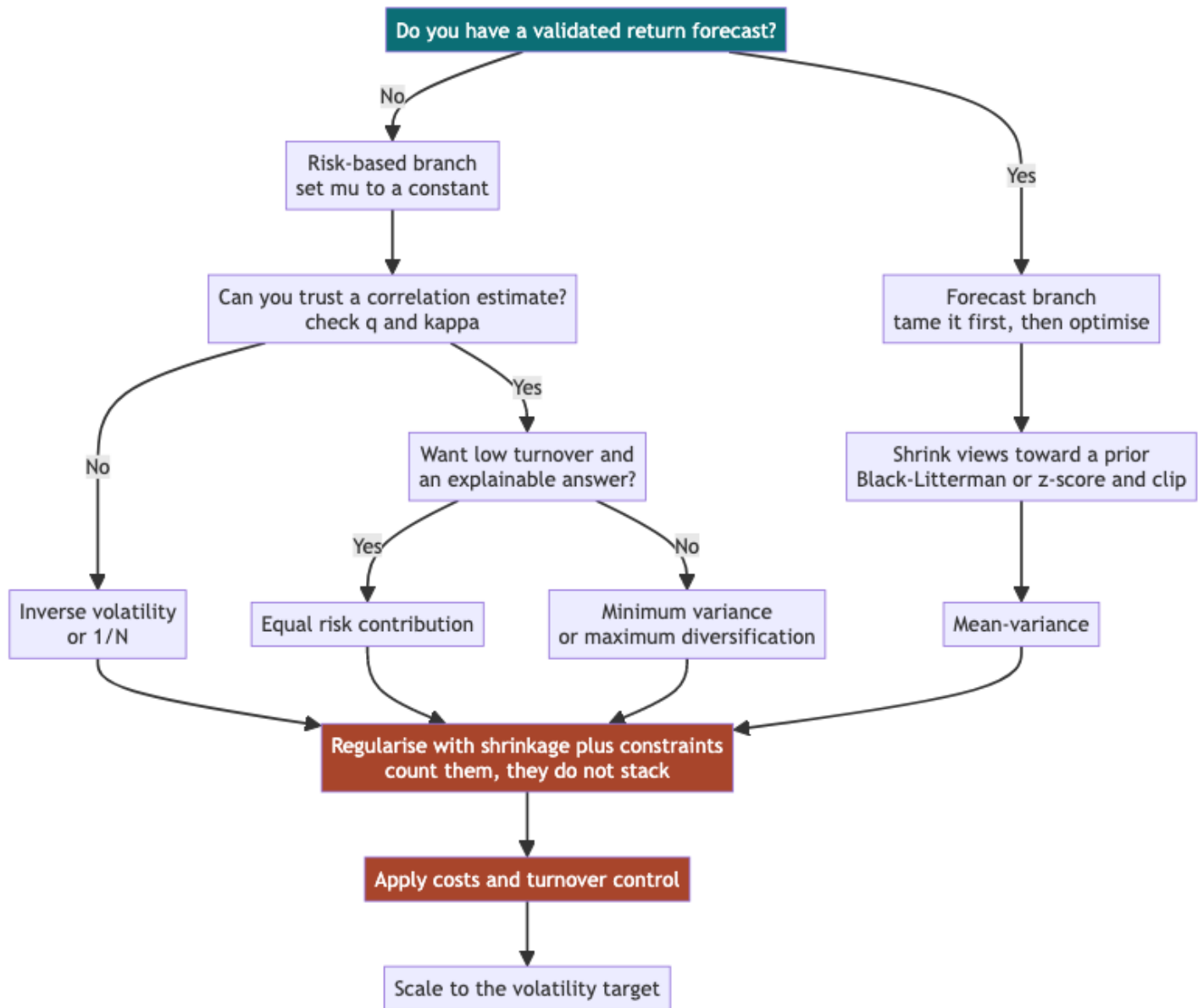
One scoping note before the list. It ranks the five slots, and data quality is not a slot: it sits upstream of the whole formula. §9.1 puts it above everything here, and I mean that — a synchronicity bug (§9.3) beats every choice below. The ordering that follows is “given clean data, which slot matters”.

1. S3, the return view — dominates the other four combined. Whether you put a forecast in, and how dispersed it is, moves outcomes more than every other slot put together. Chopra–Ziembra's 22 : 2 : 1 is the quantitative version. The binary decision “do I have a real μ or not” is the single most consequential line in your codebase.
2. S2 and S4 jointly — but only as a binary. Whether you regularise matters enormously; which regulariser you pick matters much less. Linear shrinkage, eigenvalue clipping, a long-only constraint and a gross-exposure cap all buy most of the same improvement, because §8.2 says they are the same act. Choose the one you can explain and monitor.
3. S5, leverage. Sets the volatility you actually experience, which is what gets noticed, but is orthogonal to composition and easy to change.
4. S1, the covariance estimator — matters least. Conditional on doing some regularisation, the choice between a shrunk sample matrix, a factor model, and nonlinear shrinkage is a real but second-order decision, typically worth a few percent of portfolio variance rather than a multiple of it.

Point 4 deserves emphasis because it is counterintuitive and this document is half about covariance estimation:

The covariance estimator is the least important of the five slots. The literature is largest here because the problem is mathematically interesting, not because it is where the money is. If you are choosing between nonlinear shrinkage and POET while your $\hat{\mu}$ is a raw sample mean and your gross exposure is uncapped, you are optimising the wrong slot by two orders of magnitude.

8.4 The design space as a picture



The two red nodes are the ones every branch passes through, and they are the ones most often skipped. Note also what the diagram makes visible: the estimator choice from §6 does not appear as a branch at all. It is a modifier applied inside the “regularise” node, which is §8.3’s ordering drawn rather than asserted.

8.5 Same name, different thing

Name	Meaning A	Meaning B
Risk parity	Equal risk contribution: an allocation rule (§7.6)	A levered multi-asset product, typically bond-heavy
Hierarchical risk parity	López de Prado’s tree algorithm (§7.11)	Often assumed to be ERC applied hierarchically. It is not — it never equalises risk contributions
Minimum variance	The unconstrained $\Sigma^{-1}1$ solution	The long-only constrained version, which is a different portfolio and a different estimator (§8.2)
Shrinkage	Toward a structured target matrix (§6.5)	Of individual eigenvalues (§6.6). Different operations

Name	Meaning A	Meaning B
Diversification	Number of holdings	The diversification ratio (§7.5), or the effective number of bets (§11.4). The three routinely disagree
Optimisation	The QP solve — trivial, never the problem	The whole input-estimation pipeline — hard, always the problem
Factor model	A risk model: explains covariance	A return model: predicts μ . Same word, opposite slots (S1 vs S3)

The last row causes more confusion than any other. A factor model used for risk and a factor model used for alpha share mathematics and share nothing else; one lives in S1 and the other in S3, and §8.3 says they sit at opposite ends of the importance ordering.

§8 Key takeaways

1. Five slots: covariance estimator, regulariser, return view, constraints, leverage. Every named method is a coordinate, and most names pin down only two or three of them.
2. Minimum variance, maximum diversification, inverse volatility, ERC and $1/N$ are one formula with two choices of μ — 1 or σ — and five different amounts of structure imposed on Σ . They are not a separate “risk-based” family.
3. Jagannathan and Ma’s identity is the load-bearing result: a long-only constraint is covariance shrinkage. Constraints are estimators, not concessions.
4. Because of that, regularisation does not stack. Shrinkage plus long-only plus position caps plus a turnover penalty is four shrinkages. Measure how close the result is to $1/N$ — you may already be there.
5. The importance ordering is: the return view, then whether you regularise at all, then leverage, then — last — which covariance estimator you use.
6. The covariance literature is the biggest because the problem is the most mathematically interesting, not because it is where the money is.
7. “Factor model” means a risk model in slot S1 and a return model in slot S3. These are opposite ends of the importance ordering and share only their algebra.

9. Implementation: what actually matters

9.1 Effort ordering

Readers of this literature consistently misallocate effort, because the interesting problems and the important problems are different problems. Here is the ordering I would defend, worst-first in terms of what typically goes wrong:

Rank	Decision	Why it ranks here
1	Return data correctness — survivorship, corporate actions, synchronicity, stale prices	Silently corrupts everything downstream, and produces plausible portfolios. An order of magnitude above the rest
2	Whether you have a μ at all	§2.5's 11× penalty. A binary decision worth more than every refinement below
3	Budgeting $q = N/T$	Determines whether the problem is well-posed. Free to check, expensive to ignore
4	Applying some regularisation	Enormous as a binary; small as a choice among methods (§8.3)
5	Turnover control	Frequently dominates rank 4 in live P&L, and is invisible in a frictionless backtest
6	Which covariance estimator	Real, second-order, and the subject of most of the literature

The gap between rank 1 and rank 6 is larger than most people's intuition allows. [Practice] I have never seen a portfolio-construction problem where switching covariance estimators recovered what a synchronicity bug had cost.

9.2 The data layer

Define the universe as of each date, not as of today. The single most common backtest error in this area is building a covariance matrix over the assets that exist now. Every dead ticker removed is a survivorship bias, and it flatters covariance estimates specifically: the assets that disappeared are disproportionately the ones that moved violently.

Corporate actions must be applied before returns, not after. A missed 3-for-1 split produces a −67% return, which enters the covariance squared and can single-handedly reorder the eigenvalues. [Practice] A cheap and effective guard: flag any daily return beyond ±50% for a liquid name and inspect it rather than winsorising it silently — a real −50% day is information, and a split artefact is a bug, and only inspection distinguishes them.

Decide the return convention once. Log returns aggregate over time and simple returns aggregate across assets, and a portfolio is a cross-sectional aggregation — so portfolio weights multiply simple returns. Using log returns in a covariance matrix and then treating $w'r$ as the portfolio return is a small error at daily frequency and a real one at monthly. See the companion note on Simple and Log Returns for when the distinction bites.

Choose the estimation frequency deliberately. Higher frequency buys more observations, which lowers q , which §2.4 says is the thing that matters. It also buys microstructure noise and the synchronicity problems of §9.3. [Practice] Daily is the usual compromise; weekly is right when the universe spans time zones; intraday is right only when you have properly synchronised data and a reason.

9.3 Non-synchronous and stale prices

This deserves its own subsection because it is the most damaging data problem that produces no error message and no obviously wrong number.

The mechanism. Suppose you compute daily close-to-close returns for a US stock and a Japanese stock. Tokyo closes roughly fourteen hours before New York, so the two “daily” returns cover offset windows. News arriving during New York hours on day t enters the US return for day t , but Tokyo has already closed, so it enters the Japanese return for day $t + 1$. The contemporaneous correlation therefore understates the true relationship, with part of the co-movement displaced into a lagged term you never compute.

The same thing happens without any time zones at all, whenever an asset trades infrequently. A bond or small-cap that did not trade near the close carries a stale price, so its measured return is zero on days when its true return was not. Zeros depress measured volatility and drag measured correlation toward zero.

Why it is dangerous rather than merely inaccurate. Both biases point the same way: toward zero correlation and lower volatility. In the language of §5.4, they lower $\hat{R}_i^2$ and $\hat{\sigma}_i^2$ — and both sit in the denominator of the weight. So an asset made illiquid or non-synchronous by measurement looks like a genuine diversifier with low risk, and the optimiser reaches for exactly those.

Non-synchronous and stale pricing systematically overweights the illiquid, the foreign, and the infrequently traded — the assets whose risk you can least afford to underestimate, and which will be hardest to exit. [Fact] This is a recognised contributor to the historical understatement of risk in portfolios holding private, illiquid, or appraisal-priced assets.

The frequency interaction. The bias is worse at higher frequency: measured correlation between two assets falls monotonically as the sampling interval shrinks, an effect documented by Epps (1979) and named after him. So sampling your data more finely, which §9.2 recommended for lowering q , makes this problem worse. The two considerations genuinely trade off.

Remedies, in increasing order of effort:

1. Lower the frequency for correlations. Weekly returns largely absorb a one-day timing offset. Keep daily data for volatilities, where the problem is milder and the extra observations help. This is the §6.3 split again, now for a different reason.
2. Align to a common timestamp. If you can get synchronous snapshots — 16:00 London for everything, say — do that instead of using local closes. Futures make this easy, cash equities much less so.
3. Lead-lag correction. Estimate covariance including one or more lagged cross-terms and sum them, in the spirit of the Scholes–Williams and Dimson corrections for beta. Effective, and it costs parameters.
4. Model staleness explicitly, unsmoothing an appraisal or infrequently traded series before estimating. Standard in private-asset work and rarely worth it elsewhere.

[Practice] A diagnostic worth running: count the fraction of exactly-zero returns per asset. Anything above a few percent for a supposedly liquid instrument is stale, and its risk numbers are fiction.

9.4 Budgeting the aspect ratio

Treat q as a budget you allocate rather than a number you discover.

$$q = \frac{N}{T_{\text{eff}}}, \quad T_{\text{eff}} = \begin{cases} T & \text{rolling window} \\ \frac{1+\theta}{1-\theta} & \text{EWMA with decay } \theta \end{cases}$$

[Practice] The working targets I would defend:

q	Verdict
< 0.1	Comfortable. Sample covariance plus light shrinkage is fine
$0.1 - 0.3$	Normal operating range. Shrinkage is mandatory
$0.3 - 0.7$	Difficult. Use a factor model or nonlinear shrinkage, and constrain hard
> 0.7	Do not invert anything. Inverse volatility, HRP, or reduce N

Three levers move q , and it is worth knowing which is cheapest:

- Reduce N . Usually the most effective and the least popular. Grouping 500 stocks into 40 sector baskets, allocating across baskets, and equal-weighting within them turns $q = 1.0$ into $q = 0.08$. You lose within-sector differentiation and gain a well-posed problem — often a good trade.
- Increase T . Cheap until the window is long enough that stationarity fails. [Practice] Beyond roughly five years of daily data for equity correlations you are averaging over structurally different regimes.
- Increase frequency. Lowers q fastest, and §9.3 explains the price.

9.5 Numerics

Small implementation choices here have large consequences, and most of them go one way.

Never form the inverse. `np.linalg.inv(cov) @ mu` is slower and less accurate than `np.linalg.solve(cov, mu)`, and for a symmetric positive-definite matrix `scipy.linalg.cho_solve` on a cached Cholesky factor is better than both. The difference is not cosmetic at the condition numbers of §5.7.

Work in correlation space. Split $\hat{\Sigma} = \hat{D}\hat{C}\hat{D}$, regularise $\hat{C}$, and recombine. This matters more than it looks. Shrinking Σ toward the identity pulls every volatility toward the average as a side effect, which is wrong: volatilities are the part you can estimate well (§5.3) and there is no reason to damage them. Shrinking C leaves them alone.

Choose the shrinkage target to match your data. [Practice] The identity is the wrong target for anything with a market factor, which is nearly everything. Constant correlation or a single-index target is nearly always better. §9.7 demonstrates how much this matters, and the answer is: a lot.

Check positive-definiteness, and fix it properly. Blends of estimators, pairwise-deleted missing data, and lead-lag corrections can all produce a matrix with small negative eigenvalues. The right repair is Higham's (2002) nearest-correlation-matrix projection; the cheap approximation — clip negative eigenvalues to a small floor and rescale to a unit diagonal — is adequate when the violation is tiny. Adding ϵI until Cholesky succeeds is a hack that works, and you should log how large ϵ had to be, because a large value means something upstream is broken.

Keep units consistent and assert it. Mixing annualised and per-period quantities is the most common numerical bug in this code, it produces results that are wrong by a factor of $\sqrt{252}$, and it is easy to miss because the portfolio still looks sensible. [Practice] Annualise once, at a boundary, and assert that the diagonal of your covariance matrix implies volatilities in a plausible range before you use it.

Log q and κ on every rebalance. Both are one line (§5.7). A regime where they drift is a regime where your weights are becoming unstable, and you want to know before the P&L tells you.

9.6 Constraints in practice

§8.2 established that constraints are regularisation. That does not make them interchangeable — they regularise in different directions and have different side effects.

Constraint	Form	What it does	Watch for
Long-only	$w \geq 0$	Strongest single regulariser available; equivalent to a specific shrinkage	Discards genuine short alpha. Often binds on most assets
Box	$ w_i \leq c$	Direct control of concentration	Binding on many names means the optimiser is not choosing — the cap is
Gross exposure	$\sum_i w_i \leq L$	Controls leverage and, through it, costs	The most defensible constraint to a risk committee
Turnover	$\ w - w_{\text{prev}}\ _1 \leq \tau$	Shrinks toward the current portfolio (§10.3)	Makes the problem path-dependent; backtest it as such
Cardinality	at most k non-zero	Limits operational complexity	Non-convex. Avoid unless it is a genuine requirement

[Practice] Two rules I would apply without hesitation. Count how many constraints bind at the solution. If most positions sit at their cap, the optimiser is decorative and you are running a constrained naive portfolio — which may be fine, but you should know it. And always solve the unconstrained problem too, purely as a diagnostic. The distance between constrained and unconstrained solutions measures how much your prior is doing, which is exactly the quantity §8.2 says you should be tracking.

9.7 A reference implementation

The whole pipeline, in the order it must execute. This runs; the numbers in §9.8 come from it.

```
import numpy as np
from scipy.linalg import cho_factor, cho_solve

def split_correlation(cov):
    """Separate a covariance matrix into correlations and volatilities."""
    vol = np.sqrt(np.diag(cov))
    return cov / np.outer(vol, vol), vol

def shrink_to_constant_correlation(corr, delta):
    """Pull every correlation toward the universe average. Volatilities untouched."""
    n = len(corr)
    off = ~np.eye(n, dtype=bool)
    target = np.full((n, n), corr[off].mean())
    np.fill_diagonal(target, 1.0)
    out = (1 - delta) * corr + delta * target
    np.fill_diagonal(out, 1.0)
    return out

def clip_to_psd(corr, floor=1e-8):
    """Clip negative eigenvalues, restore a unit diagonal. §9.5's cheap repair."""
    vals, vecs = np.linalg.eigh((corr + corr.T) / 2)
    if vals.min() >= floor:
        return corr
    out = vecs @ np.diag(np.clip(vals, floor, None)) @ vecs.T
    d = np.sqrt(np.diag(out))
    return out / np.outer(d, d)

def build_portfolio(returns, view=None, delta=0.3, max_weight=0.05,
                    target_vol=0.10, periods_per_year=252):
    t, n = returns.shape

    # 1. estimate, in correlation space # §9.5
    cov = np.cov(returns, rowvar=False) * periods_per_year
    corr, vol = split_correlation(cov)

    # 2. regularise the correlations only # §6.5
    corr = clip_to_psd(shrink_to_constant_correlation(corr, delta))
    cov_reg = corr * np.outer(vol, vol)

    # 3. solve. mu = 1 is minimum variance # §5.1
    view = np.ones(n) if view is None else view
    w = cho_solve(cho_factor(cov_reg), view)

    # 4. constrain, then scale to the risk target # §9.6, §1.3
    w = np.clip(w / np.abs(w).sum(), -max_weight, max_weight)
    w = w * target_vol / np.sqrt(w @ cov_reg @ w)

    lam = np.linalg.eigvalsh(corr)
    return {"weights": w, "q": n / t, "gross": np.abs(w).sum(),
            "condition_number": lam[-1] / lam[0],
            "predicted_vol": np.sqrt(w @ cov_reg @ w)}
```

Five things about this listing are load-bearing, and they are the ones people get wrong.

The matrix is split into correlations and volatilities before anything is regularised, and only the correlations are touched. This is §9.5's rule, and it is the difference between a shrinkage that helps and one that quietly damages your volatility estimates.

The shrinkage target is the constant-correlation matrix, not the identity. §9.8 shows what happens when you get this wrong, and it is not subtle.

`cho_solve` on a factorisation, not an explicit inverse. At the condition numbers of §5.7 this is a correctness issue, not a performance one.

Constraining and scaling are separate steps in that order. Clipping changes the portfolio's volatility, so the risk target must be applied after the clip, or you will not hit it. Reversing these two lines is a common bug that produces a portfolio running persistently below target. Note what `max_weight` therefore caps: the gross-normalised shape, before leverage is applied. A position sitting at the cap is delivered at `max_weight` times gross exposure, so the 5% cap above, on a book running gross 3.6, is an 18% position. A hard cap on the weights you actually hold is a different constraint and has to be imposed after the scaling.

q and κ come out with the weights. They are diagnostics you want logged on every rebalance, not things to compute when something has already gone wrong. The κ here is measured on the regularised correlation matrix, which is the object the shrinkage acts on.

Not shown, because they belong to the next section: turnover control and costs. This function returns a target, and §10 is about the fact that you should usually not trade all the way to it.

9.8 What the shrinkage choice actually buys

Running the above on simulated one-factor data — $N = 120$, $T = 750$ daily observations, so $q = 0.16$ — and sweeping the shrinkage intensity. The identity rows replace the constant-correlation target with I and change nothing else. Realised volatility is computed against the true covariance matrix, which the simulation knows and the estimator does not. These are my own numbers. The predicted-volatility column is 10.0% by construction — step 4 of §9.7 scales every portfolio to the target — so it is the realised column, and the ratio, that carry the information.

Target	δ	$\kappa(C)$	Gross	Predicted vol	Realised vol	Realised $\div$ predicted
—	0.00	178	4.35	10.0%	11.6%	1.155
Constant corr.	0.10	146	3.99	10.0%	10.7%	1.067
Constant corr.	0.30	108	3.60	10.0%	10.0%	0.999
Constant corr.	0.50	85	3.38	10.0%	9.9%	0.986
Constant corr.	0.80	64	3.20	10.0%	10.7%	1.070
Identity	0.30	62	3.38	10.0%	10.6%	1.064
Identity	0.80	10	2.18	10.0%	17.8%	1.775

Four readings, and the last one is the point of the table.

Shrinkage's main effect is not lower risk — it is an honest risk forecast. At $\delta = 0.3$ the realised-to-predicted ratio is 0.999. The unshrunk portfolio was taking 16% more risk than it reported, consistent with the $1/(1 - q) = 1.19$ of §5.6; shrinkage removed essentially all of that gap.

Gross exposure falls monotonically, from 4.35 to 3.20. Since costs scale with turnover and turnover scales with gross exposure, this is a large P&L effect that does not appear anywhere in the volatility column.

Over-shrinking has a cost, and it is mild. At $\delta = 0.8$ the target's bias starts to show, but the ratio only drifts back to 1.07. The loss function is distinctly asymmetric: too little shrinkage is much worse than too much, which is a good reason to err high.

And the trap. The identity target at $\delta = 0.8$ produces the best-conditioned matrix in the table by a wide margin — $\kappa = 10$, against 64 for the comparable constant-correlation row — and the worst risk forecast, understating realised volatility by 44%. It is worse than doing nothing at all. A practitioner monitoring the condition number as a proxy for matrix quality would pick precisely this row.

[Practice] The condition number tells you whether a matrix is numerically invertible. It tells you nothing about whether the answer is right. Shrinking toward a badly-chosen target improves κ while destroying the structure that made the estimate useful, and no amount of monitoring κ will reveal it. The only check that catches this is realised-versus-predicted risk, measured out of sample (§11.2).

§9 Key takeaways

1. Effort ordering: data correctness, then whether you have a μ , then q , then whether you regularise at all, then turnover, and only last which estimator. The literature's emphasis is roughly the reverse.
2. Non-synchronous and stale prices bias correlations and volatilities down, which makes illiquid and foreign assets look like diversifiers. The optimiser then overweights exactly the positions that are hardest to exit.
3. Sampling more finely lowers q and worsens the Epps attenuation. Resolve it by splitting: daily data for volatilities, weekly for correlations.
4. Regularise in correlation space. Shrinking Σ toward I damages volatility estimates you had no reason to doubt.
5. Never form an explicit inverse; factor once and solve. Apply constraints before scaling to the volatility target, never after.
6. Shrinkage's main benefit is an honest risk forecast, not a lower one. In the worked example it moved realised-over-predicted risk from 1.16 to 1.00 and cut gross exposure by 17%.
7. Err toward over-shrinking. The loss is markedly asymmetric — too little is far worse than too much.
8. A good condition number is not evidence of a good matrix. The best-conditioned row of §9.8's table has the worst risk forecast in it.

10. Costs, turnover, and rebalancing

10.1 Why turnover is the hidden variable

Every section so far has treated portfolio construction as a one-shot problem: given inputs, produce weights. In production it is a sequence of such problems, and the link between them is what you pay to move.

This changes the character of §2's instability entirely. In a one-shot setting, an optimiser that produces wildly different weights from slightly different inputs is merely unreliable. In a repeated setting, that same instability is turnover — and turnover is a direct, compounding, non-recoverable cost. The sensitivity we spent §2 and §5 characterising as a statistical problem shows up in the P&L as a trading bill.

Weight instability and transaction costs are not two problems. They are one problem, measured in two units. Anything that stabilises weights — shrinkage, constraints, longer windows, fewer assets — reduces costs, and anything that reduces costs is doing so by stabilising weights.

This is why §9.8's gross-exposure column mattered as much as its volatility column, and why a frictionless backtest systematically overstates the value of better estimation. The methods that look best without costs are typically the ones that move most.

10.2 Where costs come from

Three components, with different scaling behaviour. Getting the scaling right matters more than getting the levels right, because the levels are instrument-specific and the scaling determines what the optimiser does.

Component	Scales with	Typical magnitude
Spread and fees	Trade size, linearly	1–5 bp for liquid futures and large caps; far more elsewhere †
Temporary impact	Roughly the square root of participation rate	The dominant term for institutional size
Permanent impact	Trade size, linearly	Hard to measure; often folded into the above

† [Practice] Those magnitudes are order-of-book figures, not measurements; your own fills are the only authority.

[Fact] The square-root form of impact — cost per share rising roughly with the square root of the fraction of daily volume traded — is one of the more robust empirical regularities in market microstructure, holding across venues, asset classes and decades. [Contested] The exponent itself is debated, with estimates spanning roughly 0.4 to 0.6; the concavity is not in doubt, the precise power is.

For portfolio construction purposes, the useful simplification is a quadratic total cost, $\frac{1}{2}(w - w_{\text{prev}})' \Lambda_c (w - w_{\text{prev}})$ with Λ_c diagonal and its entries inversely related to liquidity. This is not the true cost function — square-root impact means total cost grows as size^{3/2}, not size² — but it makes the problem solvable in closed form, and §10.3 shows the resulting policy is qualitatively right.

10.3 Costs are the third regulariser

Add a quadratic turnover penalty to the master form and something instructive happens. The objective becomes

$$\max_w w' \mu - \frac{\gamma}{2} w' \Sigma w - \frac{\nu}{2} \|w - w_{\text{prev}}\|^2,$$

which is §10.2's quadratic cost at $\Lambda = \nu I$, so ν scales the cost of moving. Differentiating with respect to w and setting to zero:

$$\mu - \gamma \Sigma w - \nu (w - w_{\text{prev}}) = 0 \quad \implies \quad (\gamma \Sigma + \nu I) w = \mu + \nu w_{\text{prev}},$$

so

$$w = (\gamma \Sigma + \nu I)^{-1} (\mu + \nu w_{\text{prev}})$$

Look at what the cost term did. On the left it added νI to $\gamma \Sigma$ — exactly ridge regularisation, with coefficient ν/γ on Σ , which by §8.2's rearrangement is the linear shrinkage of §6.5 toward a scaled identity, up to a positive scalar that moves only leverage. On the right it added νw_{prev} to the return view. One penalty, and it landed in both slot S2 and slot S3 of §8.1.

The right-hand half is worth making exact, because “shrinks the view toward the current portfolio” is doing real work. Substitute $\mu = \gamma \Sigma w^*$, where $w^* = \frac{1}{\gamma} \Sigma^{-1} \mu$ is §1.3's frictionless optimum:

$$w = (\gamma \Sigma + \nu I)^{-1} (\gamma \Sigma w^* + \nu w_{\text{prev}}).$$

Distribute the inverse and the two matrix coefficients sum to the identity, so w is a matrix-weighted average of where you would go without costs and where you already are. Raising ν slides you from w^* toward w_{prev} . That is what §8.2 means by shrinkage toward the current portfolio, and for a quadratic penalty it is an identity rather than an analogy.

This completes the argument §8.2 began:

The three great regularisers — covariance shrinkage, position constraints, and transaction costs — are the same act performed for three different stated reasons. Shrinkage is justified statistically, constraints institutionally, and costs economically, but all three add structure that damps the small-eigenvalue directions, and their effects compound.

The practical consequence is the same warning as before, now with a third contributor. A shop that shrinks its covariance matrix, imposes a long-only constraint, caps positions, and runs a turnover penalty has applied four regularisations and will land close to $1/N$ — often without anyone having decided to. [Practice] The cheapest way to detect this is to compute the distance between your live weights and $1/N$, and track it. If it is small and falling, your sophistication is decorative.

10.4 The no-trade region

The quadratic-cost result above says to move partway toward the target. With proportional costs — a spread you pay on every share, regardless of size — the answer changes shape: there is a region around the current portfolio inside which the optimal action is to do nothing at all, and outside which you trade only to the region's edge, never to the target.

[Fact] The classical asymptotic result, from the Constantinides (1986) line of work, is that the no-trade band's half-width scales as the cube root of the proportional cost, not linearly with it. That non-linearity is the useful part: quadrupling your cost estimate widens the band by only about 60%, so the policy is far less sensitive to cost mis-estimation than it looks. Getting costs approximately right is enough; the band is forgiving.

Two implementation forms, both standard:

Buffering. Compute the target w^* ; trade only if $\|w^* - w_{\text{prev}}\|$ exceeds a threshold, and then trade to the edge rather than to w^* . Simple, robust, and it composes with everything.

Partial adjustment. Move a fixed fraction of the way, $w = w_{\text{prev}} + \eta (w^* - w_{\text{prev}})$ with $\eta \in (0, 1]$. This is §10.3's solution with its matrix coefficient collapsed to a scalar, and it is easier to reason about.

[Practice] Both work. The mistake that matters is doing neither — computing a fresh optimum each period and trading all the way to it, which is the default behaviour of every code sample in this literature including §9.7's.

10.5 Aim at where you are going

The subtlest result in this area is worth stating because it changes what you optimise, not just how fast you move toward it.

Gârleanu & Pedersen (2013) solve the multi-period problem with predictable returns and quadratic costs, and find that the optimal policy trades partway toward an aim portfolio that is not today's Markowitz portfolio. The aim is a weighted average of the current optimal portfolio and all the future ones you expect, and the weighting systematically discounts fast-decaying signals.

The intuition is clean. If a signal will have decayed by the time you finish trading into it, you will pay the full cost of establishing the position and capture only part of the return. So a fast signal deserves less weight than its standalone Sharpe ratio implies, and a slow one deserves more.

[Practice] Signal persistence, not just signal strength, belongs in the portfolio construction step. Two signals with identical information coefficients but different half-lives should receive different weights, and the difference can be large. Optimising each signal's weight on its Sharpe ratio alone and then applying costs afterward gets this wrong in a way no amount of post-hoc cost adjustment repairs.

This connects directly to the trend note's treatment of trading rate and buffering, and it is the strongest argument in this document for treating costs as part of the objective rather than as a subsequent haircut.

10.6 Rebalancing policy

Policy	Mechanism	Verdict
Calendar	Rebalance monthly, quarterly	Simple, auditable, and arbitrary. Trades when the calendar says, not when the portfolio needs it
Threshold	Rebalance when drift exceeds a band	§10.4's theory, and generally better
Cost-aware optimisation	Costs inside the objective	Best, and the only one that handles §10.5

[Practice] Two details that matter more than the choice among these. Rebalance the risk model on a different schedule from the portfolio — re-estimating $\hat{\Sigma}$ daily while trading monthly is fine and often preferable, since it gives you an up-to-date risk view without generating turnover. And never rebalance everything on the same day; staggering across the universe reduces market impact and avoids a monthly signature that others can trade against.

§10 Key takeaways

1. Weight instability and transaction costs are the same problem in different units. A frictionless backtest overstates the value of better estimation, because better estimation mostly buys stability.
2. A quadratic turnover penalty is exactly ridge regularisation of Σ plus shrinkage of the view toward the current portfolio. Costs are a regulariser, provably.
3. Shrinkage, constraints and costs are one act with three justifications. Apply all three at full strength and you have quietly built $1/N$.
4. With proportional costs, do not trade to the target — trade to the edge of a no-trade band whose width scales as the cube root of cost. That makes the policy robust to cost mis-estimation.
5. The aim portfolio is not today's optimum. Fast-decaying signals should be discounted at the construction stage, because you pay to enter a position you will not hold long enough to harvest.
6. Signal persistence belongs in portfolio construction. Weighting signals by Sharpe ratio and applying costs afterward is not equivalent and is worse.

11. Evaluation and pitfalls

11.1 The ladder

Test in this order, because each rung is cheaper than the next and can fail the method outright.

Stage	Question	Cost
0	Is $q < 1$? Is $\hat{\Sigma}$ positive definite? Is κ sane?	Free
1	Does realised risk match predicted risk out of sample?	Cheap, and the highest-value test here
2	Are the weights stable — turnover, resampling spread, concentration?	Cheap
3	Does it beat $1/N$ and inverse volatility, at matched volatility?	Moderate
4	Does it survive realistic costs and constraints?	Expensive
5	Is the improvement statistically distinguishable from zero?	Requires care (§11.5)

[Practice] Most published and internal comparisons start at stage 3 and never run stages 1 and 2. That is backwards: stage 1 catches the failure mode that actually destroys portfolios, and it does so without any return forecast, any backtest, or any assumption about the future.

11.2 The single most important test

§5.6 established that a portfolio built on $\hat{\Sigma}$ and evaluated on $\hat{\Sigma}$ will report risk that is too low by construction, and §9.8 showed a shrinkage target that made this worse while improving every other diagnostic. The test that catches both:

Compute the ratio of realised to predicted portfolio volatility, out of sample, and track it over time.

Concretely: at each rebalance, record the volatility your risk model predicted for the portfolio you actually held. Later, compute the volatility that portfolio actually realised. The ratio should be close to 1.

Ratio	Reading
≈ 1.0	The risk model is honest. This is the goal
1.1 – 1.3	Normal for a lightly regularised model. Compare against $1/(1 - q)$ — if it matches, your regularisation is doing nothing
> 1.5	Something is badly wrong: q too high, target mis-chosen (§9.8), or stale data (§9.3)
< 0.9	Over-shrunk, or the realised period was unusually calm. Less dangerous, still worth explaining

[Practice] This is one number, it needs no return forecast, it is computable from data you already store, and I would put it on the first page of any risk report ahead of the portfolio volatility itself. A model that says 10% and delivers 10% is more valuable than one that says 8% and delivers 13%.

11.3 Traps specific to portfolio construction

Evaluating on the matrix that built the portfolio. The one above. Use a genuinely out-of-sample window, not a different estimator on the same window.

Comparing at unmatched volatility. A method that produces 14% volatility will usually beat one producing 9% on raw return, and that comparison is meaningless (§1.3: leverage is a free parameter). Always scale every candidate to the same target volatility before comparing. This single discipline invalidates a surprising fraction of published comparisons.

Ignoring the universe's own selection. Testing on today's index members imports survivorship into the covariance matrix, and it flatters minimum-variance methods specifically, since the assets that vanished were the volatile ones.

Reporting one (N, T) pair. §2.4 says q governs everything, so a result at $q = 0.1$ tells you almost nothing about behaviour at $q = 0.6$. Report a curve over q , not a point. A method that wins at one aspect ratio and loses at another is common and the sweep is cheap.

Free constraints. Adding a long-only constraint improves out-of-sample results (§8.2 explains why), so a comparison in which the sophisticated method is constrained and the naive one is not measures the constraint, not the method. Constrain both identically.

Frictionless turnover. §10.1. The methods that look best without costs are disproportionately the ones that trade most.

Tuning on the test set. Shrinkage intensities, thresholds, cluster linkages and band widths are all hyperparameters. Choosing them by out-of-sample performance and then reporting that performance is the ordinary sin, wearing a matrix for a hat.

11.4 What to measure

Beyond return and volatility, five diagnostics that actually discriminate:

Realised ÷ predicted risk. §11.2. First among equals.

Turnover, as annualised two-way fraction of capital. The number that converts directly into cost.

Concentration. Report the largest weight, the gross exposure, and the Herfindahl index $\sum_i \tilde{w}_i^2$, whose reciprocal is the effective number of positions — a far more honest count than the number of non-zero weights. The normalisation is not optional and is easy to get wrong: $\tilde{w}_i = w_i / \sum_j |w_j|$, each weight over gross exposure. On a long-only book that is the same as dividing by the sum of the weights; on a levered or long-short one it is not, and the undivided reciprocal measures leverage rather than breadth. Equal weights give exactly N . A 500-name portfolio with an effective count of 12 is a 12-name portfolio.

Effective number of bets. Meucci's refinement: decorrelate the portfolio into uncorrelated risk sources, compute each one's share of total variance, and take the exponential of the entropy of that distribution. Where the Herfindahl index counts positions, this counts independent risks — and the gap between them is the honest measure of diversification. [Practice] A portfolio of forty correlated bank stocks has an effective number of positions near forty and an effective number of bets near two, and only the second number tells you what will happen in a crisis.

Transfer coefficient. How much of your signal survived construction: the risk-adjusted correlation between the portfolio you hold and the unconstrained optimum w^* ,

$$TC = \frac{w' \Sigma w^*}{\sqrt{w' \Sigma w} \sqrt{w^{*'} \Sigma w^*}}.$$

Note the Σ : a plain correlation of the weight vectors is a different and more flattering number whenever volatilities are dispersed, which is why the risk-adjusted form is the standard one. [Practice] A low value is not necessarily bad — §8.2 says constraints are estimators — but it should be a decision, not a surprise.

11.5 Statistical significance, honestly

The uncomfortable arithmetic. For returns that are roughly IID, the standard error of an estimated Sharpe ratio over n years is approximately (Lo, 2002)

$$se(\widehat{SR}) \approx \sqrt{\frac{1 + SR^2/2}{n}}.$$

At a true Sharpe of 0.5 over ten years, that is 0.34. A 95% interval runs from about -0.16 to 1.16 , which comfortably contains zero. The comparison you actually make is worse still: two methods measured over ten years each, one at 0.5 and one at 1.5, differ by 1.0 against a standard error on that difference of $\sqrt{0.34^2 + 0.46^2} \approx 0.57$ — under two standard errors, and so not distinguishable at any conventional level. That step treats the two estimates as independent, which is the conservative reading and the right one when the methods were run on different data; the closing paragraph below is about how much sharper the test becomes when they share a sample.

[Fact] A ten-year backtest cannot distinguish a Sharpe ratio of 0.5 from one of 1.5. Any claim that method A beat method B on ten years of data, presented as a comparison of levels, is a claim about noise.

There is a way out, and it is important: compare the difference, not the levels. Two portfolio construction methods run on the same universe share almost all of their market exposure, so the difference of their return series has far lower volatility than either series alone. The paired comparison can be significant when neither level is, and it is the right test for the question you are actually asking, which is “does B beat A”, not “is B good”.

[Practice] Three further disciplines worth the effort. Use a block bootstrap on the return series rather than an IID one, since both volatility and correlation cluster. Deflate for the number of configurations you tried — if you evaluated twelve shrinkage intensities, the best of twelve is biased upward and the deflated Sharpe ratio adjustment quantifies by how much. And test on several universes: portfolio construction methods are much more universe-sensitive than signal research, and a method that wins on US large caps, loses on futures, and wins on emerging markets has told you something a single number cannot.

§11 Key takeaways

1. Test in order: sanity checks, then risk-model honesty, then weight stability, then performance. Most comparisons skip straight to performance and miss the failure that matters.
 2. Realised $\div$ predicted volatility, out of sample, is the highest-value single number in this document. It needs no forecast and no backtest.
 3. Compare methods only at matched volatility. Leverage is a free parameter, so unmatched comparisons measure nothing.
 4. Report a curve over q , not a point. A result at one aspect ratio does not transfer to another.
 5. Give every candidate the same constraints. Constraints are estimators, so an asymmetric comparison measures the constraint.
 6. Effective number of bets and effective number of positions are different numbers, and the gap between them is what diversification actually means.
 7. A ten-year backtest cannot separate Sharpe 0.5 from 1.5. Compare the paired difference between methods, where the shared market exposure cancels.
-

12. Synthesis

12.1 The framework in one page

Everything in this document is three claims and one consequence.

The master form. Every portfolio construction method is

$$w \propto (\hat{\Sigma} + \Delta)^{-1} \hat{\mu} \quad \text{s.t.} \quad w \in \mathcal{C}, \quad \text{scaled to } \sigma^*.$$

Naming a method fixes some of those five slots and leaves others implicit. Minimum variance sets $\hat{\mu} = 1$; risk parity sets $\hat{\mu} \propto \sigma$ and assumes equicorrelation; $1/N$ degenerates both $\hat{\Sigma}$ and $\hat{\mu}$ at once. None of them has escaped the formula.

The mechanism. Inverting a covariance matrix means allocating in inverse proportion to estimated variance. The eigen-decomposition makes this exact: $w \propto \sum_i (v_i' \mu / \lambda_i) v_i$, so the quietest direction gets the largest bet. And the quietest direction is precisely the one the sample estimates worst — its eigenvalue is biased downward, its R_i^2 biased upward, and both errors feed the same denominator. The optimiser is most confident exactly where it has least right to be.

The diagnostic. One number governs the damage on the covariance side: $q = N/T$. Realised volatility exceeds predicted volatility by $1/(1 - q)$ — one fact with three faces, read off the Marchenko–Pastur spectrum (§5.5), off the inverse-Wishart expectation (§5.6), and asset by asset off the regression identity (§5.4). At $q = 0.5$ you take twice the risk you believe you are taking. At $q \geq 1$ the problem is not defined. Note the scope: q governs the covariance problem. The μ problem of §2.5 is governed by calendar span alone, and sampling the same years more finely does nothing for it.

The consequence. Every repair in the field adds structure that lifts the smallest eigenvalues and damps the positions taken in those directions. Covariance shrinkage does it statistically, position constraints do it institutionally, transaction costs do it economically, factor models do it structurally. For the first three that identity is exact and proved — §8.2 for constraints, §10.3 for costs; for factor models and hierarchical methods it is an argument made in §6.8 and §6.11 rather than a theorem. Either way the effects compound rather than adding four independent safeguards, and a practitioner applying all of them at full strength has built $1/N$ without deciding to.

If you remember one sentence: the optimiser is not the problem, and the covariance estimator is not the solution — the binding constraints are your return view and your aspect ratio, in that order.

12.2 What to actually run

Situation to configuration, with all five slots specified. These are my recommendations, not measured optima.

Your situation	Recommended configuration
No return forecast, $q < 0.3$	Minimum variance or ERC. Sample covariance, shrunk to constant correlation at $\delta \approx 0.3$ – 0.5 , in correlation space.
No return forecast, $0.3 < q < 0.7$	Long-only or box-constrained. Scaled to a volatility target
No return forecast, $q > 0.7$	Equal risk contribution, shrunk harder ($\delta \approx 0.5$) and box-constrained. A factor covariance model if you have credible characteristics
Validated forecast, $q < 0.3$	Inverse volatility, or HRP if the universe has real group structure. Do not invert anything. Reduce N by grouping if you can
Validated forecast, $q > 0.3$	Mean-variance. Rank-transform and clip the signal, scale it to a modest dispersion, shrink toward zero. Same covariance treatment as above. Gross-exposure cap
Large universe, thousands of names	Factor covariance model, or nonlinear shrinkage — both stay invertible by construction where the sample matrix does not. Constrain hard. Consider allocating across signal-sorted baskets rather than individual names
You trade residual relationships	Explicit factor model (§6.8), for the Woodbury inverse and the risk attribution as much as the statistics
	POET or a factor model with a thresholded residual. A standard factor model will tell you your position is riskless (§6.8)

Your situation	Recommended configuration
Universe spans time zones	Weekly returns for correlations, daily for volatilities. Check the zero-return fraction per asset first (§9.3)

Regardless of branch: split into correlations and volatilities before regularising; solve with a Cholesky factorisation rather than an inverse; constrain before scaling to the risk target; buffer your trades; and log q , κ , and realised-over-predicted risk on every rebalance.

12.3 A staged build, with gates

Each stage has a gate. Do not proceed past a failed gate — the later stages will mask the failure rather than fix it.

Stage 1 — Data. Point-in-time universe, corporate actions applied, one return convention, a documented estimation frequency. Gate: the fraction of exactly-zero returns per asset is below a threshold you chose deliberately, and every return beyond $\pm 50\%$ has been inspected rather than clipped.

Stage 2 — The naive baseline. Implement $1/N$ and inverse volatility end to end, with real rebalancing and real costs. Gate: both run in production shape, and you have their Sharpe, turnover and realised volatility. Everything after this must beat these two, at matched volatility, or it is not worth its complexity.

Stage 3 — Risk model. Sample covariance, split into correlation and volatility, shrunk toward constant correlation. Gate: out-of-sample realised $\div$ predicted volatility is within about 10% of 1.0 (§11.2). If it is not, the fault is in stage 1 or in q , not in your estimator — go back rather than reaching for a better estimator.

Stage 4 — Allocation. Minimum variance or ERC, constrained, scaled to target. Gate: beats stage 2 at matched volatility, on more than one universe, with the same constraints applied to both.

Stage 5 — Costs. Turnover buffering or a cost term in the objective. Gate: the net-of-cost improvement over stage 2 is still positive. [Practice] This gate fails more often than any other, and it is the honest place for a project to stop.

Stage 6 — Return views. Only now, and only with a signal validated independently of the portfolio. Gate: the paired difference against stage 4 is significant (§11.5), and the result survives a sweep over q .

Stages 1 through 3 are infrastructure and feel like they are not the interesting part. They are where the outcome is determined. [Practice] The characteristic failure of this project is to spend a month on stage 6 while stage 1 is quietly broken — and stage 6 will produce plausible, presentable, wrong portfolios the entire time.

12.4 Ten things I would tell someone starting today

1. Fix the data before you touch the mathematics. Point-in-time universes, corporate actions applied, synchronised prices, and a count of zero returns per asset. This outranks everything below it, it is the least interesting work in the project, and it is where the outcome is usually decided (§9.2, §9.3).
2. Check $q = N/T$ next. One line. It tells you whether your problem is well-posed, and a startling amount of published and internal work is done at q values where it is not.
3. Decide explicitly whether you have a return forecast. This is the highest-leverage line in your codebase. If the answer is no, set $\hat{\mu}$ to a constant and be glad — you have deleted the dominant error term.
4. Never skip $1/N$ and inverse volatility as baselines. They are hard to beat, they are free, and a method that cannot beat them at matched volatility has not earned its complexity.
5. Regularise in correlation space, and shrink harder than feels comfortable. Volatilities are the part you can estimate; do not damage them on the way to fixing the part you cannot. And the loss is asymmetric — §9.8 showed too little shrinkage costs much more than too much.
6. Count your regularisers. Shrinkage, long-only, position caps and a turnover penalty are four applications of one idea, and their product may be $1/N$ with extra steps.
7. Measure realised versus predicted risk, out of sample, forever. It is one number, it requires no forecast, and it is the only diagnostic that catches the failure in §9.8.
8. Compare at matched volatility, on multiple universes, over a sweep of q . Any comparison that violates these is measuring something other than what it claims.
9. Never form an explicit matrix inverse, and never trade all the way to the target.
10. Distrust a beautifully conditioned matrix. A good κ means the arithmetic will complete, not that the answer is right — and the two are routinely confused.

12.5 What is known, what is not, and what I would bet on

Known. [Fact] Estimation error dominates model error in this problem, and errors in means dominate errors in covariances by an order of magnitude. Sample eigenvalues spread in a predictable way governed by q , and the resulting distortion is correctable. Constraints and shrinkage are mathematically the same operation — exactly so for long-only minimum variance and for norm constraints, which is not the same as for constraints in general (a cardinality constraint is not a shrinkage of anything). Minimum-variance portfolios have delivered better risk-adjusted returns than cap-weighting over long samples in most equity markets; [Contested] risk parity's record is real but far more argued over, and rests on a leverage assumption that minimum variance does not need. Non-synchronous and stale pricing biases correlations downward and systematically overweights illiquid assets.

Not known. [Contested] Whether the out-of-sample advantage of sophisticated covariance estimators over simple shrinkage survives realistic costs and constraints — the published evidence is mixed and depends heavily on universe, period and constraint set. Whether hierarchical methods genuinely beat well-regularised conventional ones, or merely match them while being easier to compute at $q \geq 1$. How much of the minimum-variance premium is construction skill versus exposure to the low-volatility factor. Whether machine-learned, end-to-end allocation will add anything beyond what shrinkage already provides — the results so far are promising in-sample and thin out of it.

What I would bet on. [Hypothesis] Three things, all stated as my view rather than as findings.

That the covariance-estimation literature has reached diminishing returns while the return-view and cost sides remain comparatively neglected, because the former is mathematically tractable and the latter two are not. The field optimises where the mathematics is pleasant.

That most live portfolios are more heavily regularised than their owners believe, for the compounding reason of §10.3 — and that a meaningful fraction of “our risk model is conservative” is actually four shrinkages stacked unknowingly.

That the largest available improvement for most practitioners is not in this document's §6 at all but in §9.2 and §9.3: point-in-time universes, correct corporate actions, and synchronised prices. That is an unglamorous claim and I believe it more strongly than anything else here.

The honest summary of the whole subject is that portfolio construction is a problem where the mathematics is easy, the statistics is hard, the data work is harder, and the field's attention is allocated in almost exactly the reverse order.

Appendix A. Concepts and prerequisites

Everything the main text leans on without stopping to explain. The reader this is written for is mathematically comfortable but does not do this for a living: they hit “the tangency portfolio” or “complementary slackness” somewhere in §7, want the idea rather than a definition, and would rather not leave the document to get it.

Entries are ordered by dependency, not alphabetically — later ones use earlier ones — and grouped into five parts that are themselves in dependency order, so the appendix reads as a build-up. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. Notation follows the main text’s block exactly; where a standard formula from another field would collide with it, the collision is flagged rather than resolved by inventing new symbols.

Nothing here is needed to follow the argument of the document. A good deal of it is needed to implement it.

Index. Where each concept first bites:

Concept	First used	Concept	First used
Quadratic forms and ∇_w	§1.3	Euler’s theorem	§7.6
Positive definiteness	§2.3	Coordinate descent	§7.6
Eigenvalues and the spectrum	§2.3	Realised volatility	§5.3
The trace	§5.5	GARCH	§6.10
Rotational invariance	§6.1	Principal component analysis	§6.9
The condition number	§5.5	Correlation as a distance	§6.11
The Frobenius norm	§6.5	Hierarchical clustering	§6.11
Cholesky, and solve vs invert	§9.5	Basis points and turnover	§6.6
Sherman–Morrison–Woodbury	§6.4	Excess returns, Sharpe ratio	§2.1
IID and stationarity	§2.1	Frontier and tangency	§5.1
The bias–variance trade-off	§6.1	CAPM and reverse optimisation	§7.8
Effective sample size (Kish)	§6.3	Cap-weighted benchmarks	§7.4
Wishart and inverse-Wishart	§5.6	The low-volatility anomaly	§7.4
James–Stein shrinkage	§3	The information coefficient	§7.7
Prior, posterior, MAP	§7.8	Transfer coefficient, the law	§11.4
R^2 and adjusted R^2	§5.4	Statistical arbitrage	§3
Mahalanobis distance	§6.12	Cointegration	§6.7
The bootstrap	§7.9	Market impact, participation	§10.2
Multiple testing	§11.5	Winsorising	§6.12
Shannon entropy	§11.4	Herfindahl, effective counts	§11.4
Convexity and the QP	§3	Effective number of bets	§11.4
Lagrange, KKT, slackness	§5.2	Standard error of a Sharpe	§11.5
ℓ_1 , ℓ_2 , ridge, lasso	§7.10	The deflated Sharpe ratio	§11.5

Part I — Linear algebra and numerics. The covariance matrix is the object the whole document manipulates, and almost every claim in §2 and §5 is a claim about what happens to a matrix when you invert it. These nine entries are the vocabulary for saying that precisely.

A.1 Quadratic forms, and differentiating with respect to a vector

The idea. Portfolio variance is not the sum of the assets’ variances. It is a sum over every pair of holdings, with each covariance counted twice, and the compact way to write that double sum is $w'\Sigma w$. Expressions of that shape — a vector sandwiching a matrix — are called quadratic forms, and they are the only non-linear object in this entire document. That is also why the subject is tractable at all: differentiating a quadratic form gives a linear equation, and linear equations you can solve.

Formally. For symmetric $A \in \mathbb{R}^{N \times N}$, the quadratic form is the scalar $w'Aw = \sum_i \sum_j w_i A_{ij} w_j$. The two gradient rules needed throughout are

$$\nabla_w (w'\mu) = \mu, \quad \nabla_w (w'Aw) = (A + A')w = 2Aw \text{ for symmetric } A.$$

Differentiating a scalar by a vector produces a vector of partial derivatives, one per coordinate; setting it to zero is the first-order condition for a stationary point, and for a convex objective (A.21) that is the optimum.

Why it appears here. §1.3 obtains the master form $w^* = \gamma^{-1} \Sigma^{-1} \mu$ by applying exactly these two rules to $w'\mu - \frac{\gamma}{2} w' \Sigma w$ and setting the result to zero. §10.3 runs the same manoeuvre with a turnover penalty attached, and §7.6's marginal risk contribution $(\Sigma w)_i / \sigma_p$ is this derivative pushed through a square root.

Deeper. Petersen & Pedersen, *The Matrix Cookbook* (2012), sec. 2 — the standard lookup table for identities of this kind.

A.2 Positive definiteness and semi-definiteness

The idea. A covariance matrix cannot be arbitrary. Whatever portfolio you form, its variance has to come out non-negative, because variance is an average of squares. That single requirement — “every portfolio has non-negative variance” — is positive semi-definiteness. Strict positive definiteness adds that no portfolio has exactly zero variance, which is the condition for the matrix to be invertible and therefore for the master form to mean anything.

Formally. Symmetric A is positive semi-definite (PSD) if $x'Ax \geq 0$ for every $x \in \mathbb{R}^N$, and positive definite (PD) if $x'Ax > 0$ for every $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (PSD) or > 0 (PD). A true covariance matrix is always PSD; the sample covariance S from T observations has rank at most $\min(N, T - 1)$, so it is PD only when $T > N$ and merely PSD — singular — otherwise.

Why it appears here. This is the formal content of §2.4's “at $q \geq 1$ the sample covariance matrix is singular and the optimiser is meaningless”: with $N \geq T$ there exist non-zero w with $w'Sw = 0$ exactly, and the optimiser will find one and lever it without limit. §9.5's positive-definiteness check and Higham's (2002) repair are about restoring the property after blending or patching has destroyed it.

Deeper. Horn & Johnson, *Matrix Analysis*, 2nd ed. (CUP, 2012), chapter 7.

A.3 Eigenvalues, eigenvectors, and the spectrum

The idea. A symmetric matrix does something geometrically simple that is hidden by its entries: there is a set of mutually perpendicular directions in which the matrix acts purely by stretching, with no rotation. Those directions are the eigenvectors, the stretch factors are the eigenvalues, and together they are the spectrum. For a covariance matrix this is not an abstraction. Each eigenvector is a portfolio, and its eigenvalue is that portfolio's variance — which is why §2.3 can talk about “eigenportfolios” without any extra machinery.

Formally. For symmetric Σ , the spectral theorem guarantees an orthonormal basis of eigenvectors $v_1, \dots, v_N$ (so $v_i'v_j = 0$ for $i \neq j$ and $v_i'v_i = 1$) with real eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$ satisfying $\Sigma v_i = \lambda_i v_i$. Collecting them into $V = [v_1 \dots v_N]$ and $\Lambda = \text{diag}(\lambda_i)$,

$$\Sigma = V \Lambda V' = \sum_{i=1}^N \lambda_i v_i v_i', \quad \Sigma^{-1} = V \Lambda^{-1} V' = \sum_{i=1}^N \frac{1}{\lambda_i} v_i v_i'.$$

Because v_i has unit length, $v_i' \Sigma v_i = \lambda_i$: the eigenvalue is the variance of the portfolio v_i . Note what inversion does — it leaves every direction alone and replaces each λ_i by $1/\lambda_i$, reversing the order of importance completely.

Why it appears here. This is the mechanism of the whole document. §2.3 decomposes $w^* \propto \sum_i (v_i' \mu / \lambda_i) v_i$ and reads off that the smallest eigenvalue receives the largest bet; §5.5 describes how the sample spreads the λ_i apart; §6.6 and §6.7 repair the spectrum directly. The single sentence to carry is that the eigenvalues of Σ are the variances of N uncorrelated portfolios that together span everything you can hold.

Deeper. Strang, *Introduction to Linear Algebra*, 5th ed. (2016), chapter 6 for the geometry; Golub & Van Loan, *Matrix Computations*, 4th ed. (2013), chapter 8 for how it is actually computed.

A.4 The trace, and trace preservation

The idea. The trace adds up a matrix's diagonal, which for a covariance matrix means adding up the individual asset variances. The useful fact is that this same number is also the sum of the eigenvalues — so total variance can be counted two ways, asset by asset or eigen-portfolio by eigen-portfolio, and the two agree. That makes the trace a budget: an estimator that redistributes variance across directions should not create or destroy any, and “preserving the trace” is the constraint that enforces it.

Formally. $\text{tr}(A) = \sum_i A_{ii}$, and for symmetric A , $\text{tr}(A) = \sum_i \lambda_i$. The trace is invariant under similarity and cyclic under products, $\text{tr}(AB) = \text{tr}(BA)$. For a correlation matrix every diagonal entry is 1, so $\text{tr}(C) = N$ and the eigenvalues of any correlation matrix average to exactly 1.

Why it appears here. §6.7's eigenvalue clipping replaces the whole noise bulk by its common mean “preserving the trace” — that is what stops the repair from quietly changing the portfolio's overall risk level. §5.5's remark that the fit must allow for “outlier eigenvalues absorbing part of the trace” is the same budget seen from the other side: if the market mode carries eigenvalue 66, only $N - 66$ of the total is left for the bulk, and the Marchenko–Pastur edge must be fitted to what remains.

Deeper. Horn & Johnson, *Matrix Analysis*, 2nd ed. (CUP, 2012), sec. 1.2.

A.5 Rotational invariance

The idea. Some estimators have an opinion about which combinations of assets are risky; others only have an opinion about how much risk there is in each direction the sample already found. The second kind is called rotationally invariant, and the name is literal: if you relabel your assets by an arbitrary orthogonal mixing, the estimator's output gets mixed the same way rather than changing character. Such an estimator can only ever adjust eigenvalues — it has no information with which to move an eigenvector.

Formally. An estimator $\hat{\Sigma}(\cdot)$ applied to returns is rotationally invariant if $\hat{\Sigma}(UR) = U\hat{\Sigma}(R)U'$ for every orthogonal U ($U'U = I$) and every return matrix R . Any such estimator has the form $\hat{\Sigma} = \sum_i \tilde{\lambda}_i v_i v_i'$ where the v_i are the sample eigenvectors and only the map $\lambda_i \mapsto \tilde{\lambda}_i$ is chosen.

Why it appears here. §6.1's three-family table uses this as the top-level division: nonlinear shrinkage (§6.6), eigenvalue clipping (§6.7) and shrinkage toward a scaled identity keep the sample's eigenvectors and are therefore rotationally invariant, while factor models and constant correlation assert where risk lives and are not. The distinction tells you what an estimator can possibly fix — a rotationally invariant one cannot repair a badly estimated leading eigenvector, because it never touches one.

Deeper. Bun, Bouchaud & Potters (2017), *Physics Reports* 666, sec. 5 — the class is developed there under exactly this name.

A.6 The condition number

The idea. Every matrix you invert has an amplification factor: feed the linear system an input that is wrong by one part in a thousand, and the answer can come back wrong by that factor times one part in a thousand. The condition number is that factor. It is a property of the matrix alone, computable before you have any data error at all, which is what makes it a usable early-warning signal rather than a post-mortem.

Formally. For symmetric positive definite Σ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_N > 0$, the (spectral) condition number is

$$\kappa(\Sigma) = \frac{\lambda_1}{\lambda_N} \geq 1,$$

and for w solving $\Sigma w = \mu$, a relative perturbation of size ϵ in Σ or μ changes w by at most about $\kappa(\Sigma) \epsilon$ in relative terms. A singular matrix has $\kappa = \infty$. As a rule of thumb, you lose roughly $\log_{10} \kappa$ decimal digits of accuracy, so at $\kappa \approx 10^{16}$ double precision retains none.

Why it appears here. §5.7 makes κ one of the two numbers that belong in production monitoring, and §5.5 shows that pure noise at $q = 0.9$ manufactures $\kappa = 1442$ in a matrix whose truth is the identity. But §9.8 is the entry that matters most: the best-conditioned row of its table has the worst risk forecast in it, so κ certifies that the arithmetic will complete and says nothing whatever about whether the answer is right.

Deeper. Trefethen & Bau, *Numerical Linear Algebra* (SIAM, 1997), lectures 12 and 18.

A.7 The Frobenius norm

The idea. To say one matrix is close to another you need a way to measure the gap, and the simplest is to treat both as long lists of numbers and take the ordinary Euclidean distance between the lists. That is the Frobenius norm. Its virtue is that it is differentiable and leads to closed-form answers; its vice is that it weights every entry equally, which is not how a portfolio experiences error.

Formally. For $A \in \mathbb{R}^{N \times N}$,

$$\|A\|_F^2 = \sum_i \sum_j A_{ij}^2 = \text{tr}(A'A) = \sum_i \lambda_i(A'A),$$

so for symmetric A it is the sum of squared eigenvalues. It is the norm induced by the inner product $\langle A, B \rangle = \text{tr}(A'B)$, which is what makes minimising it a least-squares problem.

Why it appears here. §6.5's shrinkage intensity δ is the one minimising $\mathbb{E}\|\hat{\Sigma}_{\text{LW}} - \Sigma\|_F^2$, and the same section's most important caveat is a criticism of that choice: portfolio variance depends on Σ^{-1} , which weights small-eigenvalue directions enormously, while the Frobenius loss weights them like any other entry. That mismatch is why the Frobenius-optimal δ is usually somewhat too small for portfolio use, and why §9.8 finds the loss function asymmetric in favour of over-shrinking.

Deeper. Ledoit & Wolf (2004), "Honey, I Shrunk the Sample Covariance Matrix," where the loss is set up explicitly.

A.8 Cholesky factorisation, and why you solve instead of inverting

The idea. You almost never want Σ^{-1} itself. What you want is the vector w with $\Sigma w = \mu$, and computing the inverse and then multiplying is the slow, inaccurate way to get it. The fast, accurate way factorises Σ once into a triangular piece and its transpose, after which solving is two sweeps of back-substitution. A positive definite matrix admits such a factorisation uniquely, which is why the check of A.2 and this algorithm are the same check.

Formally. For symmetric positive definite Σ there is a unique lower-triangular G with positive diagonal such that

$$\Sigma = GG',$$

costing about $N^3/3$ operations — half a general LU factorisation. Solving $\Sigma w = \mu$ is then $Gy = \mu$ followed by $G'w = y$, each $O(N^2)$. Forming Σ^{-1} explicitly costs roughly three times as much and, worse, incurs a second round of rounding error in the multiplication that follows. Cholesky also fails — no positive diagonal exists — precisely when the matrix is not positive definite, which makes it a free definiteness test.

Why it appears here. §9.5 states the rule flatly: never form the inverse. §9.7 implements it with `cho_factor / cho_solve`, and its commentary calls this "a correctness issue, not a performance one" — at the condition numbers of §5.7 the extra digits lost to an explicit inverse are digits you do not have.

Deeper. Golub & Van Loan, *Matrix Computations*, 4th ed. (Johns Hopkins, 2013), sec. 4.2.

A.9 The Sherman–Morrison–Woodbury identity

The idea. If you have already inverted a matrix and then someone adds a low-rank correction to it — a market factor, a handful of industry factors, a single common correlation — you should not start the inversion over. Woodbury's identity says the corrected inverse is the old inverse plus a patch, and the patch requires inverting only something as small as the correction's rank. For a factor model this turns an $N \times N$ problem into a $K \times K$ one, which is the difference between feasible and infeasible at N in the thousands.

Formally. For invertible A ($N \times N$) and Ω ($K \times K$) and any $F \in \mathbb{R}^{N \times K}$,

$$(A + F\Omega F')^{-1} = A^{-1} - A^{-1}F(\Omega^{-1} + F'A^{-1}F)^{-1}F'A^{-1}.$$

The Sherman–Morrison special case is $K = 1$. Applied to the factor model $\hat{\Sigma} = F\Omega F' + \Psi$ with Ψ diagonal, $A^{-1} = \Psi^{-1}$ is free and the only genuine inversion is of a $K \times K$ matrix.

Why it appears here. §6.8 cites exactly this as the practical advantage of explicit factor models — "the inverse has a closed form ... that never requires an $N \times N$ decomposition" — and §12.2 recommends a factor model for very large

universes “for the Woodbury inverse ... as much as the statistics”. §6.4’s constant-correlation matrix is the rank-one case: $C = (1 - \bar{\rho})I + \bar{\rho} \mathbf{1}\mathbf{1}'$ inverts in closed form with no decomposition at all.

Deeper. Golub & Van Loan, *Matrix Computations*, 4th ed. (2013), sec. 2.1.4; Hager, “Updating the Inverse of a Matrix,” *SIAM Review* 31(2) (1989), 221–239.

Part II — Probability and estimation. Part I described what the matrix is. This part describes what goes wrong when you have to estimate one from T observations, which is the entire subject of §2 and §5.

A.10 IID, and stationarity

The idea. Every estimator here averages over history, and averaging only estimates something if the thing being averaged held still while you averaged. Two assumptions do that work, and they are routinely conflated. Independence says today tells you nothing about tomorrow. Stationarity says the statistical rules do not change with the calendar. You can have the second without the first — volatility clustering is exactly that — and the second is the one you actually need.

Formally. Observations $r_1, \dots, r_T$ are IID if they are mutually independent and share one distribution. A process is strictly stationary if the joint law of $(r_t, \dots, r_{t+k})$ is invariant to shifting t ; covariance (weak) stationarity — the version used in practice — requires only that $\mathbb{E}[r_t]$, $\text{Var}(r_t)$ and $\text{Cov}(r_t, r_{t-k})$ exist and do not depend on t . IID implies stationarity; the converse fails badly for returns.

Why it appears here. §2.1’s demonstration is run on data that are “IID, multivariate normal, stationary, with no fat tails, no regime changes” precisely so that the damage it measures can only be estimation error — the argument would be worthless otherwise. §6.2 lists IID with finite fourth moments as the sample covariance’s assumption, and §9.4’s warning that “beyond roughly five years of daily data for equity correlations you are averaging over structurally different regimes” is the statement that stationarity, not sample size, is what caps T .

Deeper. Hamilton, *Time Series Analysis* (Princeton, 1994), chapter 3. The companion note on Market Regimes is about what to do when stationarity fails.

A.11 The bias–variance trade-off

The idea. There are two ways an estimator can be wrong. It can be systematically off — pointing at the wrong place on average — or it can be erratic, jumping around the right place from sample to sample. Total error is the sum of both, and they trade against each other: imposing structure moves you toward a wrong-but-steady answer, and refusing to impose any leaves you with a right-on-average but wildly unsteady one. There is no rule saying zero bias is best. Usually it is not.

Formally. For an estimator $\hat{\vartheta}$ of a scalar ϑ , the mean squared error decomposes as

$$\text{MSE}(\hat{\vartheta}) = \underbrace{(\mathbb{E}[\hat{\vartheta}] - \vartheta)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{\vartheta})}_{\text{variance}},$$

with the cross term vanishing identically. The decomposition extends entrywise to matrices under the Frobenius loss of A.7.

Why it appears here. §6.1 declares this the one axis organising the whole of §6 — “how much structure am I willing to impose in exchange for less estimation error” — with the sample covariance at the zero-bias, maximum-variance end and a single number applied to everything at the other. The document’s recurring instruction to shrink harder than feels comfortable (§9.8, §12.4) is the claim that in this problem the variance term is almost always the binding one.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 7.3.

A.12 Effective sample size, in the Kish sense

The idea. When observations are weighted unequally, the number of rows in your data stops being the number of observations you actually have. A weighting scheme that puts most of its mass on a handful of days has, statistically, only a handful of days in it. The effective sample size is the count of equally weighted observations that would give an estimator the same sampling variance — and it can be dramatically smaller than T without anything in the code hinting at it.

Formally. For non-negative weights $u_1, \dots, u_T$, Kish’s effective sample size is

$$T_{\text{eff}} = \frac{(\sum_t u_t)^2}{\sum_t u_t^2},$$

which equals T when all weights are equal and falls otherwise. For the EWMA of §6.3, $u_t \propto \theta^t$ for $t = 0, 1, 2, \dots$, so $\sum u_t = 1/(1 - \theta)$ and $\sum u_t^2 = 1/(1 - \theta^2)$, giving

$$T_{\text{eff}} = \frac{(1 - \theta^2)}{(1 - \theta)^2} = \frac{1 + \theta}{1 - \theta}.$$

Why it appears here. That last line is where §6.3's $T_{\text{eff}} = (1 + \theta)/(1 - \theta)$ comes from, and it is the source of the trap that section calls out: at RiskMetrics' $\theta = 0.94$ the effective sample is 32 observations, so a 100-asset EWMA correlation matrix has $q_{\text{eff}} = 3.1$ and is meaningless however many years you feed it. §9.4 makes N/T_{eff} , not N/T , the quantity to budget.

Deeper. Kish, *Survey Sampling* (Wiley, 1965), sec. 8.2, where the formula originates in the context of unequal-probability survey designs.

A.13 The Wishart and inverse-Wishart distributions

The idea. If a single sample variance has a chi-squared distribution, the matrix generalisation — the sampling distribution of a whole sample covariance matrix — is the Wishart. It is the answer to “what does S look like across repeated samples”, and its most useful property for this document is not about S at all but about S^{-1} : the inverse of an unbiased estimate is not an unbiased estimate of the inverse, and the Wishart says by exactly how much it is off.

Formally. If $r_1, \dots, r_T$ are IID $\mathcal{N}(0, \Sigma)$ in $\mathbb{R}^N$, then $A = \sum_t r_t r_t' \sim W_N(\Sigma, T)$, the Wishart distribution with scale Σ and T degrees of freedom. It has $\mathbb{E}[A] = T\Sigma$, so $S = A/T$ is unbiased. The distribution of A^{-1} is the inverse-Wishart, and provided $T > N + 1$,

$$\mathbb{E}[A^{-1}] = \frac{\Sigma^{-1}}{T - N - 1} \quad \Rightarrow \quad \mathbb{E}[S^{-1}] = \frac{T}{T - N - 1} \Sigma^{-1} \approx \frac{1}{1 - q} \Sigma^{-1}.$$

Estimating the mean costs one degree of freedom and changes nothing to leading order. The inverse-Wishart also serves as the conjugate prior for Σ in Bayesian work (A.15), which is a separate use of the same family.

Why it appears here. This is §5.6 in one line. The sample covariance is unbiased and its inverse is not, by a factor of about $1/(1 - q)$ — and the optimiser uses the inverse. §12.1 calls that “one fact with three faces”, the other two being the Marchenko–Pastur integral of §5.5 and the inflated $\hat{R}_i^2$ of §5.4; the inverse-Wishart expectation is the face that states it for the whole matrix at once. Note the condition $T > N + 1$: it is A.2's invertibility requirement reappearing as the condition for the expectation to exist.

Deeper. Muirhead, *Aspects of Multivariate Statistical Theory* (Wiley, 1982), chapter 3 — the standard reference for both distributions and the derivation of the expectation above.

A.14 James–Stein shrinkage

The idea. In 1961 James and Stein proved something that still looks like a mistake. If you are estimating three or more means at once and you are scored on total squared error, the obvious estimator — use each sample mean for its own quantity — is inadmissible: pulling all of them toward a common point beats it, always, no matter what the true means are. The individual estimates get worse in exchange for the collection getting better. Every shrinkage estimator in this document, including Ledoit–Wolf, is a descendant of that result.

Formally. Observe $\hat{\mu} \sim \mathcal{N}(\mu, \sigma^2 I_N)$ with $N \geq 3$. The James–Stein estimator shrinking toward the origin is

$$\hat{\mu}^{\text{JS}} = \left(1 - \frac{(N - 2) \sigma^2}{\|\hat{\mu}\|^2}\right) \hat{\mu},$$

and it has strictly smaller total mean squared error than $\hat{\mu}$ for every μ . Shrinking toward the grand mean instead of the origin replaces $N - 2$ by $N - 3$. The shrinkage is data-dependent: noisy, dispersed estimates get pulled hard, tight ones barely at all.

Why it appears here. §6.5's linear shrinkage is this idea applied to a matrix, with the intensity derived rather than guessed; §3's Era IV records Jorion (1986) applying it to the means a decade before it reached the covariance. It is also the cleanest rebuttal to the instinct that unbiasedness is a virtue: §2's whole argument is that the unbiased choice here is the bad one.

Deeper. Efron & Morris, “Stein's Paradox in Statistics,” *Scientific American* 236(5) (1977), 119–127 — six pages, no prerequisites, and it makes the result feel inevitable rather than paradoxical.

A.15 Prior, posterior, the Bayesian update, and MAP

The idea. A Bayesian treats the unknown parameter as itself uncertain. You start with a distribution describing what you believed before seeing data (the prior), the data supply a likelihood, and the two combine into a distribution describing what you believe after (the posterior). The posterior is always a compromise between the two, weighted by their relative precision — which is why “shrinkage” and “a prior” turn out to be the same act described in two vocabularies. If you then want a single number rather than a distribution, taking the posterior’s mode gives the MAP estimate.

Formally. With prior $p(\vartheta)$ and likelihood $p(\text{data} \mid \vartheta)$, Bayes’ rule gives the posterior $p(\vartheta \mid \text{data}) \propto p(\text{data} \mid \vartheta) p(\vartheta)$. The maximum a posteriori estimate is $\hat{\vartheta}_{\text{MAP}} = \arg \max_{\vartheta} p(\vartheta \mid \text{data})$, equivalently the maximiser of $\log p(\text{data} \mid \vartheta) + \log p(\vartheta)$ — a penalised likelihood, with the prior supplying the penalty. In the Gaussian conjugate case the posterior mean is a precision-weighted average of prior mean and sample estimate.

Why it appears here. §7.8’s Black–Litterman is precisely this: equilibrium returns Π as the prior, stated views as the likelihood, μ_{BL} as the posterior. §8.2’s equivalence chain then closes the loop — a ridge penalty $\nu \|w\|^2$ is a MAP estimate under a zero-mean Gaussian prior on w of variance proportional to $1/\nu$, which is why “add a penalty”, “impose a prior” and “shrink the covariance” name one operation. §7.9’s resampling is described as an attempt to approximate the same decision without the apparatus.

Deeper. Gelman, Carlin, Stern, Dunson, Vehtari & Rubin, Bayesian Data Analysis, 3rd ed. (CRC, 2013), chapters 1–2.

A.16 The coefficient of determination, and its adjusted form

The idea. R^2 answers “what fraction of this variable’s variation did my regressors account for?” Its notorious defect is that it can only go up when you add regressors, even regressors made of pure noise — with enough of them you can explain anything perfectly and have learned nothing. The adjusted version charges rent for each regressor, and the size of that rent is the whole story of §5.4.

Formally. For a regression with T observations and k regressors plus an intercept, $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$, the fraction of variance explained. Under the null that no regressor has any relationship to the target, $\mathbb{E}[R^2] = k/(T - 1)$ — not zero. The adjusted version corrects for the degrees of freedom consumed,

$$1 - \bar{R}^2 = (1 - R^2) \frac{T - 1}{T - k - 1},$$

which restores $\mathbb{E}[1 - \bar{R}^2] = 1$ under that null.

Why it appears here. §5.4 defines R_i^2 as the coefficient of determination from regressing asset i on all $N - 1$ others, so $k = N - 1$ and the rent is $(T - 1)/(T - N)$. The overfitting inflation $\mathbb{E}[\hat{R}_i^2] = (N - 1)/(T - 1) \approx q$ is the formula above read straight off, and since asset i ’s weight carries $1/(1 - R_i^2)$ in its denominator, that inflation is exactly the $1/(1 - q)$ weight inflation the rest of §5 derives by two other routes.

Deeper. Greene, Econometric Analysis, 8th ed. (Pearson, 2018), sec. 3.5 — the derivation of both the expectation and the adjustment.

A.17 Mahalanobis distance

The idea. Euclidean distance is the wrong way to ask how unusual an observation is, because it treats a 3% move in a volatile asset and a 3% move in a quiet one as equally surprising, and treats two correlated assets both moving together as twice as surprising as one moving alone. Mahalanobis distance fixes both by measuring in units of the data’s own covariance: it is the Euclidean distance you would get after whitening the data so that every direction has unit variance and no direction is correlated with another.

Formally. For an observation r with mean μ and covariance Σ ,

$$d_{\mathcal{M}}(r)^2 = (r - \mu)' \Sigma^{-1} (r - \mu),$$

a quadratic form (A.1) in the inverse covariance. Under multivariate normality $d_{\mathcal{M}}^2$ is chi-squared with N degrees of freedom, which supplies a calibrated threshold for “unusual”. It is invariant to any non-degenerate linear recombination of the assets.

Why it appears here. §6.12's robust estimators down-weight observations by this distance — fitting a multivariate t does so automatically — and the same quantity is what makes a portfolio-level outlier detectable when no single asset looks extreme. It is also the same algebra as the maximum squared Sharpe ratio $c = \mu' \Sigma^{-1} \mu$ of §5.2, which is worth noticing: the optimiser's own objective is a Mahalanobis length.

Deeper. Mardia, Kent & Bibby, *Multivariate Analysis* (Academic Press, 1979), sec. 1.6 and chapter 3.

A.18 The bootstrap, and the block bootstrap

The idea. You want the sampling distribution of a statistic — how much it would move across alternative histories — and you have one history. The bootstrap's answer is to resample your own data with replacement, recompute the statistic on each resample, and read the spread. It stops working the moment the data have memory, because resampling one observation at a time destroys exactly the serial structure you were relying on. The block bootstrap is the repair: resample stretches of consecutive observations instead of single points.

Formally. Given data $r_1, \dots, r_T$ and a statistic $\hat{\vartheta}$, draw B resamples of size T with replacement, compute $\hat{\vartheta}^{(1)}, \dots, \hat{\vartheta}^{(B)}$, and use their empirical distribution as an estimate of the sampling distribution. The moving-block bootstrap instead draws blocks of consecutive observations of length ℓ and concatenates T/ℓ of them, preserving dependence up to lag ℓ ; the stationary bootstrap randomises ℓ geometrically so that the resampled series is itself stationary. Choosing ℓ trades bias (too short, dependence lost) against variance (too long, too few distinct blocks).

Why it appears here. §7.9's resampled efficiency bootstraps returns, re-optimises on each draw, and averages the weight vectors — and §7.9's better recommendation is to keep the same machinery as a diagnostic, reading the spread of asset 12's weight across resamples rather than the average. §11.5 insists the block version be used for significance testing, “since both volatility and correlation cluster”: an IID bootstrap of financial returns will report standard errors that are too small.

Deeper. Efron & Tibshirani, *An Introduction to the Bootstrap* (Chapman & Hall, 1993); Politis & Romano, “The Stationary Bootstrap,” *JASA* 89(428) (1994), 1303–1313.

A.19 Multiple testing and selection bias

The idea. A 5% significance level means that one test in twenty comes back positive on pure noise. Run twenty configurations and report the best one, and you have manufactured a significant result out of nothing — and this happens whether or not you thought of yourself as running twenty tests. The maximum of many noisy quantities is biased upward, and the bias grows with the number of things you looked at, including the ones you tried and discarded and forgot.

Formally. For M independent test statistics under a global null, the chance of at least one rejection at level α is $1 - (1 - \alpha)^M$, which is 0.64 at $M = 20$, $\alpha = 0.05$. The classical corrections control the family-wise error rate (Bonferroni: test each at α/M) or the false discovery rate (Benjamini–Hochberg). The quantity that matters for a maximum is its expected value: for M independent standard normals, $\mathbb{E}[\max_m Z_m] \approx \sqrt{2 \ln M}$, so the best of a hundred tries scores about 3.0 by luck alone.

Why it appears here. §11.3 names this “the ordinary sin, wearing a matrix for a hat” — shrinkage intensities, thresholds, linkage choices and band widths are all hyperparameters, and tuning them on the evaluation window and then reporting that window's performance is a multiple-testing failure regardless of what it is called. §11.5's instruction to deflate is the quantitative form (A.46). §2.2's error-maximisation argument is the same phenomenon acting on portfolio directions rather than on model configurations.

Deeper. Harvey, Liu & Zhu, “...and the Cross-Section of Expected Returns,” *Review of Financial Studies* 29(1) (2016), 5–68 — the finance-specific version, which argues the field's usual $t > 2$ hurdle should be roughly $t > 3$.

A.20 Shannon entropy

The idea. Entropy measures how spread out a distribution is — not how far from some centre, but how many outcomes genuinely share the probability. A distribution concentrated on one outcome has zero entropy; a uniform one over n outcomes has the maximum. Its exponential is therefore an effective count: the number of equally likely outcomes that would be this spread out. That reading is the whole reason entropy appears in a portfolio document.

Formally. For a discrete distribution $p_1, \dots, p_n$ with $p_i \geq 0$ and $\sum_i p_i = 1$,

$$H(p) = - \sum_{i=1}^n p_i \ln p_i \in [0, \ln n],$$

with $0 \ln 0 = 0$ by convention. $H = 0$ when one p_i is 1, and $H = \ln n$ when all are equal. The perplexity $\exp(H)$ lies in $[1, n]$ and equals n exactly when the distribution is uniform.

Why it appears here. §11.4's effective number of bets is $\exp(H)$ applied to the shares of portfolio variance carried by uncorrelated risk sources (A.44), and the "effective count" reading is what makes it comparable to the reciprocal Herfindahl of A.43. The two constructions are cousins: both take a vector of shares and return a number of things, and both answer n when the shares are equal.

Deeper. Cover & Thomas, Elements of Information Theory, 2nd ed. (Wiley, 2006), chapter 2.

Part III — Optimisation. The document insists the optimisation was never the hard part, which is true and is not a reason to be vague about it. These five entries are what you need to read §5.2, §7.6, §7.10 and §8.2 without taking anything on trust.

A.21 Convexity, and the quadratic programme

The idea. A convex problem is one with no false summits: any local optimum is the global one, so a solver that walks downhill cannot be trapped. That property divides optimisation problems that are effectively solved from ones that are not, and mean-variance sits comfortably on the easy side. A quadratic programme is the specific convex form this document keeps producing — a quadratic objective with linear constraints — for which reliable solvers have existed for decades.

Formally. A set is convex if it contains the segment between any two of its points; a function J is convex if $J(\alpha x + (1 - \alpha)y) \leq \alpha J(x) + (1 - \alpha)J(y)$ for $\alpha \in [0, 1]$. A twice-differentiable J is convex exactly when its Hessian is positive semi-definite, so $w' \Sigma w$ is convex precisely because Σ is PSD (A.2). A quadratic programme is

$$\min_w \frac{1}{2} w' \Sigma w - w' \mu \quad \text{s.t.} \quad w \in \mathcal{C},$$

with $\mathcal{C}$ defined by linear equalities and inequalities — §8.1's constraint set, in other words. It is convex whenever Σ is PSD, and solvable in polynomial time. Jensen's inequality — $\mathbb{E}[J(X)] \geq J(\mathbb{E}[X])$ for convex J — is the probabilistic companion, and matrix inversion is convex in the relevant sense, which is why §5.6 can assert a bias in $\mathbb{E}[S^{-1}]$ before computing it.

Why it appears here. §3 records that Markowitz “posed the problem as a quadratic programme” in 1952, and §1.3's remark that the optimisation is “a first-year exercise” is a statement about convexity. It also draws the boundary of what is easy: §9.6's cardinality constraint (at most k non-zero positions) is non-convex, which is why that row of the table says avoid it.

Deeper. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), chapters 2–4 — free at web.stanford.edu/~boyd/cvxbook.

A.22 Lagrange multipliers, KKT conditions, and complementary slackness

The idea. Constrained optimisation has one governing intuition: at the optimum, the direction you would like to move is exactly cancelled by the constraints holding you back. A Lagrange multiplier is the size of that cancelling force, and it has a price interpretation — how much the objective would improve if the constraint were relaxed by one unit. For inequality constraints there is an extra wrinkle that turns out to be the most consequential fact in §8: a constraint either binds, in which case it has a positive price, or it is slack, in which case its price is zero. Never both. That is complementary slackness, and it is what lets you convert a constrained solution into an unconstrained solution on a modified problem.

Formally. For $\min_w J(w)$ subject to $g_j(w) \leq 0$ and $h_m(w) = 0$, form the Lagrangian $\mathcal{L}(w, \zeta, \xi) = J(w) + \sum_j \zeta_j g_j(w) + \sum_m \xi_m h_m(w)$. The Karush–Kuhn–Tucker conditions, necessary at any optimum under mild regularity and sufficient when the problem is convex, are

$$\begin{aligned} \nabla_w \mathcal{L} &= 0 && \text{(stationarity)} \\ g_j(w) &\leq 0, \quad h_m(w) = 0 && \text{(primal feasibility)} \\ \zeta_j &\geq 0 && \text{(dual feasibility)} \\ \zeta_j g_j(w) &= 0 \text{ for every } j && \text{(complementary slackness)} \end{aligned}$$

The last line is the product of two non-negative numbers being zero, so at least one of them is: either the constraint is tight ($g_j = 0$) or its multiplier vanishes.

Why it appears here. §5.2 uses equality multipliers to trace the efficient frontier, and every scalar in that section (a, b, c, d) falls out of the resulting linear system. Far more importantly, §8.2's load-bearing result is a KKT argument: solving minimum variance subject to $w \geq 0$ gives multipliers $\zeta \geq 0$ on the non-negativity constraints, and complementary slackness makes the constrained solution identical to the unconstrained solution on $\tilde{\Sigma} = \Sigma - \zeta 1' - 1 \zeta'$. Assets that wanted to be shorted have their covariances reduced — which is shrinkage. That is Jagannathan & Ma (2003), and it is why the document can say constraints are estimators rather than concessions.

Deeper. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), chapter 5 — sec. 5.5 is the KKT conditions and sec. 5.6 the sensitivity interpretation of the multipliers.

A.23 The ℓ_1 and ℓ_2 norms; ridge and lasso

The idea. Two ways to say “keep the answer small”, with completely different personalities. The ℓ_2 norm penalises the squares of the weights, so it hates one large position and is fairly relaxed about many small ones — it spreads. The ℓ_1 norm penalises absolute values, so it is equally unhappy about a position of size $2c$ and two of size c , and its optimum tends to sit at corners where many weights are exactly zero — it selects. Ridge and lasso are the regression names for the same two penalties, and the geometry is identical.

Formally. $\|w\|_1 = \sum_i |w_i|$ and $\|w\|_2 = (\sum_i w_i^2)^{1/2}$. Adding $\nu\|w\|_2^2$ to a quadratic objective (ridge) leaves it a QP and, by §8.2, is exactly the same as replacing Σ by $\Sigma + (\nu/\gamma)I$. Adding $\nu\|w\|_1$ (lasso) keeps the problem convex but non-differentiable at zero; splitting $w = w^+ - w^-$ with both parts non-negative makes the constraint linear and restores a QP. In portfolio terms $\|w\|_1$ is gross exposure, so an ℓ_1 constraint and a leverage cap are the same object.

Why it appears here. §7.10 is built on this pair, and DeMiguel and co-authors (2009) show the resulting family nests $1/N$, minimum variance and shrinkage. §8.2 closes the circle: an ℓ_2 penalty on w , a ridge on Σ , a Gaussian prior centred at zero, and linear shrinkage to a scaled identity are four names for one operation. And §10.3 shows a quadratic turnover penalty is the same thing again, centred on w_{prev} rather than on zero.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 3.4 — including the picture of the two constraint regions that explains why one sets coefficients to zero and the other does not.

A.24 Euler’s theorem, and why risk contributions add up

The idea. “Asset i contributes 8% of portfolio risk” sounds like an accounting convention chosen for convenience. It is not — it is forced. Portfolio volatility has the property that doubling every weight doubles it exactly, and for any function with that property there is a unique decomposition into per-asset contributions that sum to the whole. Risk parity is only well posed because that decomposition exists.

Formally. A function f is homogeneous of degree one if $f(\alpha w) = \alpha f(w)$ for $\alpha > 0$. Euler’s theorem then gives $\sum_i w_i \partial f / \partial w_i = f(w)$. Portfolio volatility $\sigma_p(w) = \sqrt{w' \Sigma w}$ is homogeneous of degree one, and its partial derivatives are the marginal contributions of §7.6, so

$$\frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}, \quad \text{RC}_i = w_i \frac{(\Sigma w)_i}{\sigma_p}, \quad \sum_{i=1}^N \text{RC}_i = \frac{w' \Sigma w}{\sigma_p} = \sigma_p.$$

Note that variance would not work: it is homogeneous of degree two, and its Euler decomposition sums to $2 w' \Sigma w$.

Why it appears here. §7.6’s equal-risk-contribution portfolio sets $\text{RC}_i = \text{RC}_j$ for all pairs, which is only a meaningful target because the RC_i exhaust σ_p with nothing left over. It is also why §11.4 can report risk shares as percentages of a whole rather than as loose indicators.

Deeper. Roncalli, *Introduction to Risk Parity and Budgeting* (Chapman & Hall, 2013), chapter 2 — the decomposition and the algorithms that use it, written out.

A.25 Cyclical coordinate descent

The idea. Some problems have no closed-form solution but become trivial if you freeze all the variables but one. Coordinate descent exploits that: cycle through the coordinates, optimising each in turn with the others held fixed, and repeat until nothing moves. It is unglamorous, has no tuning parameters, and for the problems it suits it beats far cleverer methods, because each step is a scalar solve rather than a matrix operation.

Formally. To minimise $J(w)$ over $w \in \mathbb{R}^N$, repeatedly sweep $i = 1, \dots, N$ setting

$$w_i \leftarrow \arg \min_x J(w_1, \dots, w_{i-1}, x, w_{i+1}, \dots, w_N),$$

until the sweep changes w by less than a tolerance. For a convex J that is smooth plus a separable non-smooth part — a sum of per-coordinate terms such as $\nu\|w\|_1$ — the iteration converges to a global optimum. It fails when the non-smooth part couples coordinates, which is the standard cautionary case.

Why it appears here. §7.6 notes that the ERC portfolio “is found numerically, and cyclical coordinate descent converges reliably”: with all other weights fixed, the equal-risk condition for w_i reduces to a scalar quadratic with one positive root, so each update is closed form. The same algorithm is what makes lasso (A.23) practical at scale.

Deeper. Roncalli, Introduction to Risk Parity and Budgeting (2013), sec. 2.2 for the risk-parity version; Friedman, Hastie & Tibshirani, “Regularization Paths for Generalized Linear Models via Coordinate Descent,” Journal of Statistical Software 33(1) (2010), for the general machinery.

Part IV — Time series and clustering. Two smaller toolkits: the first for estimators that let Σ move over time (§6.3, §6.10), the second for estimators that impose group structure on it (§6.11, §7.11).

A.26 Realised volatility

The idea. Volatility is not observed, but unlike expected return it is nearly observable: chop a period into many small intervals, square each return, add them up, and you have a measurement of that period's variation that gets sharper the finer you chop. This is the structural asymmetry the document keeps returning to. Precision in a mean depends on how long you watch; precision in a variance depends on how finely you watch, and only the second is something you can buy.

Formally. Over a period divided into n sub-intervals with returns $r_1, \dots, r_n$, the realised variance is

$$RV = \sum_{j=1}^n r_j^2,$$

which converges to the period's integrated variance as $n \rightarrow \infty$ for a continuous price process. In practice the convergence stops and reverses at high frequency, where microstructure noise — bid-ask bounce, discrete ticks — dominates the signal and inflates RV; the standard remedies are to sample at five minutes rather than at every tick, or to use a noise-robust estimator.

Why it appears here. §5.3's table rests on it: volatilities are “the most forecastable quantity in finance” and correlations are not, which is why §9.5 insists on regularising in correlation space and leaving D alone, and why §6.3 recommends EWMA for volatilities and long windows for correlations. §9.3's Epps effect is the same frequency dial turned the other way, where finer sampling hurts.

Deeper. Andersen, Bollerslev, Diebold & Labys, “Modeling and Forecasting Realized Volatility,” *Econometrica* 71(2) (2003), 579–625.

A.27 GARCH

The idea. Volatility clusters: violent days follow violent days. GARCH is the minimal model of that, and its content is one recursion — today's variance is a blend of a long-run level, yesterday's surprise, and yesterday's variance. Two of those three terms make it a smoothed average of past squared returns, which is EWMA; the third is what distinguishes it, because it pulls the forecast back toward a long-run mean instead of drifting wherever the data went.

Formally. GARCH(1,1) models the conditional variance σ_t^2 of a return ε_t with mean zero as

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \omega > 0, \quad \alpha_1, \beta_1 \geq 0,$$

stationary when $\alpha_1 + \beta_1 < 1$, with unconditional variance $\omega/(1 - \alpha_1 - \beta_1)$ and forecasts decaying toward it at rate $\alpha_1 + \beta_1$. Setting $\omega = 0$ and $\alpha_1 + \beta_1 = 1$ recovers the EWMA of §6.3 with $\theta = \beta_1$, which is why RiskMetrics is sometimes called integrated GARCH. (The coefficients α_1, β_1 here are the universal GARCH notation and have nothing to do with §5.4's residual alpha α_i or regression betas β_{ij} .)

Why it appears here. §6.10's DCC fits a univariate GARCH to each asset to obtain $\hat{D}_t$ before modelling correlation separately — the split of §5.3 made into an algorithm. The mean-reversion property is also the honest answer to why EWMA is not enough: an EWMA forecast at any horizon is today's estimate, while GARCH knows that calm follows storms.

Deeper. Bollerslev, “Generalized Autoregressive Conditional Heteroskedasticity,” *Journal of Econometrics* 31(3) (1986), 307–327; and Engle (2002) for the multivariate step.

A.28 Principal component analysis

The idea. Given many correlated series, find the single combination that captures the most variation, then the best combination orthogonal to it, and so on. The result is a new set of coordinates in which the data are uncorrelated and ordered by importance — and for returns, the first coordinate is almost always “everything, long”, the market. PCA is not a separate technique from the eigendecomposition of A.3; it is the eigendecomposition of the covariance matrix, given a statistical reading.

Formally. With $\Sigma = V\Lambda V'$ as in A.3, the i th principal component is the portfolio v_i , its variance is λ_i , and the components are mutually uncorrelated: $v_i' \Sigma v_j = 0$ for $i \neq j$. The share of total variance explained by the first K components is

$\sum_{i \leq K} \lambda_i / \sum_i \lambda_i$, the denominator being $\text{tr}(\Sigma)$ by A.4. Truncating at K gives the best rank- K approximation to Σ in Frobenius norm.

Why it appears here. §6.9's statistical factor models take the leading K principal components as factors, with K chosen by the Marchenko–Pastur edge — which is the honest version of the usual scree-plot eyeballing, since §5.5 tells you in advance where noise ends. §11.4's effective number of bets (A.44) uses the same decomposition for a different purpose: not to reduce dimension but to count independent risks. The known weakness is that principal components rotate over time and resist interpretation, which §6.9 lists as its main failure mode.

Deeper. Jolliffe, *Principal Component Analysis*, 2nd ed. (Springer, 2002), chapters 1–3.

A.29 The correlation-to-distance metric

The idea. Clustering algorithms need distances, and a correlation is not one: it runs the wrong way (high means close), it is not zero for identical objects, and it does not obey the triangle inequality. The standard conversion fixes all three at once, and it is not an arbitrary rescaling — the resulting quantity is literally the Euclidean distance between the two assets' return series after each has been demeaned and scaled to unit length.

Formally. For correlation $\rho_{ij} \in [-1, 1]$, define

$$d_{ij} = \sqrt{2(1 - \rho_{ij})} \in [0, 2].$$

This is a proper metric: $d_{ii} = 0$, symmetric, and it satisfies the triangle inequality. The reason is direct — if x and y are the two demeaned return series normalised so $\|x\| = \|y\| = 1$, then $x'y = \rho_{ij}$ and $\|x - y\|^2 = 2 - 2\rho_{ij} = d_{ij}^2$. Perfectly correlated assets sit at distance 0, uncorrelated ones at $\sqrt{2}$, perfectly anti-correlated at 2.

Why it appears here. §6.11 and §7.11 both open with this conversion, and it is the first step of hierarchical risk parity. Note the consequence that anti-correlated assets are treated as maximally far apart, which is right for a taxonomy and arguably wrong for a portfolio, where a strong hedge is a close relative. Using $\sqrt{2(1 - |\rho_{ij}|)}$ instead is the common variant, and it is a genuine modelling choice rather than a detail.

Deeper. Mantegna, “Hierarchical Structure in Financial Markets,” *European Physical Journal B* 11 (1999), 193–197 — the paper that introduced the metric to finance.

A.30 Hierarchical clustering, and linkage

The idea. Given distances between objects, repeatedly merge the two closest into a group, then treat that group as an object and continue, until everything is one tree. The output is not a partition but a nested hierarchy — a dendrogram — which you can cut at any height to get however many clusters you want. The one genuinely arbitrary decision is how to measure the distance between two groups once they contain more than one member, and that choice is called the linkage.

Formally. Agglomerative clustering starts with N singletons and merges the nearest pair at each of $N - 1$ steps. Given clusters $\mathcal{A}$ and $\mathcal{B}$, the common linkages are

$$\begin{array}{ll} \text{single:} & \min_{i \in \mathcal{A}, j \in \mathcal{B}} d_{ij} \\ \text{complete:} & \max_{i \in \mathcal{A}, j \in \mathcal{B}} d_{ij} \\ \text{average:} & \frac{1}{|\mathcal{A}||\mathcal{B}|} \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{B}} d_{ij} \\ \text{Ward:} & \text{merge that minimises the increase in within-cluster variance} \end{array}$$

Single linkage chains — it will happily string a long thin cluster through the space — while complete and Ward produce compact, roughly equal-sized groups. The cost is $O(N^2 \log N)$.

Why it appears here. §6.11 uses the tree to impose block structure on the correlation matrix and §7.11 uses it to order and bisect the universe without ever inverting anything. Both sections flag the same two weaknesses, and they are real: the linkage is “a free parameter with real consequences and no principled selection rule”, and the tree is unstable — small perturbations to $\hat{C}$ can reorganise it entirely, which propagates straight into weights. A tree also cannot represent an asset that genuinely belongs to two groups, which is most of them.

Deeper. Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, 2nd ed. (Springer, 2009), sec. 14.3.12; Murtagh & Contreras, "Algorithms for Hierarchical Clustering: An Overview," *WIREs Data Mining and Knowledge Discovery* 2(1) (2012), 86–97.

Part V — Finance. The domain vocabulary. Several of these are not deep — a basis point is a unit — but a reader who has to guess at them will misread §10 and §11 entirely, and the last few carry more content than their names suggest.

A.31 Basis points, and two-way turnover

The idea. Two units of account that the cost and evaluation sections use without ceremony. A basis point exists because percentages are too coarse for trading costs: a spread of “0.03%” is awkward to say and easy to mistype, and “3 bp” is neither. Turnover exists because “how much did you trade” needs a denominator, and the convention is capital — but the convention also double-counts by design, since every rebalance both sells something and buys something, and you pay on both.

Formally. One basis point (bp) is 10^{-4} , so 1% is 100 bp and 5 bp is 0.05%. Two-way turnover over a rebalance is

$$\text{TO} = \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|,$$

the total absolute weight traded, buys and sells together, expressed as a fraction of capital and annualised by multiplying by the number of rebalances per year. Selling one asset entirely and buying another entirely is $\text{TO} = 2$, or 200%; the one-way convention would call the same trade 100%, which is why the qualifier is not decoration. Cost per rebalance is then roughly $\text{TO} \times (\text{cost in bp})$, with the caveats of A.41.

Why it appears here. §10.2 quotes spreads and fees as “1–5 bp for liquid futures and large caps”, §11.4 asks for turnover “as annualised two-way fraction of capital ... the number that converts directly into cost”, and §6.6 justifies nonlinear shrinkage by “a few basis points of variance reduction”. Getting the one-way/two-way convention wrong is a factor-of-two error in every cost estimate downstream.

Deeper. Any institutional trading-cost analysis primer; the conventions are industry usage rather than theory.

A.32 Excess returns, and the Sharpe ratio

The idea. Return alone ranks nothing, because you can manufacture return with leverage. The Sharpe ratio is the fix: return per unit of volatility, measured above what you could have earned doing nothing. Everything in the ratio is deliberate. Subtracting the risk-free rate makes it a reward for taking risk rather than for owning money; dividing by volatility makes it invariant to leverage, so two managers running the same strategy at different sizes get the same score.

Formally. With r_f the risk-free rate, the excess return is $r - r_f$, and this is what the document’s μ always holds. For a portfolio with excess return mean $w'\mu$ and volatility $\sigma_p = \sqrt{w'\Sigma w}$,

$$\text{SR} = \frac{w'\mu}{\sqrt{w'\Sigma w}}.$$

It is scale-invariant — $\text{SR}(\alpha w) = \text{SR}(w)$ for $\alpha > 0$ — and is conventionally annualised by multiplying a per-period Sharpe by $\sqrt{\text{periods per year}}$, which is valid only under serial independence.

Why it appears here. §5.1’s Problem A maximises it, and its scale invariance is why the tangency portfolio’s leverage is undetermined — §1.3’s separation result appearing as a property of the objective. §2.1’s headline finding is a Sharpe ratio comparison (5.32 reported against 0.43 available), and §11.3’s rule to compare methods only at matched volatility is the same invariance used defensively.

Deeper. Lo (2002), “The Statistics of Sharpe Ratios,” including what serial correlation does to the $\sqrt{\text{periods}}$ annualisation.

A.33 The efficient frontier, and the tangency portfolio

The idea. Plot every portfolio you could hold as a point with volatility on one axis and expected return on the other. The cloud has a left boundary — for each level of return, the lowest risk that achieves it — and that boundary is the efficient frontier. Now add cash, which sits on the vertical axis at zero risk. Mixing cash with any risky portfolio traces a straight line from that point, and the best such line is the one that just touches the frontier. The touching point is the tangency portfolio, and its slope is the highest Sharpe ratio available. That is where the name comes from, and the main text never says it.

Formally. With $a = \mathbf{1}'\Sigma^{-1}\mathbf{1}$, $b = \mathbf{1}'\Sigma^{-1}\mu$, $c = \mu'\Sigma^{-1}\mu$, $d = ac - b^2$, the frontier is the hyperbola $\sigma^2(m) = (am^2 - 2bm + c)/d$ in (σ, m) space, with left vertex at the GMV portfolio $(1/\sqrt{a}, b/a)$. The tangency portfolio is

$$w_{\text{tan}} = \frac{\Sigma^{-1}\mu}{\mathbf{1}'\Sigma^{-1}\mu} = \frac{\Sigma^{-1}\mu}{b},$$

and the line from the origin through it — the capital market line — has slope $\sqrt{c} = \sqrt{\mu'\Sigma^{-1}\mu}$, the maximum attainable Sharpe ratio.

Why it appears here. §5.2 derives all of this, and §5.2's two-fund separation says every efficient portfolio is an affine mix of the GMV and tangency portfolios. The reason it matters practically is §2.1's flat realised frontier: sliding along the frontier means sliding toward the $\Sigma^{-1}\hat{\mu}$ end, and §5.2 identifies that end as the noisiest object in the problem. The reported $\sqrt{\hat{c}}$ was the 5.32 of §2.1.

Deeper. Markowitz (1952) for the frontier; Tobin (1958) for the separation that introduces the tangency point.

A.34 CAPM, equilibrium returns, and reverse optimisation

The idea. If everyone solves the same mean-variance problem with the same inputs, then in equilibrium the sum of everyone's holdings must be the market itself — so the market portfolio is the tangency portfolio, and expected returns must be whatever makes that true. Run the logic backwards and you get a genuinely useful trick: instead of forecasting returns and deriving weights, take the observed market weights and derive the returns that would justify them. Those are the equilibrium returns, and they are a defensible neutral starting point that required no forecasting skill whatsoever.

Formally. CAPM states $\mathbb{E}[r_i] - r_f = \beta_i (\mathbb{E}[r_{\text{mkt}}] - r_f)$ with $\beta_i = \text{Cov}(r_i, r_{\text{mkt}}) / \text{Var}(r_{\text{mkt}})$. Reverse optimisation inverts the master form of §1.3: if w_{mkt} is optimal at risk aversion γ , then

$$\Pi = \gamma \Sigma w_{\text{mkt}},$$

the vector of implied equilibrium excess returns. This is $\mu = \gamma \Sigma w$ solved for μ rather than for w , and it requires no estimation of means at all — only Σ , the observed weights, and a scalar.

Why it appears here. §7.8's Black–Litterman starts from exactly this Π and updates it with views (A.15), and §8.2 records that with no views the posterior collapses back to reverse-optimised market weights. It is also the general technique behind §9.6's advice to “always solve the unconstrained problem too” — reverse-optimising any portfolio tells you what its holder must believe, which is often more informative than asking them.

Deeper. Sharpe, “Capital Asset Prices,” *Journal of Finance* 19(3) (1964), 425–442; and Black & Litterman (1992) for the reverse-optimisation step in use.

A.35 Cap-weighted benchmarks

The idea. The default way to build an index is to hold every constituent in proportion to its market value. This is not one convention among many — it has two properties nothing else has. It is the only weighting that everyone can hold simultaneously, since it is what the market actually consists of; and it needs no rebalancing as prices move, because a stock that doubles also doubles its own weight. Zero turnover and universal capacity are why it is the benchmark, and both are properties of the arithmetic rather than of any performance claim.

Formally. Asset i 's weight is $w_i = \text{Cap}_i / \sum_j \text{Cap}_j$ where Cap_i is shares outstanding times price (usually adjusted for free float). Because Cap_i moves with price, the weights self-maintain between index reconstitutions. Under CAPM (A.34) this portfolio is the tangency portfolio, so cap-weighting is mean-variance optimal exactly when CAPM holds.

Why it appears here. §7.4 reports that long-only minimum-variance portfolios “have outperformed cap-weighted benchmarks on a risk-adjusted basis over long samples in most equity markets”, and §12.5 repeats it as a [Fact]. The comparison is the sharp one because of the last sentence above: beating cap-weighting on risk-adjusted terms is evidence against CAPM, which is why the finding is interesting rather than merely commercial. Note also what it costs — every alternative weighting, including $1/N$, requires rebalancing that cap-weighting does not, and §7.2 warns that comparisons routinely omit it.

Deeper. Sharpe, “Indexed Investing: A Prosaic Way to Beat the Average Investor” (1976, reprinted widely) for the arithmetic argument; the free-float and reconstitution mechanics are in any index provider's methodology document.

A.36 The low-volatility anomaly

The idea. CAPM says more risk earns more return. Empirically the relationship is far flatter than predicted, and over long samples has often been inverted: portfolios of low-volatility, low-beta stocks have delivered similar or better returns than high-volatility ones, with far less risk. This is one of the most persistent embarrassments in asset pricing, and it matters here for an unflattering reason — any strategy that systematically prefers quiet assets will pick it up whether or not it meant to.

Formally. Sorting stocks on trailing volatility or on market beta and going long the lowest quintile, short the highest, has produced positive risk-adjusted returns in most equity markets over multi-decade samples. The leading explanations are leverage constraints (investors who want more return and cannot borrow bid up high-beta stocks instead) and benchmarking (a manager measured against a cap-weighted index treats low-beta stocks as risky).

Why it appears here. §7.4 and §12.5 both flag this as the standing [Contested] caveat on minimum-variance performance: much of the measured outperformance may be exposure to this factor rather than skill in construction, and §3's Era VI says the same about the risk-based turn generally. The practical consequence is a test — if you cannot distinguish your construction method's contribution from a low-volatility tilt, you have not measured construction.

Deeper. Frazzini & Pedersen, "Betting Against Beta," *Journal of Financial Economics* 111(1) (2014), 1–25; Black, Jensen & Scholes (1972) for the original flat-line finding.

A.37 The information coefficient

The idea. How good is a forecast? Not "how often is the sign right" and not "what is its R^2 ", but the cross-sectional correlation between what you predicted and what happened. It is a useful measure precisely because it is scale-free: it grades the ranking, which is what a cross-sectional portfolio consumes, and ignores the calibration, which is a separate and easier problem. The sobering part is the range. A genuinely valuable equity signal has an IC of perhaps 0.03 to 0.05, and anything above 0.10 out of sample should be assumed broken until proven otherwise.

Formally. For forecasts $\hat{\mu}_i$ and realised returns r_i across N assets in one period,

$$\text{IC} = \text{Corr}(\hat{\mu}_i, r_i),$$

usually computed on ranks (Spearman) rather than levels, and averaged over periods. Its relationship to a return forecast is direct: under a standard normalisation, $\mathbb{E}[r_i | \hat{\mu}_i] \approx \text{IC} \cdot \sigma_i \cdot z_i$ where z_i is the cross-sectional z-score of the signal — which is exactly the industry recipe §7.7 describes.

Why it appears here. §7.7's tamed mean-variance converts a signal to cross-sectional z-scores, clips at ± 2 or ± 3 , and multiplies by "a modest assumed information coefficient". That multiplication is what sets the dispersion of $\hat{\mu}$, and §7.7 is emphatic that dispersion, not accuracy, is what determines the damage. §10.5 then notes that two signals with identical ICs but different half-lives deserve different weights, which is the point where IC alone stops being sufficient.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed. (McGraw-Hill, 1999), chapter 6.

A.38 The transfer coefficient, and the fundamental law of active management

The idea. Grinold's law is an attempt to say what a manager's performance should be, from two numbers: how good each forecast is, and how many independent forecasts they make. Skill times the square root of breadth. The original version assumes you can hold whatever the forecasts imply, which nobody can — so the extended version adds a third factor measuring how much of the intended portfolio survived the constraint set. That factor is the transfer coefficient, and it turns "our constraints cost us something" from a complaint into a number.

Formally. With IC as in A.37 and breadth BR the number of independent bets per year, the information ratio satisfies $\text{IR} \approx \text{IC} \sqrt{\text{BR}}$. The generalised form inserts the transfer coefficient,

$$\text{IR} \approx \text{TC} \cdot \text{IC} \sqrt{\text{BR}}, \quad \text{TC} = \frac{w' \Sigma w^*}{\sqrt{w' \Sigma w} \sqrt{w^{*'} \Sigma w^*}},$$

the correlation between the portfolio actually held and the unconstrained optimum w^* , measured in risk-adjusted terms — that is, with Σ in the inner product rather than a plain correlation of the weight vectors. Typical long-only institutional values are 0.3 to 0.6, meaning most of the signal is discarded at the construction step.

Why it appears here. §11.4 lists TC as one of five diagnostics that actually discriminate, with the right reading attached: a low value “is not necessarily bad — §8.2 says constraints are estimators — but it should be a decision, not a surprise.” Read alongside §9.6’s advice to always solve the unconstrained problem too, this is the number that quantifies the answer.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed. (1999), chapter 6; Clarke, de Silva & Thorley, “Portfolio Constraints and the Fundamental Law of Active Management,” *Financial Analysts Journal* 58(5) (2002), 48–66, for the transfer coefficient itself.

A.39 Statistical arbitrage and relative-value books

The idea. A relative-value position is a bet that two things which normally move together have drifted apart and will reconverge. You are long one and short the other, so the shared exposure — the market, the sector — cancels, and what is left is a small, uncorrelated, high-turnover spread. Statistical arbitrage is that done systematically across hundreds of pairs or residuals at once. The reason it appears in a covariance document is that it lives entirely inside the part of the matrix that most risk models throw away.

Formally. In the language of §5.4, a relative-value book holds positions in residual space: asset i hedged against its replicating portfolio, with the position sized by α_i/s_i^2 where $s_i^2 = \sigma_i^2(1 - R_i^2)$ is the residual variance. Because R_i^2 is high by construction — the whole point is to pick assets that are well replicated — such books operate exactly where the $1/(1 - R_i^2)$ amplifier is largest.

Why it appears here. §6.8 names the failure precisely: a factor model with diagonal Ψ asserts that residuals are uncorrelated, so “two airlines have correlated residuals after every standard factor, and a pairs strategy trading that residual will be told by its risk model that the position is riskless.” §6.7 adds the same warning for eigenvalue clipping, which destroys tightly cointegrated structure. §3’s Era III makes the general point: what factor structure discards is exactly what a statistical-arbitrage desk trades. If this is your business, §6.9’s POET is the recommendation.

Deeper. Avellaneda & Lee, “Statistical Arbitrage in the US Equities Market,” *Quantitative Finance* 10(7) (2010), 761–782.

A.40 Cointegration

The idea. Two prices can each wander without limit while the gap between them stays bounded — tethered, not anchored. That is cointegration, and it is a much stronger statement than correlation. Correlation is about co-movement of returns over some window and can be high by accident; cointegration is about a long-run equilibrium relationship in levels, which implies the spread must eventually revert. It is the formal object underneath every pairs trade.

Formally. Series that are individually non-stationary with a unit root (integrated of order one) are cointegrated if some linear combination of them is stationary: there exists $u \neq 0$ with $u'p_t$ stationary, where p_t stacks the levels. The Engle–Granger procedure tests this by regressing one series on the other and testing the residual for a unit root; the Johansen procedure handles several series and multiple cointegrating relationships at once. Cointegration implies an error-correction representation, in which returns respond to the current size of the disequilibrium.

Why it appears here. §6.7 warns that eigenvalue clipping “destroys genuine small-eigenvalue structure such as tightly cointegrated pairs — which is fatal if that structure is what you trade.” That sentence is the crux of the tension in §6: the smallest eigenvalues are where the noise is and where the real spread relationships are, and no rotationally invariant estimator (A.5) can tell them apart. If you hold cointegrated positions, the aggressive spectral repairs of §6.6 and §6.7 are removing your book, not your noise.

Deeper. Engle & Granger, “Co-integration and Error Correction,” *Econometrica* 55(2) (1987), 251–276; Hamilton, *Time Series Analysis* (1994), chapters 19–20.

A.41 Market impact, and the participation rate

The idea. Trading moves the price against you, and it does so in a specific, robustly measured shape: cost per share grows roughly with the square root of how much of the day’s volume you are trying to take. Square root, not linear — which means large trades are cheaper per share than a linear model predicts, but total cost still grows faster than size. The variable that governs everything is not how many dollars you trade but what fraction of the available liquidity you consume.

Formally. With x the order size and ADV the average daily volume, the participation rate is $POV = x/ADV$, and the standard impact model is

$$\text{cost per share} \approx \text{const} \times \sigma \times \sqrt{POV},$$

with σ the asset's daily volatility. Total cost is that times x , so it grows as $x^{3/2}$ — superlinear, which is what makes size genuinely costly. Impact is conventionally split into temporary (price recovers after you stop) and permanent (it does not).

Why it appears here. §10.2 states the square-root form as [Fact] with the exponent [Contested] between roughly 0.4 and 0.6, then makes the modelling choice that matters: for construction purposes it uses a quadratic cost $\frac{1}{2}(w - w_{\text{prev}})' \Lambda_c (w - w_{\text{prev}})$ instead, which is not the true cost function but is what makes §10.3's closed form possible. §10.4's proportional costs are a third shape, producing a no-trade band — three shapes, three qualitatively different policies, which is why §10.2 insists the scaling matters more than the level. (§10.2's Λ is a diagonal cost matrix, not the notation block's eigenvalue matrix.)

Deeper. Almgren & Chriss (2000), “Optimal Execution of Portfolio Transactions”; Tóth et al., “Anomalous Price Impact and the Critical Nature of Liquidity,” *Physical Review X* 1 (2011), 021006, for the square-root law's empirical status.

A.42 Winsorising

The idea. An extreme observation can dominate a variance estimate all by itself, because variance squares things. Winsorising limits the damage by pulling outliers in to a threshold rather than removing them: the observation still counts, it just stops counting extra. That distinction matters — deleting a point changes the sample size and can bias the estimate, while capping it keeps every observation and only bounds its influence.

Formally. Given a chosen quantile level u (commonly 0.01), winsorising replaces every observation below the u quantile with that quantile's value and every observation above the $1 - u$ quantile with that one. Truncation or trimming discards them instead. In cross-sectional signal work the same operation is usually applied to z-scores with a fixed cap — clip at ± 2 or ± 3 — rather than to an empirical quantile.

Why it appears here. It plays two opposite roles and the document separates them carefully. §6.12 lists it as a robust covariance technique and then argues against it — in markets the outliers are usually the information, and §9.2's rule is to flag any return beyond $\pm 50\%$ and inspect it, since a real crash and a missed split look identical to a winsoriser. §7.7's clipping of signal z-scores is the role where it is unambiguously right, because there the extremes are estimation error by construction.

Deeper. Huber & Ronchetti, *Robust Statistics*, 2nd ed. (Wiley, 2009), chapters 1–2.

A.43 The Herfindahl index, and effective counts

The idea. Borrowed from industrial organisation, where it measures how concentrated an industry is: square every firm's market share and add them up. The useful property is what the reciprocal means. If n things share the total equally, the sum of squared shares is $1/n$ and the reciprocal is n — so the reciprocal always reads as “the number of equally-sized things this is equivalent to”, however unequal the actual distribution is. That is the same “effective count” construction as Kish's sample size (A.12) and the perplexity of A.20, and noticing that they are one idea saves learning three.

Formally. For shares $p_i \geq 0$ summing to one, $\text{HHI} = \sum_i p_i^2 \in [1/n, 1]$, and the effective number is $1/\text{HHI} \in [1, n]$. Applied to a portfolio, the shares must be gross-normalised — $p_i = |w_i| / \sum_j |w_j|$ — because on a levered or long-short book the raw squared weights measure leverage rather than breadth.

Why it appears here. §11.4 puts this among the five diagnostics worth reporting, with the observation that lands hardest: “a 500-name portfolio with an effective count of 12 is a 12-name portfolio.” It is the natural companion to §7.4's warning that unconstrained minimum variance is notoriously concentrated, and to Clarke, de Silva & Thorley's (2011) characterisation of exactly which assets survive. But it counts positions, and §11.4's next diagnostic exists because positions are not what diversifies you.

Deeper. Hirschman, “The Paternity of an Index,” *American Economic Review* 54(5) (1964), 761 — a one-page note on the index's own history.

A.44 The effective number of bets

The idea. Forty bank stocks is forty positions and one bet. The effective number of bets fixes what the Herfindahl index cannot see, by counting independent risks rather than holdings. The recipe is to rotate the portfolio into uncorrelated components, ask what share of total variance each one carries, and then apply the effective-count construction to those shares. Because the components are uncorrelated by construction, the resulting number cannot be inflated by holding the same bet many times.

Formally. Decorrelate the portfolio's risk — by PCA (A.28) on Σ , or by Meucci's minimum-torsion rotation, which stays closer to the original assets. Write p_i for the fraction of portfolio variance carried by the i th uncorrelated component, so $\sum_i p_i = 1$. Then

$$\text{ENB} = \exp(H(p)) = \exp\left(-\sum_{i=1}^N p_i \ln p_i\right) \in [1, N],$$

the perplexity of A.20. It equals N when every independent risk source carries an equal share, and 1 when one source carries everything.

Why it appears here. §11.4 calls the gap between this and the effective number of positions “the honest measure of diversification”, with the forty-bank-stocks example: effective positions near forty, effective bets near two, “and only the second number tells you what will happen in a crisis.” It is also the right counterweight to §7.5's diversification ratio, which §7.5 admits can be gamed by adding near-duplicates — a manoeuvre that raises the DR and leaves ENB flat.

Deeper. Meucci (2009), “Managing Diversification,” where the measure is introduced; Meucci, Santangelo & Deguest (2015) for the minimum-torsion rotation.

A.45 The standard error of a Sharpe ratio

The idea. A Sharpe ratio is an estimate, and like any estimate it has a confidence interval — which in this case is embarrassingly wide. The intuition is that a Sharpe ratio is essentially a t-statistic divided by the square root of the sample length, so its precision is governed by calendar span and nothing else. Sampling the same three years more finely does not help. This is A.26's asymmetry again, seen from the performance-measurement side.

Formally. For IID returns, the estimated Sharpe ratio over n years has

$$\text{se}(\widehat{\text{SR}}) \approx \sqrt{\frac{1 + \text{SR}^2/2}{n}},$$

so distinguishing a true Sharpe of 0.5 from zero at 95% confidence needs roughly 16 years, and distinguishing 0.5 from 1.5 needs longer still. Serial correlation inflates this further; Lo (2002) gives the corrected form, and positively autocorrelated returns — common in illiquid books — make the naive annualisation overstate the Sharpe ratio as well as its precision.

Why it appears here. §11.5 works the arithmetic: at a true Sharpe of 0.5 over ten years the standard error is 0.34, and comparing two methods at 0.5 and 1.5 gives a difference of 1.0 against a standard error of 0.57 — under two standard errors, and therefore not a result. The escape §11.5 offers is to compare the paired difference of the two return series rather than their levels, since two construction methods on the same universe share nearly all their market exposure and the difference has far lower volatility than either leg.

Deeper. Lo (2002), “The Statistics of Sharpe Ratios,” *Financial Analysts Journal* 58(4), 36–52.

A.46 The deflated Sharpe ratio

The idea. If you try many configurations and report the best, the number you report is not an estimate of that configuration's quality — it is the maximum of a sample of noisy quantities, and maxima are biased upward. The deflated Sharpe ratio asks the honest question: given that I ran M trials, how large would the best result have been by luck alone, and does mine beat that? It converts “my best backtest scored 1.4” into a testable statement, and the answer is often that 1.4 is unremarkable.

Formally. Combine the multiple-testing arithmetic of A.19 with the standard error of A.45. If M independent trials each have a Sharpe ratio with standard error se , the expected best of them under a null of no skill is about $\text{se} \sqrt{2 \ln M}$, so the benchmark to beat rises with $\sqrt{\ln M}$ rather than staying at zero. The deflated Sharpe ratio is the probability that the observed maximum exceeds this null-adjusted threshold, and Bailey and López de Prado's version additionally corrects for skewness, excess kurtosis and the effective number of independent trials, which is smaller than M when configurations are correlated.

Why it appears here. §11.5 prescribes it directly: “if you evaluated twelve shrinkage intensities, the best of twelve is biased upward and the deflated Sharpe ratio adjustment quantifies by how much.” Note how easily the trial count grows here — shrinkage intensity, target matrix, window length, rebalance frequency, constraint set, linkage method — which is §11.3's “ordinary sin, wearing a matrix for a hat.”

Deeper. Bailey & López de Prado, “The Deflated Sharpe Ratio,” *Journal of Portfolio Management* 40(5) (2014), 94–107; Harvey & Liu, “Backtesting,” *Journal of Portfolio Management* 42(1) (2015), 13–28.

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